

TRAJECTORY OPTIMIZATION

-LECTURE NOTE

Sang-Young Park

Astroynamics & Control Lab.

Yonsei University

2011 Fall

A Brief History of Optimal Control Theory

Simple Optimizayion Problems

- The shortest path joining two points: a straight line segment.
- Isoperimetric problem (i.e., Dido's problem) to find the plane curve of a given length that encloses the largest possible area : the curve is a circle.

1638 Galileo posed two shape problems;

- The shape of a heavy chain suspended between two points (the catenary).
- The shape of a wire such that a bead sliding along its end-part in minimum time (the brachystochrone).

1662 Fermat found that Light chooses the path in minimum time

1685 Newton was the first to solve the shape of a body with minimum drag. He did not publish the result until 1694.

1696 Johann Bernoulli (the Netherlands, professor of univ. of Basel) Challenged the Brachystochrone problem

6 people; Johann Bernoulli, Jakob Bernoulli, Leibniz, l'Hopital, Tschirnhaus, Newton.

1744 Euler(1707-1783) published the collected works (when he was 13)

"Nothing at all takes places in the universe in which some rule of maximum or minimum does not appear."

Euler formulated the problem in general terms.

; Finding the curve $q(t)$ over the internal $a \leq t \leq b$ with given values $q(a)$, $q(b)$ which minimize

$$J = \int_a^b L(t, q(t), \dot{q}(t)) dt \quad (1)$$

For some given function $L(t, q(t), \dot{q}(t))$

a necessary condition of optimality for the curve q

$$\frac{d}{dt} \frac{\partial L}{\partial \dot{q}} = \frac{\partial L}{\partial q} \quad (2)$$

→ solution technique; essential geometric

To compute left side of (2), one first evaluate $\frac{\partial L}{\partial \dot{q}}$ treating $\dot{q}$ as an independent variable (not $\dot{q}(t)$, foundation of time), then plugs in $q(t)$ and $\dot{q}(t)$ for q and $\dot{q}$, and finally differentiates with respect to t .

$$\frac{d}{dt} \left[\frac{\partial L}{\partial u}(q(t), \dot{q}(t), t) \right] = \frac{\partial L}{\partial q}(q(t), \dot{q}(t), t) \quad (2')$$

where $L(q, u, t)$ is a function of $q \in \mathbf{R}^n, u \in \mathbf{R}^n, t \in \mathbf{R}^n$ (i.e. $u \equiv \dot{q}(t)$).

- 1755 Lagrange(1736-1813) described an analytical approach (when he was 19)
 → based on “variations” of the optimal curve and using his “undetermined multipliers”
 → which led directly to Euler’s necessary condition Euler adopted this approach and rename the subject “the calculus of variations”
 Euler - Lagrange equation based on first - order variations, yields only a stationary condition

- 1786 Legendre(1752-1833) studied the second variation and produced a second - order necessary condition of optimality(minimum).

$$\frac{\partial^2 L}{\partial \dot{q}^2}(t, q(t), \dot{q}(t)) \geq 0 \quad (3)$$

Legendre derived it for the scalar case, but it was later extended to the vector case by Clebsch(1833-1872).

→ now known as Legendre - Clebsch condition.

$$\frac{\partial^2 L}{\partial \dot{q}_i \partial \dot{q}_j} \geq 0, \quad 1 \leq i, j < n \quad ; \text{ the Hessian matrix}$$

<Motivation> The Legendre condition is clearly the second order necessary condition for a minimum of a function, but eq(2) does not look at all like the first order condition for a minimum of that same function.

=> It is natural to ask whether there might be a way to relate the two conditions.

Is it possible that both can be expressed as necessary conditions for a minimum of one and the same function?

=> The maximum principle.

[(Solution)

Let us look at another way of writing (2)

$$\mathbf{H}(q, u, p, t) \equiv \langle p, u \rangle - L(q, u, t) \quad (\text{i})$$

$$\dot{q}(t) \equiv u$$

$$\text{Define } p(t) \equiv \frac{\partial L}{\partial u}(q(t), \dot{q}(t), t) \quad (\text{ii})$$

$$\frac{\partial \mathbf{H}}{\partial p} = u$$

$$\dot{q}(t) = u = \frac{\partial \mathbf{H}}{\partial p} \Rightarrow \dot{q}(t) = \frac{\partial \mathbf{H}}{\partial p} \quad (\text{iii})$$

$$\text{Eq(2')} \quad \frac{d}{dt} \left[\frac{\partial L}{\partial u} \right] = \frac{\partial L}{\partial q}, \quad \frac{d}{dt} [p] = \frac{\partial L}{\partial q}, \quad \frac{\partial \mathbf{H}}{\partial q} = - \frac{\partial L}{\partial q}$$

$$\frac{dp}{dt} = - \frac{\partial \mathbf{H}}{\partial q} \quad (\text{iv})$$

$$\text{We know } \frac{\partial \mathbf{H}}{\partial u} = p - \frac{\partial L}{\partial u}$$

So, equation (ii) gives

$$\frac{\partial \mathbf{H}}{\partial u} = 0 \quad (\text{v})$$

The system of eqs usually written more concisely as

$$\frac{dq}{dt} = \frac{\partial \mathbf{H}}{\partial p}, \quad \frac{dp}{dt} = - \frac{\partial \mathbf{H}}{\partial q}, \quad \frac{\partial \mathbf{H}}{\partial u} = 0 \quad (\text{vi})$$

is exactly equivalent to (2) under completely general condition if $\mathbf{H}$ is defined as (i).

$\mathbf{H}$ is called control Hamitonian

Since $\mathbf{H} = -L + a \text{ linear function of } u$, eq (3) is completely equivalent to

$$\frac{\partial^2 \mathbf{H}}{\partial u^2} \leq 0 \quad \left(\text{i.e., } \frac{\partial^2 \mathbf{H}}{\partial \dot{q}^2} \leq 0 \right) \quad (\text{vii})$$

$\Rightarrow$ check eq(v) and eq(vii)

$$\frac{\partial \mathbf{H}}{\partial u} = 0, \frac{\partial^2 \mathbf{H}}{\partial u^2} \leq 0$$

$\mathbf{H}$ must have a maximum as a function of u .

End]

Before 1838, William Roman Hamilton(1805-1865) had been reformulating the equations of mechanics as a variational principle (least action).

He introduced the function (now known as “Hamiltonian function”)

$$H(t, p, q) \equiv \langle p, \dot{q} \rangle - L(q, \dot{q}, t) \quad (4)$$

$$\text{where } p = \frac{\partial L}{\partial \dot{q}}(t, q, \dot{q}) \quad (5)$$

treat $\dot{q}$ not as an independent variable, but as a function of t, q, p .

$$\rightarrow \dot{q}(t) = \frac{\partial H}{\partial p}(t, p(t), q(t)), \quad \dot{p}(t) = -\frac{\partial H}{\partial q}(t, p(t), q(t)) \quad (6)$$

if and only if the Euler-Lagrange equation(2) is satisfied.

Eq(6) resembles eq(vi), but is not all the same. The difference is that in Hamilton’s definition, $\dot{q}$ is supposed to be treated not as an independent variable but as a function of q, p, t , defined implicitly by the eq (5).

It is easy to see that, if the map $(q, \dot{q}, t) \rightarrow (q, p, t)$ defined by (5) can be inverted, i.e. if we can solve (5) for $\dot{q}$ as a function of p, q, t , then (6) is equivalent to (vi).

Indeed, it is clear that

$$H(q, p, t) \equiv \mathbf{H}(q, u(q, p, t), t)$$

$$\frac{\partial H}{\partial q} = \frac{\partial \mathbf{H}}{\partial q} + \frac{\partial \mathbf{H}}{\partial u} \frac{\partial u}{\partial q} \quad (6')$$

$$\text{Since } \frac{\partial \mathbf{H}}{\partial u} = 0 \text{ for } u = u(q, p, t) \Rightarrow \frac{\partial H}{\partial q} = \frac{\partial \mathbf{H}}{\partial q}$$

Note: It should be clear that the Hamiltonian reformulation of the Euler-

Lagrange equations in terms of the “control Hamiltonian” is at least as natural as the classical one, and perhaps even simpler. Moreover, the control formulation has at least one obvious advantage, namely,

(A1) the control version of the Hamilton equations is equivalent to the Euler-Lagrange system under completely general conditions, whereas the classical version only makes sense when the transformation eq(5) can be inverted, at least locally, to solve for $\dot{q}$ as a function of p, q, t .

Eqs(v) and (vii) suggest something. Clearly, H must have a maximum as a function of u . So we state this as a conjecture.

Conjecture M1 ; Besides eq(vi), an additional necessary condition for optimality is that H , as a function of u , have a maximum at $\dot{q}(t)$ for each t .

1838 Jacobi(1804-1851) showed that it could be more compactly written in terms of what is now known as the Hamilton-Jacobi equation (basis of “dynamic programming” developed by Bellman over 100 years later).

$$\phi_t(t, q) + H(t, \phi_q(t, q), q(t)) = 0 \quad (7)$$

If $\phi(t, q)$ is a twice continuously differentiable solution of this equation, then equation (6) defined a regular optimal trajectory, provided that

(a) the strict Legendre-Clebsch condition is satisfied

$$H_{pp} < 0$$

(b) there are no points conjugate to a along trajectory (the Jacobi condition)

Weierstrauss (1815-1897) considered strong variations (δq small, but no restriction on $\delta \dot{q}$). He considered the problem of minimizing an integral I of the form

$I = \int_a^b L(q(s), \dot{q}(s)) ds$ for Lagrangian such that $L(q, \dot{q})$ is positively homogeneous with respect to the velocity $\dot{q}$ (that is, $L(q, \alpha \dot{q}) = \alpha L(q, \dot{q})$ for all $q, \dot{q}$ all $\alpha (\geq 0)$ and does not depend on

time)

He introduced the “excess function”

$$E(q, u, \bar{u}) = L(q, \bar{u}) - \frac{\partial L}{\partial u}(q, u) \cdot \bar{u} \quad (8)$$

Depending on three sets of independent variables q, u and $\bar{u}$.

He then proved his side condition:

For a curve $s \rightarrow q^*(s)$ to be solution of the minimum problem,
The function E has to be $E \geq 0$ when evaluated for $q \rightarrow q^*(s)$,
 $u \rightarrow \dot{q}^*(s)$ and a completely arbitrary $\bar{u}$ (is arbitrary value of u).

Note : Homogeneity property $\Rightarrow L(q, u) = \frac{\partial L}{\partial u}(q, u) \cdot u$

So, the “excess function” equally can be written

$$\begin{aligned} E(q, u, \bar{u}) &= L(q, \bar{u}) - \frac{\partial L}{\partial u}(q, u) \cdot \bar{u} - \left[L(q, u) - \frac{\partial L}{\partial u}(q, u) \cdot u \right] \\ &= [L(q, \bar{u}) - \langle p, \bar{u} \rangle] - [L(q, u) - \langle p, u \rangle] \\ &= -\mathbf{H}(q, \bar{u}, p) - (-\mathbf{H}(q, u, p)) = \mathbf{H}(q, u, p) - \mathbf{H}(q, \bar{u}, p) \end{aligned}$$

He essentially discovered “maximum principle”, however he did it using a obscured procedure and then missed some profound implication of his discovery. Weierstrass’ condition, expressed in terms of the control Hamiltonian, simply says that

(MAX) along an optimal curve $t \rightarrow q(t)$, if we define $p(t)$ via (ii), then for every t , the value $u = \dot{q}(t)$ must maximize the (control) Hamiltonian $\mathbf{H}(q(t), u, p(t), t)$ as a function of u .

(Max) can be simplified considerably. Indeed, the requirement that $p(t)$ be defined via (ii) is now redundant: if $\mathbf{H}(q(t), u, p(t), t)$, regarded as a

function of u , has a maximum at $u = \dot{q}(t)$, then $\frac{\partial H}{\partial u}(q(t), \dot{q}(t), p(t), t) = 0$,

so $p(t)$ has to be given by (ii). Moreover, the vanishing of

$\frac{\partial H}{\partial u}(q(t), \dot{q}(t), p(t), t)$ is also one of the conditions of (vi). So we can

state (vi) and (MAX) together.

Then,

(Necessary Condition for Optimality: NCO) If a curve $q(t)$ is a solution of the maximization problem, then there has to exist a function $p(t)$ such that the following three conditions hold for all t

$$\begin{aligned}\dot{q}(t) &= \frac{\partial \mathbf{H}}{\partial p} (q(t), \dot{q}(t), p(t), t) \\ \dot{p}(t) &= -\frac{\partial \mathbf{H}}{\partial q} (q(t), \dot{q}(t), p(t), t) \\ \mathbf{H}(q(t), \dot{q}(t), p(t), t) &= \max_u \mathbf{H}(q(t), u, p(t), t)\end{aligned}\quad (9)$$

$\mathbf{H}$ must have a maximum as a function of u .

NCO encapsulates in one single statement the combined power of the Euler-Lagrange necessary conditions and Weierstrass side condition as well as the Legendre condition.

With control Hamiltonian, Weierstrass's side condition becomes much simpler. The control Hamiltonian does not require the introduction of an "excess function". One does not need to include a formula specifying how $p(t)$ is defined, since eq (9) does this automatically.

If you compare (NCO) with all the other necessary conditions that we had written earlier, quite amazingly the derivatives with respect to the u variable are gone.

Principles of Mathematical Guessing¹ suggest that the existence of the $\frac{\partial L}{\partial u}$ is not needed. $\rightarrow$ Then there is no longer any reason to insist that the range of values of u be the whole space; any subset of $\mathbf{R}^n$ would do, since the minimization that occurs in eq(9) makes sense over any set.

Conjecture M2 ;

(NCO) should still be a necessary condition for optimality even for problems where $u = \dot{q}(t)$ is restricted to belong to some subset U of $\mathbf{R}^n$, and $L(q, u, t)$ is not required to be differentiable with respect to u .

¹ Principles of Mathematical Guessing

: If a statement is proved under some specific restrictions but turns out not to involve these restrictions at all, chances are that the restrictions are not needed and the statement is valid even without them.

Now that we have liberated ourselves from the constraint that L be differentiable with respect to u .

u (i.e., $\dot{q}(t)$) can be anything?

→ Instead of letting $\dot{q}$ be u , allow $\dot{q}(t)$ to be even more arbitrary. (It ought to be possible for u (i.e., $\dot{q}$) to be anything, and NCO will still work.)

→ After applying the Principle of Mathematical Guessing again,

we can write $\dot{q} = f(q, u, t)$
a general function

→ $\langle p, u \rangle = \langle p, f(q, u, t) \rangle$

Conjecture M3 ;

(NCO) should still be a necessary condition for optimality even for problems where q is restricted to satisfy a differential equation $\dot{q} = f(q, u, t)$, with the “control function” $t \rightarrow u(t)$ taking values in some set U and allowed to be a “completely arbitrary” U -valued function of t , and the Hamiltonian H now being defined by

$$H(q, u, p, t) = \langle p, f(q, u, t) \rangle - L(q, u, t) \quad (10)$$

A minor modification is needed to make Conjecture M3 to be the Pontryagin maximum principle.

Consider the restriction of the class of admissible function $\dot{q}(t)$ to a subset of $\mathbf{R}^n$, specially so that they also satisfy the set of equations.

$$g(t, q(t), \dot{q}(t)) = 0, \quad t \in [a, b] \quad (11)$$

⇒ a general set of differential-algebraic equations

⇒ sufficient conditions were imposed to ensure that

$\dot{q} = f(t, q(t))$ satisfying equation (11) : problem of Lagrange

since it was solved by the use of Lagrange multipliers.

⇒ The complete solution set the scene for parameterizing the degrees of freedom implicit in the Lagrange problem by

considering constraints of the form

$$\dot{q} = f(t, q(t), u(t)) \quad (12)$$

$\downarrow u(t)$; “controls”, which can be chosen at each $t \in (a, b)$ yielding the optimal control problem.

optimal control problem :

Find $u(t)$ on (a, b) to minimize

$$J = \int_a^b L(t, q(t), \dot{q}(t)) dt$$

Subject to $\dot{q} = f(t, q(t), u(t))$

$$t \in (a, b)$$

$$u(t) \in \Omega \subset R^m$$

$$q(a), q(b) \text{ given}$$

1962 Pontryagin

established the necessary conditions in his famous “maximum principle”

$$\dot{q}(t) = \frac{\partial H}{\partial p} \quad , \quad \dot{p}(t) = -\frac{\partial H}{\partial q} \quad (13)$$

$$H(t, p, q, \dot{q}) = \max_{u \in \Omega} \mathbf{H}(t, p, q, u)$$

$$u^* = \max_u \mathbf{H}(t, p, q, u)$$

Introduction to Optimal Control

Objectives:

We will be given a dynamic system, contain constraints, and a performance index to be minimized (or maximized).

We have to determine the control that achieves the objectives.

Dynamic system : equation of motion

$$\dot{\vec{x}} = f(\vec{x}, \vec{u}, t)$$

Constraints : state constraints, control constraints

Objective function or performance index cost

$$J(\vec{x}) = \phi(\vec{x}(t_f), t_f) + \int_0^{t_f} L(\vec{x}, \vec{u}, t) dt$$

ex.) $\dot{x} = f(x, u)$

$$J = \frac{1}{2} \int_0^{t_f} (x^2 + u^2) dt$$

$$\text{or } J = \frac{1}{2} \int_0^{t_f} (x^2 + 10u^2) dt$$

→ In this case, control is more important.

Minimum effort : $J = \int_0^{t_f} u^2 dt$

Minimum energy : $J = \int_0^{t_f} (\text{energy of the system}) dt$

Minimum fuel : $J = \int_0^{t_f} |u| dt$

Minimum time : $J = \int_0^{t_f} 1 dt = t_f$

$$J = \int_0^{t_f} 100 dt = 100t_f, \text{ same meaning}$$

Mayer problem : $J = \phi(x(t_f), t_f)$

Simple Optimal Control Problem (Example)

$$\text{Minimize } J = \frac{1}{2} \int_0^{\infty} (x^2 + u^2) dt$$

$$\begin{aligned} \dot{x} &= u, & x(0) &= 1 \text{ given} & \rightarrow \dot{x} &= -kx \\ u &= -kx \end{aligned}$$

Find k !

This case is a linear quadratic problem.

Solution

$$\dot{x} = -kx$$

$$x(t) = x(0)e^{-kt}$$

$$\begin{aligned} J(k) &= \frac{1}{2} \int_0^{\infty} (x^2(0)e^{-2kt} + k^2 x^2(0)e^{-2kt}) dt \\ &= \frac{1}{2} x^2(0) \int_0^{\infty} e^{-2kt} (1 + k^2) dt \\ &= \frac{1}{2} x^2(0) (1 + k^2) \frac{1}{-2k} [e^{-2kt}]_0^{\infty} \\ &= \frac{1}{2} x^2(0) (1 + k^2) \frac{1}{2k} \\ &= \frac{1}{4} \frac{(1 + k^2)}{k} \end{aligned}$$

$$\frac{\partial J}{\partial k} = 0 \quad (\text{necessary condition})$$

$$\rightarrow \frac{1}{4} \frac{2k^2 - (1 + k^2)}{k^2} = 0$$

$$\Rightarrow k^2 = 1$$

$$\Rightarrow k = \pm 1$$

$$\frac{\partial^2 J}{\partial k^2} > 0 \quad (\text{sufficient condition for a minimum})$$

$$\frac{\partial^2 J}{\partial k^2} = \frac{\partial}{\partial k} \left(\frac{-1}{4k^2} + \frac{1}{4} \right) = \frac{1}{2k^3} > 0$$

$$\Rightarrow k^* = 1$$

$$\Rightarrow J^* = \frac{1}{2}$$

k is not function of time $\rightarrow$ parameter optimization

When u is not used and $\dot{x}$ is not restricted,

$$J = \frac{1}{2} \int_0^\infty (x^2 + \dot{x}^2) dt$$

$$x(0) = 1$$

This become a standard calculus of variations that does not treat control variables.

Optimal Control Problem

1) Establish an optimal control problem

performance index(measure)
dynamics system
constraints

2) Solve the problem

to determine a control function that minimizes or maximizes
the criterion (performance index)

Two strategies:

① The minimum (or maximum) principle (Pontryagin)

Nonlinear two-point boundary-value problem (TPBVP)

② Dynamic programming (Bellman)

Fundamental equation (Hamiltonian-Jacobi-Bellman equation).

In the method dynamic programming, an optimal policy is found by employing the intuitively appealing concept called the principle of optimality².

Many complex aerospace problems that are not amenable to classical techniques have been solved by using optimal theory. The optimal control law, if it can be obtained, usually requires a digital computer for implementation (except for the linear regulator problem), and all of the states must be available for feedback to the controller.

² Principle of optimality:

Whatever the initial state and initial decision are, the remaining decision must constitute an optimal policy with regard to the state resulting from the first decision.

Ex.) If a-b-e is the optimal path from a to e, the b-e is the optimal path from b to e.

The Calculus of Variations

<Functionals>

- A function f is a rule of correspondence that assigns each element $\vec{q}$ in a certain set $\mathbf{D}$ to a unique element in a set $\mathbf{R}$

$\mathbf{D}$: domain of f

$\mathbf{R}$: range of f

ex.) $f(\vec{q}) = \sqrt{q_1^2 + q_2^2 + \cdots + q_n^2}$

$q_1, \cdots, q_n$; coordinate of a point in n-dimensional Euclidean space.

$f(\vec{q})$ is the real number assigned by f is the distance of the point $\vec{q}$ from the origin.

- A functional J is rule of correspondence that assigns each function $\vec{x}$ in a certain set Ω to a unique real number $\mathbf{S}$

Ω : domain of J

$\mathbf{S}$: range of J

→ A functional is a “function of a function”

ex.)
$$\underset{\text{functional}}{J(x)} = \int_{t_0}^{t_f} \underset{\text{function}}{x(t)} dt$$

J is a unique real number.

J is the area under the $x(t)$ curve.

<Linearity of functional>

J is a linear functional if and only if it satisfies the principle of homogeneity.

$$J(\alpha \vec{x}) = \alpha J(\vec{x}) \quad (\text{for all } \vec{x} \in \Omega, \text{ real number } \alpha, \alpha \vec{x} \in \Omega)$$

and the principle of additivity

$$J(\vec{x}_1 + \vec{x}_2) = J(\vec{x}_1) + J(\vec{x}_2) \quad (\text{for } \vec{x}_1, \vec{x}_2, \text{ and } \vec{x}_1 + \vec{x}_2 \in \Omega)$$

ex.) i) $J(x) = \int_{t_0}^{t_f} x(t) dt$; linear functional

Homogeneity

$$J(\alpha x) = \int_{t_0}^{t_f} \alpha x(t) dt = \alpha \int_{t_0}^{t_f} x(t) dt \Rightarrow J(\alpha x) = \alpha J(x)$$

Additivity

$$J(x_1) = \int_{t_0}^{t_f} x_1(t) dt, \quad J(x_2) = \int_{t_0}^{t_f} x_2(t) dt$$

$$J(x_1 + x_2) = \int_{t_0}^{t_f} [x_1(t) + x_2(t)] dt = \int_{t_0}^{t_f} x_1(t) dt + \int_{t_0}^{t_f} x_2(t) dt = J(x_1) + J(x_2)$$

ii) $J(x) = \int_{t_0}^{t_f} x^2(t) dt$; non-linear functional

Homogeneity?

$$\begin{aligned} J(\alpha x) &= \int_{t_0}^{t_f} \alpha^2 x^2(t) dt \\ &= \alpha^2 \int_{t_0}^{t_f} x^2(t) dt \\ &= \alpha^2 J(x) \end{aligned}$$

$$\Rightarrow J(\alpha x) \neq \alpha J(x)$$

<Increment of functional>

- Function f

$\vec{q}, \vec{q} + \Delta\vec{q}$ are elements of f

Then, increment of f ($\equiv \Delta f$)

$$\Delta f \equiv f(\vec{q} + \Delta\vec{q}) - f(\vec{q})$$

Δf depends on both $\vec{q}$ and $\Delta\vec{q} \rightarrow \Delta f(\vec{q}, \Delta\vec{q})$

- Functional J

$\vec{x}, \vec{x} + \delta\vec{x}$ are functions for J
 $\delta\vec{x}$; the variation of the function $\vec{x}$

Then, increment of J ($\equiv \Delta J$)

$$\Delta J \equiv J(\vec{x} + \delta\vec{x}) - J(\vec{x})$$

ΔJ depends on both $\vec{x}$ and $\delta\vec{x} \rightarrow \Delta J(\vec{x}, \delta\vec{x})$

ex.)

$$\begin{aligned} J(x) &= \int_{t_0}^{t_f} x^2(t) dt \\ \Delta J &= J(x + \delta x) - J(x) \\ &= \int_{t_0}^{t_f} (x + \delta x)^2 dt - \int_{t_0}^{t_f} x^2 dt \\ &= \int_{t_0}^{t_f} [2x\delta x + (\delta x)^2] dt \end{aligned}$$

<The variation of functional>

- The increment of a function f of n variables

$$\Delta f(\vec{q}, \Delta \vec{q}) = \underbrace{df(\vec{q}, \Delta \vec{q})}_{\text{a linear function of } \Delta \vec{q}} + g(\vec{q}, \Delta \vec{q}) \cdot \|\Delta \vec{q}\|$$

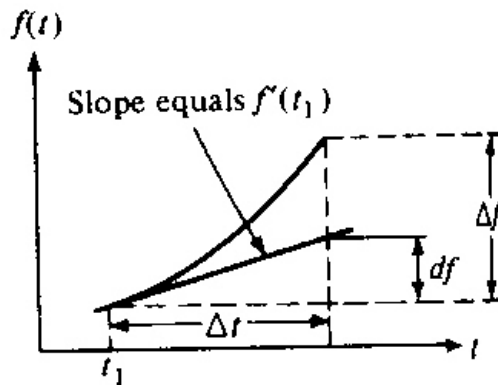
If $\lim_{\|\Delta \vec{q}\| \rightarrow 0} \{g(\vec{q}, \Delta \vec{q})\} = 0$, then, f is said to be differentiable at $\vec{q}$ and df is a

differential of f at the point $\vec{q}$.

If f is a differentiable function of one variable t , differential can be written

$$df(t, \Delta t) = f'(t) \Delta t \quad ; \quad \text{first-order approximation to } \Delta f$$

where $f'(t)$; derivative of f at t



- The increment of a functional

$$\Delta J(\vec{x}, \delta \vec{x}) = \underbrace{\delta J(\vec{x}, \delta \vec{x})}_{\text{linear term in } \delta \vec{x}} + g(\vec{x}, \delta \vec{x}) \cdot \|\delta \vec{x}\|$$

If $\lim_{\|\delta \vec{x}\| \rightarrow 0} \{g(\vec{x}, \delta \vec{x})\} = 0$

then, J is said to be differentiable on $\vec{x}$ and δJ is the variation of J evaluated for the function $\vec{x}$, δJ is the linear approximation to the difference in the functional J caused by $\vec{x}$, $\vec{x} + \delta \vec{x}$.

<Maxima and Minima of functional>

- A function f with domain $\mathbf{D}$ has a relative extremum at the point $\vec{q}^*$

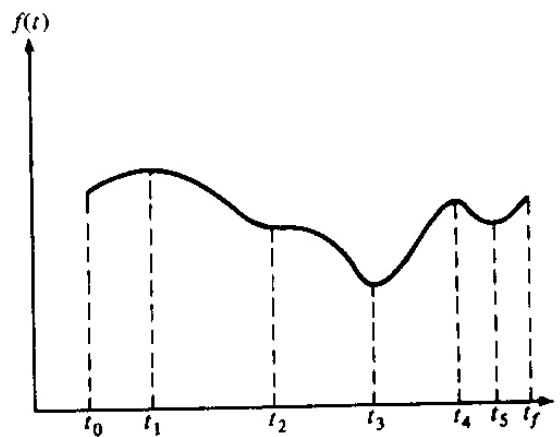
If there is $\varepsilon > 0$ ($\|\vec{q} - \vec{q}^*\| < \varepsilon$ for all point $\vec{q}$ in $\mathbf{D}$) such that the increment of f has the same sign.

$$f(\vec{q}^*) \text{ is a relative minimum} \quad \text{if } \Delta f = f(\vec{q}) - f(\vec{q}^*) > 0 \quad (\text{a})$$

$$f(\vec{q}^*) \text{ is a relative maximum} \quad \text{if } \Delta f = f(\vec{q}) - f(\vec{q}^*) < 0 \quad (\text{b})$$

If equation (a) is satisfied for arbitrarily large ε , then $f(\vec{q}^*)$ is a global or absolute minimum.

If equation (b) is satisfied for arbitrarily large ε , then $f(\vec{q}^*)$ is a global or absolute maximum.



stationary points	; t_1, t_2, t_3, t_4, t_5
extreme points	; t_1, t_3, t_4, t_5
relative minima	; t_0, t_3, t_5
relative maxima	; t_1, t_4, t_f
global minima	; t_3
global maxima	; t_1

● A functional J with domain Ω has a relative extremum at $\vec{x}^*$ (extremal), if there is an $\varepsilon > 0$ ($\|\vec{x} - \vec{x}^*\| < \varepsilon$ for all point $\vec{x}$ in Ω) such that the increment of J has the same sign.

$$J(\vec{x}^*) \text{ is a relative minimum if } \Delta J = J(\vec{x}) - J(\vec{x}^*) > 0 \quad (c)$$

$$J(\vec{x}^*) \text{ is a relative maximum if } \Delta J = J(\vec{x}) - J(\vec{x}^*) < 0 \quad (d)$$

If equation (c) is satisfied for arbitrarily large ε , then $J(\vec{x}^*)$ is a global or absolute minimum.

If equation (d) is satisfied for arbitrarily large ε , then $J(\vec{x}^*)$ is a global or absolute maximum.

Let $\vec{x}^*$ be the extremal and $J(\vec{x}^*)$ is the extremum, then $\Delta J(\vec{x}^*)$ must have the same sign for every $\vec{x}$ near $\vec{x}^*$ (i.e., every $\delta\vec{x}$).

If $\delta\vec{x}$ is sufficiently small,

$$\text{sign } \Delta J = \text{sign } \delta J$$

where ΔJ ; total variation

δJ ; first variation

$$\Delta J(\vec{x}, \delta\vec{x}) = \delta J(\vec{x}, \delta\vec{x}) + g(\vec{x}, \delta\vec{x}) \cdot \|\delta\vec{x}\|$$

linear term in $\delta\vec{x}$

$$\text{If } \lim_{\|\delta\vec{x}\| \rightarrow 0} \{g(\vec{x}, \delta\vec{x})\} \rightarrow 0, \quad \Delta J \approx \delta J$$

$$\begin{aligned} \Delta J(\vec{x}^*, \delta\vec{x}) &= J(\vec{x}^* + \delta\vec{x}) - J(\vec{x}^*) \\ &= J(\vec{x}^*) + \frac{\partial J}{\partial \vec{x}} \Big|_{\vec{x}^*} \delta\vec{x} + \frac{1}{2} \frac{\partial^2 J}{\partial \vec{x}^2} \Big|_{\vec{x}^*} (\delta\vec{x})^2 + \dots - J(\vec{x}^*) \\ &= \frac{\partial J}{\partial \vec{x}} \Big|_{\vec{x}^*} \delta\vec{x} + \frac{1}{2} \frac{\partial^2 J}{\partial \vec{x}^2} \Big|_{\vec{x}^*} (\delta\vec{x})^2 + H.O.T \end{aligned}$$

$$\text{where } \frac{\partial J}{\partial \vec{x}} \Big|_{\vec{x}^*} \delta\vec{x} \equiv \delta J(\vec{x}^*, \delta\vec{x}); \text{ first variation}$$

$$\frac{1}{2} \frac{\partial^2 J}{\partial \vec{x}^2} \Big|_{\vec{x}^*} (\delta\vec{x})^2 \equiv \delta^2 J; \text{ second variation}$$

δJ is a linear function of $\delta\vec{x}$

<The fundamental theorem of the calculus of variation>

- The fundamental theorem (necessary condition) used in finding extreme values of functions.

⇒ the differential must be zero at an extreme point(except extrema at the boundaries of closed region).

- In variational problems, the analogous theorem exists.

⇒ the variation must be zero on an extremal curve, if there are no bounds imposed on the curves.

Let $\vec{x}$ be a vector function of t in the class Ω .

$J(\vec{x})$ be a differentiable function of $\vec{x}$.

Assume $\vec{x}$ in Ω are not constrained by any boundaries.

∴ Fundamental theorem of the calculus of variations

If $\vec{x}^*$ is extremal and $J(\vec{x})$ is a smooth functional(differentiable) and $\vec{x}$ is unconstrained in the interior, the variation of J must vanish on $\vec{x}^*$ that is $\delta J(\vec{x}^*, \delta \vec{x}) = 0$ for all admissible $\delta \vec{x}$.

<proof of contradiction>

Assume that $\vec{x}^*$ is an extremal and that $\delta J(\vec{x}^*, \delta \vec{x}) \neq 0$

→ the increment ΔJ can be made to change sign in an arbitrarily small neighborhood of $\vec{x}^*$.

→ contradiction of relative extremum for J at $\vec{x}^*$

; the increment of $J(= \Delta J)$ has the same sign at extremal $\vec{x}^*$.

(remember : Let $\vec{x}^*$ be the extremal and $J(\vec{x}^*)$ is the extremum, then $\Delta J(\vec{x}^*)$ must have the same sign for every $\vec{x}$ near $\vec{x}^*$.

If $\delta \vec{x}$ is sufficiently small, $\text{sign } \Delta J = \text{sign } \delta J$)

Proof in detail)

The increment is

$$\begin{aligned}\Delta J(\bar{x}^*, \delta \bar{x}) &= J(\bar{x}^* + \delta \bar{x}) - J(\bar{x}^*) \\ &= \delta J(\bar{x}^*, \delta \bar{x}) + g(\bar{x}^*, \delta \bar{x}) \cdot \|\delta \bar{x}\| \\ \text{as } \|\delta \bar{x}\| \rightarrow 0, \quad g(\bar{x}^*, \delta \bar{x}) &\rightarrow 0\end{aligned}$$

$$\Rightarrow \Delta J \approx \delta J$$

① Let us select the variation

$$\begin{aligned}\delta \bar{x}(t) &= \alpha \delta \bar{x}^{(1)}(t) \\ \alpha > 0 \text{ and } \|\alpha \delta \bar{x}^{(1)}\| &< \varepsilon\end{aligned}$$

suppose that

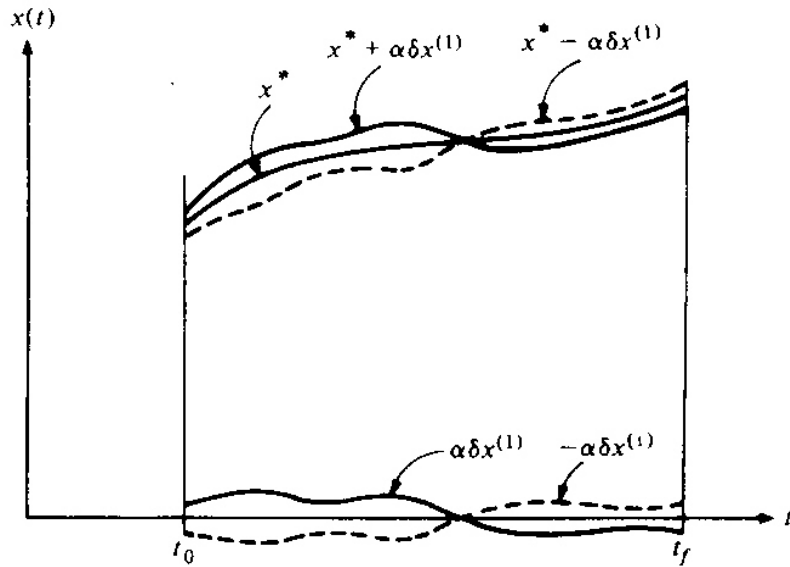
$$\delta J(\bar{x}^*, \alpha \delta \bar{x}^{(1)}) < 0 \quad (\text{Note } \delta J(\bar{x}^*, \delta \bar{x}) = \left. \frac{\partial J}{\partial \bar{x}} \right|_{\bar{x}^*} \delta \bar{x})$$

Since δJ is a linear functional of $\delta \bar{x}$, the principle of homogeneity gives

$$\delta J(\bar{x}^*, \alpha \delta \bar{x}^{(1)}) = \left. \frac{\partial J}{\partial \bar{x}} \right|_{\bar{x}^*} \alpha \delta \bar{x}^{(1)} = \alpha \delta J(\bar{x}^*, \delta \bar{x}^{(1)}) < 0 \Rightarrow \delta J(\bar{x}^*, \delta \bar{x}^{(1)}) < 0$$

$$\Rightarrow \delta J(\bar{x}^*, \alpha \delta \bar{x}^{(1)}) = \delta J(\bar{x}^*, \delta \bar{x}) < 0 \quad (\text{note } \delta \bar{x} = \alpha \delta \bar{x}^{(1)})$$

$$\rightarrow \Delta J(\bar{x}^*, \delta \bar{x}) < 0$$



② Let us select the variation

$$\begin{aligned}\delta\vec{x}(t) &= -\alpha\delta\vec{x}^{(l)}(t) \\ \alpha > 0, \quad \left\| -\alpha\delta\vec{x}^{(l)} \right\| &< \varepsilon\end{aligned}$$

The principle of homogeneity

$$\begin{aligned}\delta J(\vec{x}^*, -\alpha\delta\vec{x}^{(l)}) &= -\alpha\delta J(\vec{x}^*, \delta\vec{x}^{(l)}) > 0 \quad (\text{since } \delta J(\vec{x}^*, \delta\vec{x}^{(l)}) < 0) \\ \Rightarrow \delta J(\vec{x}^*, -\alpha\delta\vec{x}^{(l)}) &= \delta J(\vec{x}^*, \delta\vec{x}) > 0 \quad (\text{note } \delta\vec{x} = -\alpha\delta\vec{x}^{(l)}) \\ \Rightarrow \Delta J(\vec{x}^*, \delta\vec{x}) &> 0\end{aligned}$$

③ from ① and ②

If $\delta J(\vec{x}^*, \delta\vec{x}) \neq 0$

then, $\Delta J(\vec{x}^*, \alpha\delta\vec{x}^{(l)}) = \Delta J(\vec{x}^*, \delta\vec{x}) < 0$

where $\delta\vec{x} = \alpha\delta\vec{x}^{(l)}$, because $\delta\vec{x}$ can be arbitrary near $\vec{x}^*$

$$\Delta J(\vec{x}^*, -\alpha\delta\vec{x}^{(l)}) = \Delta J(\vec{x}^*, \delta\vec{x}) > 0$$

where $\delta\vec{x} = -\alpha\delta\vec{x}^{(l)}$, because $\delta\vec{x}$ can be arbitrary near $\vec{x}^*$

→ contradiction of the assumption that $\vec{x}^*$ is extremal (then, ΔJ has the same sign)

→ Hence, if $\vec{x}$ is an extremal, it is necessary that $\delta J(\vec{x}^*, \delta\vec{x}) = 0$ for arbitrary $\delta\vec{x}$ when $\vec{x}$ in Ω are not bounded.

4.2 Functionals of a single function

<The simplest variation problem>

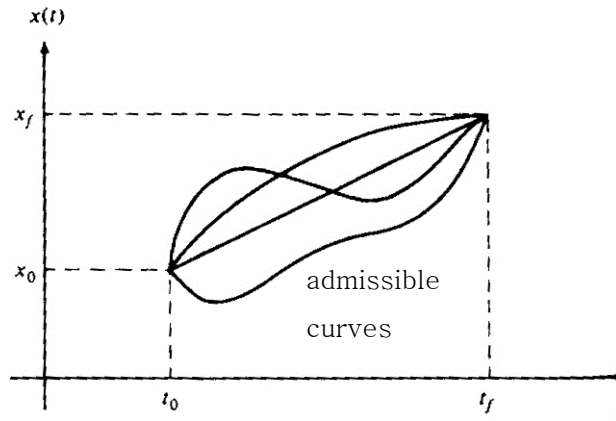
Class Ω		Type of Problem
x must be continuous	C	$J = \int g(x, t) dt$
x must be continuous with continuous first derivatives ($x \in D^1$)	D^1	$J = \int g(x, \dot{x}, t) dt$
x must be continuous with continuous first and second derivatives ($x \in D^2$)	D^2	$J = \int g(x, \dot{x}, \ddot{x}, t) dt$

① $J(x) = \int_{t_0}^{t_f} g(x(t), \dot{x}(t), t) dt$

$x(t_0), x(t_f)$ are given, t_0 and t_f are fixed

J is a functional of the function x .

Find the function x^* (extremal) for which the J has a relative extremum.



Curves in class Ω which also satisfy the end conditions are admissible.

The search begins by finding the curves that satisfy the fundamental theorem.

$$\begin{aligned}
\Delta J(x^*, \dot{x}^*, t; \delta x, \delta \dot{x}) &= J(x^* + \delta x, \dot{x}^* + \delta \dot{x}, t) - J(x^*, \dot{x}^*, t) \\
&= \int_{t_0}^{t_f} g(x, \dot{x}, t) dt - \int_{t_0}^{t_f} g(x^*, \dot{x}^*, t) dt \\
&\quad \text{Taylor series expansion about } x^*, \dot{x}^* \\
&= \int_{t_0}^{t_f} \left[g(x^*, \dot{x}^*, t) + \frac{\partial g}{\partial x} \Big|_{x^*, \dot{x}^*} \delta x + \frac{\partial g}{\partial \dot{x}} \Big|_{x^*, \dot{x}^*} \delta \dot{x} + \text{H.O.T.} \right] dt - \int_{t_0}^{t_f} g(x^*, \dot{x}^*, t) dt \\
&\quad \text{H.O.T.} = \frac{1}{2} \left[\frac{\partial^2 g}{\partial x^2} \Big|_{x^*, \dot{x}^*} (\delta x)^2 + 2 \frac{\partial^2 g}{\partial x \partial \dot{x}} \Big|_{x^*, \dot{x}^*} \delta x \delta \dot{x} + \frac{\partial^2 g}{\partial \dot{x}^2} \Big|_{x^*, \dot{x}^*} (\delta \dot{x})^2 \right] + \theta [(\delta x)^2, (\delta \dot{x})^2]
\end{aligned}$$

We extract the first variation in ΔJ that are linear in δx and $\delta \dot{x}$

$$\begin{aligned}
\Delta J \approx \delta J(x, \delta x) &= \int_{t_0}^{t_f} \left[\frac{\partial g}{\partial x} \Big|_{x^*, \dot{x}^*} \delta x + \frac{\partial g}{\partial \dot{x}} \Big|_{x^*, \dot{x}^*} \delta \dot{x} \right] dt \\
&\quad \text{integration by part } \int_{t_0}^{t_f} u \dot{v} dt = uv \Big|_{t_0}^{t_f} - \int_{t_0}^{t_f} \dot{u} v dt \\
&= \frac{\partial g}{\partial \dot{x}} \delta x \Big|_{t_0}^{t_f} + \int_{t_0}^{t_f} \left[\frac{\partial g}{\partial x} \delta x - \frac{d}{dt} \left(\frac{\partial g}{\partial \dot{x}} \right) \delta x \right] dt \\
&\quad \delta x(t_0) = 0, \delta x(t_f) = 0 \text{ fixed (natural bounday condition)} \\
&= \int_{t_0}^{t_f} \left[\frac{\partial g}{\partial x} - \frac{d}{dt} \left(\frac{\partial g}{\partial \dot{x}} \right) \right] \delta x(t) dt \underset{\text{fundamental theorem}}{=} 0
\end{aligned}$$

$\delta x(t)$ is arbitrary.

$$\therefore \frac{\partial g}{\partial x}(x^*, \dot{x}^*, t) - \frac{d}{dt} \left[\frac{\partial g}{\partial \dot{x}}(x^*, \dot{x}^*, t) \right] = 0$$

A necessary condition for x^* to be an extremal

Euler-Lagrange equation when $x(t_0), x(t_f)$ are given

Euler-Lagrange equation

Generally, nonlinear, ordinary, time-varying, second-order differential equation.

→ usually cannot be solve analytically.

→ numerical integration is used.

→ the boundary condition are split.

$x(t_0), x(t_f)$ are known, instead of having $[x(t_0), \dot{x}(t_0)]$ or $[x(t_f), \dot{x}(t_f)]$

→ nonlinear, two-point boundary value problem (TPBVP)

This problem is difficult because of the combination of split boundary values and the nonlinearity of the differential equations.

$$\textcircled{2} \quad J(x) = \int_{t_0}^{t_f} g(x(t), \dot{x}(t), \ddot{x}(t), t) dt$$

t_0 and t_f are fixed

$x(t_0), x(t_f), \dot{x}(t_0), \dot{x}(t_f)$ are given

$$\begin{aligned} \delta J &= \int_{t_0}^{t_f} \left[\frac{\partial g}{\partial x} \delta x + \frac{\partial g}{\partial \dot{x}} \delta \dot{x} + \frac{\partial g}{\partial \ddot{x}} \delta \ddot{x} \right] dt \\ &\quad \text{integration by part} \quad \int_{t_0}^{t_f} \left(\frac{\partial g}{\partial \dot{x}} \delta \dot{x} \right) dt = \frac{\partial g}{\partial \dot{x}} \delta x \Big|_{t_0}^{t_f} - \int_{t_0}^{t_f} \frac{d}{dt} \left(\frac{\partial g}{\partial \dot{x}} \right) \delta x dt \\ &\quad \text{integration by part} \quad \int_{t_0}^{t_f} \left(\frac{\partial g}{\partial \ddot{x}} \delta \ddot{x} \right) dt = \frac{\partial g}{\partial \ddot{x}} \delta \dot{x} \Big|_{t_0}^{t_f} - \int_{t_0}^{t_f} \frac{d}{dt} \left(\frac{\partial g}{\partial \ddot{x}} \right) \delta \dot{x} dt \\ &= \frac{\partial g}{\partial \ddot{x}} \delta \dot{x} \Big|_{t_0}^{t_f} + \frac{\partial g}{\partial \dot{x}} \delta x \Big|_{t_0}^{t_f} + \int_{t_0}^{t_f} \left[\frac{\partial g}{\partial x} \delta x - \frac{d}{dt} \left(\frac{\partial g}{\partial \dot{x}} \right) \delta x - \frac{d}{dt} \left(\frac{\partial g}{\partial \ddot{x}} \right) \delta \dot{x} \right] dt \\ &\quad \text{integration by part} \quad \int_{t_0}^{t_f} \left(\frac{d}{dt} \left(\frac{\partial g}{\partial \ddot{x}} \right) \delta \dot{x} \right) dt = \frac{d}{dt} \left(\frac{\partial g}{\partial \ddot{x}} \right) \delta x \Big|_{t_0}^{t_f} - \int_{t_0}^{t_f} \frac{d^2}{dt^2} \left(\frac{\partial g}{\partial \ddot{x}} \right) \delta x dt \\ &= - \frac{d}{dt} \left(\frac{\partial g}{\partial \ddot{x}} \right) \delta x \Big|_{t_0}^{t_f} + \int_{t_0}^{t_f} \left[\frac{\partial g}{\partial x} \delta x - \frac{d}{dt} \left(\frac{\partial g}{\partial \dot{x}} \right) \delta x + \frac{d^2}{dt^2} \left(\frac{\partial g}{\partial \ddot{x}} \right) \delta x \right] dt \\ &= \int_{t_0}^{t_f} \left[\frac{d^2}{dt^2} \left(\frac{\partial g}{\partial \ddot{x}} \right) - \frac{d}{dt} \left(\frac{\partial g}{\partial \dot{x}} \right) + \frac{\partial g}{\partial x} \right] \delta x dt \\ &= 0 \end{aligned}$$

$$\therefore \frac{d^2}{dt^2} \left(\frac{\partial g}{\partial \ddot{x}} \right) - \frac{d}{dt} \left(\frac{\partial g}{\partial \dot{x}} \right) + \frac{\partial g}{\partial x} = 0 \quad ; \text{ Euler-Lagrange equation}$$

$$\textcircled{3} \quad J(x) = \int_{t_0}^{t_f} g(x(t), \dot{x}(t), \ddot{x}(t), \ddot{\ddot{x}}(t), t) dt$$

t_0 and t_f are fixed

$x, \dot{x}, \ddot{x}$ are given at both the end points.

$$\therefore -\frac{d^3}{dt^3}\left(\frac{\partial g}{\partial \ddot{x}}\right) + \frac{d^2}{dt^2}\left(\frac{\partial g}{\partial \dot{x}}\right) - \frac{d}{dt}\left(\frac{\partial g}{\partial \dot{x}}\right) + \frac{\partial g}{\partial x} = 0$$

Generalization

$$\therefore (-1)^n \frac{d^n}{dt^n} \left(\frac{\partial g}{\partial x} \right) + (-1)^{n-1} \frac{d^{n-1}}{dt^{n-1}} \left(\frac{\partial g}{\partial \dot{x}} \right) + \dots + \frac{\partial g}{\partial x} = 0$$

Example (Low-Thrust Orbit Transfer (2D))

$$m\ddot{\vec{r}} = -m\nabla\phi + m\vec{u}$$

$\vec{r} = r\hat{e}_r$ (polar coordinate), $\vec{u}$: Thrust acceleration (control)

$$\phi = \phi(\vec{r}) \quad \text{gravitational potential, } \nabla\phi(\vec{r}) = \frac{\partial\phi(\vec{r})}{\partial\vec{r}} = \frac{\partial\phi}{\partial r}\hat{e}_r + \frac{1}{r}\frac{\partial\phi}{\partial\theta}\hat{e}_\theta$$

$$J = \frac{1}{2} \int_0^T (\vec{u}^T \vec{u}) dt \quad \text{minimum-control-effort, } r(0), r(T) \text{ and } T \text{ are specified}$$

Find $\vec{u}$ to minimize J

$$J(\vec{r}, \ddot{\vec{r}}) = \frac{1}{2} \int_0^T (\ddot{\vec{r}} + \nabla\phi)^T (\ddot{\vec{r}} + \nabla\phi) dt = \frac{1}{2} \int_{t_0}^{t_f} (\ddot{\vec{r}} + \nabla\phi)^2 dt$$

$$\Rightarrow J(\vec{r}, \ddot{\vec{r}}) = \int_0^T g(\vec{r}, \ddot{\vec{r}}) dt$$

Necessary Condition

$$\frac{d^2}{dt^2} \left(\frac{\partial g}{\partial \ddot{\vec{r}}} \right) - \frac{d}{dt} \left(\frac{\partial g}{\partial \dot{\vec{r}}} \right) + \frac{\partial g}{\partial \vec{r}} = 0$$

$$\frac{\partial g}{\partial \vec{r}} = \frac{\partial}{\partial \vec{r}} (\ddot{\vec{r}} + \nabla\phi)^2 = 2(\ddot{\vec{r}} + \nabla\phi) \frac{\partial}{\partial \vec{r}} (\nabla\phi(\vec{r})) = 2(\ddot{\vec{r}} + \nabla\phi) \nabla^2\phi(\vec{r})$$

$$\frac{\partial g}{\partial \dot{\vec{r}}} = 0$$

$$\frac{\partial g}{\partial \ddot{\vec{r}}} = \frac{\partial}{\partial \ddot{\vec{r}}} (\ddot{\vec{r}} + \nabla\phi)^2 = 2(\ddot{\vec{r}} + \nabla\phi) \frac{\partial}{\partial \ddot{\vec{r}}} (\ddot{\vec{r}}) = 2(\ddot{\vec{r}} + \nabla\phi)$$

$$\Rightarrow \frac{d^2}{dt^2}(\ddot{\vec{r}} + \nabla\phi) + \nabla^2\phi(\vec{r})(\ddot{\vec{r}} + \nabla\phi) = 0$$

$$\Rightarrow \ddot{\vec{u}} + \nabla^2\phi\vec{u} = 0, \quad E-L \text{ eq.}$$

$\vec{r}(0), \vec{r}(T), \dot{\vec{r}}(0), \dot{\vec{r}}(T), T$ are specified

$\Rightarrow$ Two-point Boundary-value Problem (TPBVP)

$$\Rightarrow \vec{u}^*(t)$$

$$m\ddot{\vec{r}} = -m\nabla\phi + m\vec{u}$$

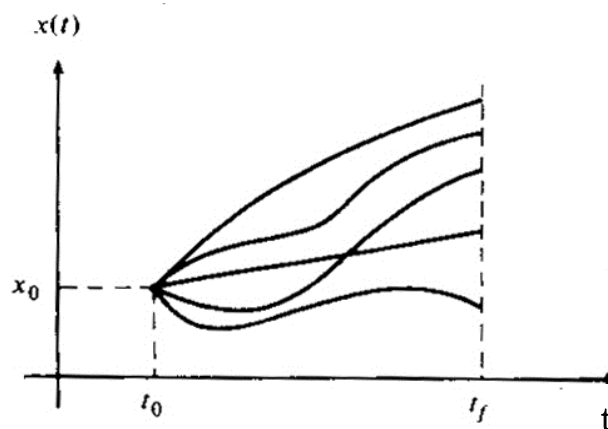
$$\Rightarrow \vec{r}^*(t)$$

< Final time specified, $x(t_f)$ free >

Find the necessary condition for a function to be an extremal for the functional

$$J(x) = \int_{t_0}^{t_f} g(x(t), \dot{x}(t), t) dt$$

t_f is specified, $x(t_0)$ is given, $x(t_f)$ is free



$$\begin{aligned}
\delta J &= \int_{t_0}^{t_f} \left[\frac{\partial g}{\partial x} \delta x + \frac{\partial g}{\partial \dot{x}} \delta \dot{x} \right] dt \\
&= \frac{\partial g}{\partial \dot{x}} \delta x \Big|_{t_0}^{t_f} + \int_{t_0}^{t_f} \left[\frac{\partial g}{\partial x} - \frac{d}{dt} \left(\frac{\partial g}{\partial \dot{x}} \right) \right] \delta x dt \\
&\qquad \qquad \qquad \frac{\partial g}{\partial \dot{x}} \delta x \Big|_{t_0}^{t_f} = \frac{\partial g}{\partial \dot{x}} \delta x(t_f) - \frac{\partial g}{\partial \dot{x}} \delta x(t_0) = \frac{\partial g}{\partial \dot{x}} \delta x(t_f) \\
&= \frac{\partial g}{\partial \dot{x}} \Big|_{t=t_f} \delta x(t_f) + \int_{t_0}^{t_f} \left[\frac{\partial g}{\partial x} - \frac{d}{dt} \left(\frac{\partial g}{\partial \dot{x}} \right) \right] \delta x dt \\
&= 0
\end{aligned}$$

For an extremal $x^*(t)$, we know $\delta J(\bar{x}^*, \delta \bar{x}) = 0$. Let us next show that

the integral of $\int_{t_0}^{t_f} \left[\frac{\partial g}{\partial x} - \frac{d}{dt} \left(\frac{\partial g}{\partial \dot{x}} \right) \right] \delta x dt$ must be zero on an extremal. Suppose that the curve $x^*(t)$ is an extremal for the free end point problem. Now consider a fixed end point problem with the same functional, the same initial and final times, and with specified end points that are the same as for the extremal $x^*(t)$ in the free end point problem. The curve $x^*(t)$ must be an extremal for this fixed end point problem; therefore, $x^*(t)$ must be a solution of the Euler equation, and the integral term must be zero on an extremal. In other words, an extremal for a free and end point problem is also an extremal for the fixed end point problem with the same end points, and the same functional; thus, regardless of the boundary conditions, the Euler equation must be satisfied.

$$\therefore \frac{\partial g}{\partial x}(x, \dot{x}, t) - \frac{d}{dt} \left[\frac{\partial g}{\partial \dot{x}}(x, \dot{x}, t) \right] = 0 \quad \text{Euler-Lagrange equation } 0 \leq t \leq t_f$$

$$\frac{\partial g}{\partial \dot{x}} \Big|_{t=t_f} = 0 \quad \text{at } t = t_f \quad \text{transversality condition}$$

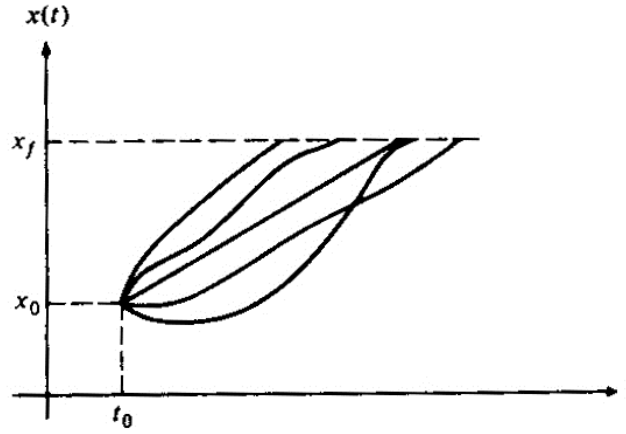
(natural boundary condition)

< Final time free, $x(t_f)$ specified >

Find the necessary condition for a function to be an extremal for the functional

$$J(x) = \int_{t_0}^{t_f} g(x(t), \dot{x}(t), t) dt$$

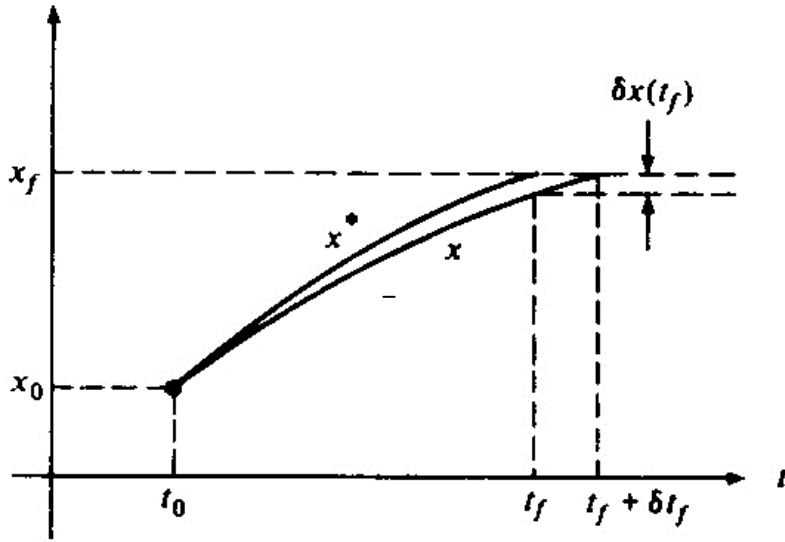
t_0 is fixed, t_f is free, $x(t_0)$ and $x(t_f)$ are given



$$\begin{aligned} \Delta J &= \int_{t_0}^{t_f + \delta t_f} g(x, \dot{x}, t) dt - \int_{t_0}^{t_f} g(x^*, \dot{x}^*, t) dt \\ &= \int_{t_0}^{t_f} g(x, \dot{x}, t) dt + \int_{t_f}^{t_f + \delta t_f} g(x, \dot{x}, t) dt - \int_{t_0}^{t_f} g(x^*, \dot{x}^*, t) dt \\ &= \int_{t_0}^{t_f} [g(x, \dot{x}, t) - g(x^*, \dot{x}^*, t)] dt + \int_{t_f}^{t_f + \delta t_f} g(x, \dot{x}, t) dt \end{aligned}$$

$$\begin{aligned} \int_{t_f}^{t_f + \delta t_f} g(x, \dot{x}, t) dt &= g(x(t_f), \dot{x}(t_f), t_f) \delta t_f \\ &= \left[g(x^*(t_f), \dot{x}^*(t_f), t_f) + \frac{\partial g}{\partial x} \Big|_{x^*, \dot{x}^*} \delta x(t_f) + \frac{\partial g}{\partial \dot{x}} \Big|_{x^*, \dot{x}^*} \delta \dot{x}(t_f) + H.O.T \right] \delta t_f \\ &\approx g(x^*(t_f), \dot{x}^*(t_f), t_f) \delta t_f \end{aligned}$$

$$\begin{aligned} \delta J &= \int_{t_0}^{t_f} \left[\frac{\partial g}{\partial x} \Big|_{x^*, \dot{x}^*} \delta x + \frac{\partial g}{\partial \dot{x}} \Big|_{x^*, \dot{x}^*} \delta \dot{x} \right] dt + g(x^*(t_f), \dot{x}^*(t_f), t_f) \delta t_f \\ &= \frac{\partial g}{\partial \dot{x}} \delta \dot{x} \Big|_{t_0}^{t_f} + \int_{t_0}^{t_f} \left[\frac{\partial g}{\partial x} - \frac{d}{dt} \left(\frac{\partial g}{\partial \dot{x}} \right) \right] \delta x dt + g(x^*, \dot{x}^*, t_f) \delta t_f \end{aligned}$$



$$-\delta x(t_f) \approx \dot{x}(t_f) \delta t_f$$

assume $\dot{x}(t_f) = \dot{x}^(t_f)$*

$$\rightarrow -\delta x(t_f) \approx \dot{x}^*(t_f) \delta t_f$$

$$\rightarrow \delta x(t_f) \approx -\dot{x}^*(t_f) \delta t_f$$

$\delta x(t_f)$ depends on δt_f and is neither zero nor free.

The dependence of $\delta x(t_f)$ on δt_f must be linearly approximated because

δJ consists of the first-order terms in the increment of ΔJ

$$\delta J = \left. \frac{\partial g}{\partial \dot{x}} \right|_{x^*, \dot{x}^*} \delta x(t_f) - \left. \frac{\partial g}{\partial \dot{x}} \right|_{x^*, \dot{x}^*} \delta x(t_0) + g(x^*, \dot{x}^*, t_f) \delta t_f + \int_{t_0}^{t_f} \left[\frac{\partial g}{\partial x} - \frac{d}{dt} \left(\frac{\partial g}{\partial \dot{x}} \right) \right]_{x^*, \dot{x}^*} \delta x(t) dt$$

$$\delta x(t_f) \rightarrow -\dot{x}^*(t_f) \delta t_f$$

$$= \left[-\frac{\partial g}{\partial \dot{x}} \dot{x}^*(t_f) + g \right] \delta t_f + \int_{t_0}^{t_f} \left[\frac{\partial g}{\partial x} - \frac{d}{dt} \left(\frac{\partial g}{\partial \dot{x}} \right) \right]_{x^*, \dot{x}^*} \delta x(t) dt, \quad \delta t_f, \delta x(t) \text{ are arbitrary}$$

$$= 0$$

$$\therefore \frac{\partial g}{\partial x} - \frac{d}{dt} \left(\frac{\partial g}{\partial \dot{x}} \right) = 0$$

Euler-Lagrange equation

$$g(x^*, \dot{x}^*, t_f) - \frac{\partial g}{\partial \dot{x}}(x^*, \dot{x}^*, t_f) \dot{x}^*(t_f) = 0 \quad \text{at } t = t_f \quad \text{Transversality condition}$$

Example $J = \int_0^{t_f} (\dot{x}^2 + x) dt$

$$x(0) = 0, x(t_f) = 1, \quad t_f \text{ is free}$$

$$g(x, \dot{x}, t_f) = (\dot{x}^2 + x)$$

① Euler-Lagrange equation

$$\frac{\partial g}{\partial x} - \frac{d}{dt} \left(\frac{\partial g}{\partial \dot{x}} \right) = 0$$

$$\rightarrow -2\ddot{x} + 1 = 0$$

$$\rightarrow \ddot{x} = \frac{1}{2}$$

$$\rightarrow x = At^2 + Bt + C$$

$$\rightarrow x(0) = 0 \rightarrow C = 0$$

$$\rightarrow x(t_f) = At_f^2 + Bt_f = 1$$

$$\rightarrow \ddot{x} = 2A \Rightarrow A = \frac{1}{4}$$

② Transversality condition

$$g - \frac{\partial g}{\partial \dot{x}} \dot{x} = 0 \quad \text{at } t = t_f$$

$$\dot{x}^2 + x - 2\dot{x}^2 = 0$$

$$x - \dot{x}^2 = 0$$

$$\dot{x} = 2At + B = \frac{1}{2}t + B$$

$$\dot{x}^2 = \frac{1}{4}t^2 + Bt + B^2$$

$$x = \dot{x}^2 \Big|_{t=t_f} \Rightarrow \frac{1}{4}t_f^2 + Bt_f = \frac{1}{4}t_f^2 + Bt_f + B^2$$

$$B^2 = 0 \rightarrow B = 0$$

$$\therefore x^*(t) = \frac{1}{4}t^2$$

$$x(t_f) = 1$$

$$\frac{1}{4}t_f^2 = 1$$

$$\rightarrow t_f = \pm 2$$

$$\therefore t_f = 2$$

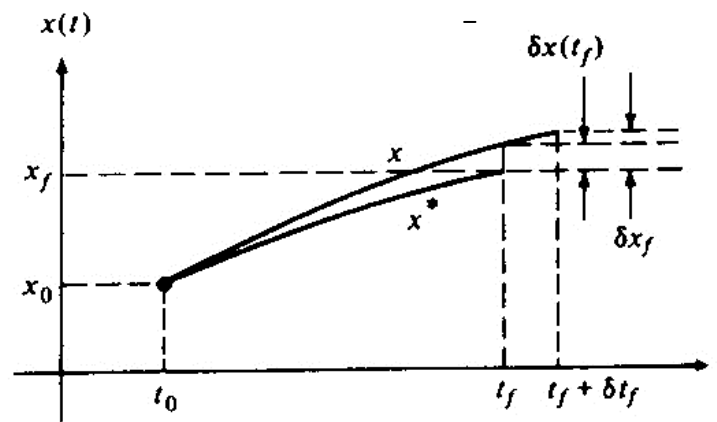
< Problems with both the final time t_f and $x(t_f)$ free >

Find a necessary condition that must be satisfied by an extremal for a function of the form.

$$J(x) = \int_{t_0}^{t_f} g(x(t), \dot{x}(t), t) dt$$

t_0 and $x(t_0)$ are specified,

t_f and $x(t_f)$ are free



$\delta x(t_f)$; the difference at $t = t_f$.

the state variation at t_f .

δx_f ; the difference at the end point of the two curves.

the state variation when t_f and $x(t_f)$ are free.

In general, $\delta x(t_f) \neq \delta x_f$

$$\delta x_f \approx \delta x(t_f) + \dot{x}(t_f) \delta t_f$$

$$\dot{x}(t_f) \approx \dot{x}^*(t_f)$$

$$\delta x(t_f) \approx \delta x_f - \dot{x}^*(t_f) \delta t_f$$

$$\begin{aligned} \Delta J &= \int_{t_0}^{t_f + \delta t_f} g(x, \dot{x}, t) dt - \int_{t_0}^{t_f} g(x^*, \dot{x}^*, t) dt \\ &= \int_{t_0}^{t_f} [g(x, \dot{x}, t) - g(x^*, \dot{x}^*, t)] dt + \int_{t_f}^{t_f + \delta t_f} g(x, \dot{x}, t) dt \end{aligned}$$

a) $x^*(t_f)$ and t_f^* are not related ($\delta x(t_f)$ and δx_f are arbitrary)

$$\left. \begin{aligned} \frac{\partial g}{\partial \dot{x}} &= 0 \\ g - \frac{\partial g}{\partial \dot{x}} \dot{x}^* &= 0 \end{aligned} \right\} \quad \text{at } t = t_f$$

$$\Rightarrow g(x^*, \dot{x}^*, t_f) = 0 \quad \text{transversality condition}$$

b) $x^*(t_f)$ and t_f^* are related by $x^*(t_f) = \theta(t_f)$



$$\begin{aligned}
\dot{\theta}(t_f) &\approx \frac{\delta x_f}{\delta t_f} \\
\rightarrow \delta x_f &\approx \dot{\theta}(t_f) \delta t_f \\
\delta J &= \frac{\partial g}{\partial \dot{x}} \dot{\theta}(t_f) \delta t_f + \left[g - \frac{\partial g}{\partial \dot{x}} \dot{x}^*(t_f) \right] \delta t_f + \int_{t_0}^{t_f} \left[\frac{\partial g}{\partial x} - \frac{d}{dt} \left(\frac{\partial g}{\partial \dot{x}} \right) \right] \delta x dt \\
&= \left[g - \frac{\partial g}{\partial \dot{x}} (\dot{x}^* - \dot{\theta}(t_f)) \right] \delta t_f + \int_{t_0}^{t_f} \left[\frac{\partial g}{\partial x} - \frac{d}{dt} \left(\frac{\partial g}{\partial \dot{x}} \right) \right] \delta x dt \\
&= 0
\end{aligned}$$

$$\therefore \frac{\partial g}{\partial x} - \frac{d}{dt} \left(\frac{\partial g}{\partial \dot{x}} \right) = 0 \quad \text{Euler-Lagrange equation}$$

$$g + \frac{\partial g}{\partial \dot{x}} \left[\dot{\theta} - \dot{x}^* \right] \Big|_{x^*(t_f), \dot{x}^*(t_f), t_f} = 0 \quad \text{Transversality condition}$$

$$\Rightarrow g(x^*(t_f), \dot{x}^*(t_f), t_f) + \frac{\partial g}{\partial \dot{x}}(x^*(t_f), \dot{x}^*(t_f), t_f) [\dot{\theta}(t_f) - \dot{x}^*(t_f)] = 0$$

Example (see Problem 4-13 on p.181)

Determine an extremal for the functional

$$J(x) = \int_0^{t_f} \sqrt{1 + \dot{x}^2(t)} dt$$

$$x(0) = 2, \quad x(t_f) \text{ terminates on the curve } \theta(t) = -4t + 5.$$

t_f and $x(t_f)$ are free.

sol.)

$$g = \sqrt{1 + \dot{x}^2(t)}$$

$$x(t_f) = \theta(t_f) = -4t_f + 5$$

Euler-Lagrange equation

$$\frac{\partial g}{\partial x} - \frac{d}{dt} \left(\frac{\partial g}{\partial \dot{x}} \right) = 0$$

$$\rightarrow \frac{d}{dt} \left(\frac{\dot{x}}{(1 + \dot{x}^2)^{1/2}} \right) = 0$$

$$\rightarrow \frac{\ddot{x}}{(1 + \dot{x}^2)^{1/2}} - \frac{1}{2} \dot{x} \frac{2\dot{x}\ddot{x}}{(1 + \dot{x}^2)^{3/2}} = 0$$

$$\rightarrow \frac{\ddot{x}}{(1+\dot{x}^2)^{1/2}} \left[1 - \frac{\dot{x}^2}{(1+\dot{x}^2)} \right] = 0$$

$$\ddot{x}(t) = 0 \quad \left(\because 1 - \frac{\dot{x}^2}{(1+\dot{x}^2)} \neq 0 \right)$$

$$\Rightarrow x^*(t) = C_1 t + C_2, \quad (1 + \dot{x}^2 > 0) \quad (i)$$

from boundary condition

$$x(0) = 2 \rightarrow C_2 = 2$$

The transversality condition is

$$g + \frac{\partial g}{\partial \dot{x}} (\dot{\theta} - \dot{x}) = 0 \quad \text{at } t = t_f$$

$$\rightarrow \frac{\dot{x}}{(1+\dot{x}^2)^{1/2}} [-4 - \dot{x}] + (1 + \dot{x}^2)^{1/2} = 0$$

$$\rightarrow -4\dot{x} - \dot{x}^2 + 1 + \dot{x}^2 = 0$$

$$\rightarrow -4\dot{x} + 1 = 0 \quad \text{at } t = t_f \quad (ii)$$

from (i) & (ii)

$$\dot{x}(t) = C_1 \rightarrow C_1 = \frac{1}{4}$$

The value of t_f is found from

$$x(t_f) = \theta(t_f)$$

$$\frac{1}{4}t_f + 2 = -4t_f + 5$$

$$\frac{17}{4}t_f = 3$$

$$t_f = \frac{12}{17}$$

$\therefore$ the solution

$$x^*(t) = \frac{1}{4}t + 2$$

$$t_f = \frac{12}{17}$$

Summary

1) Definition and phenomena

Functional, Extremal, Extremum

Relative minimum & maximum

$$\text{sign } \Delta J = \text{sign } \delta J$$

2) Fundamental theorem of calculus of variation

$$\delta J(\bar{x}^*, \delta x) = 0 \quad \text{for extremum}$$

$$3) \quad J(x) = \int_{t_0}^{t_f} g(x(t), \dot{x}(t), t) dt, \quad t_0, x(t_0) \text{ are given,}$$

The admissible curve $x(t)$ are smooth

$$\begin{aligned} \delta J(x^*, \delta x) &= \frac{\partial g}{\partial \dot{x}} \delta x_f + \left[g - \frac{\partial g}{\partial \dot{x}} \dot{x} \right] \delta t_f + \int_{t_0}^{t_f} \left[\frac{\partial g}{\partial x} - \frac{d}{dt} \left(\frac{\partial g}{\partial \dot{x}} \right) \right] \delta x dt \\ &= 0 \end{aligned}$$

We have found that regardless of the boundary conditions the Euler-Lagrange equation must be satisfied.

$$\therefore \frac{\partial g}{\partial x}(x^*(t), \dot{x}^*(t), t) - \frac{d}{dt} \left[\frac{\partial g}{\partial \dot{x}}(x^*(t), \dot{x}^*(t), t) \right] = 0$$

i) t_f and $x(t_f)$ are specified

$$\text{then, } \delta t_f = 0 \quad \text{and} \quad \delta x_f = \delta x(t_f) = 0$$

ii) t_f is specified, $x(t_f)$ is free

$$\text{then, } \delta t_f = 0 \quad \text{and} \quad \delta x_f = \delta x(t_f) \neq 0$$

$$\left. \frac{\partial g}{\partial \dot{x}} \right|_{t=t_f} = 0$$

iii) t_f is free, $x(t_f)$ is specified

then, $\delta t_f \neq 0$ and $\delta x_f = 0$

$$g - \frac{\partial g}{\partial \dot{x}} \dot{x}^* \Big|_{t=t_f} = 0$$

iv) t_f and $x(t_f)$ are free

then, $\delta t_f \neq 0, \delta x_f \neq 0$

a) $x^*(t_f)$ and t_f^* are not related

$$\frac{\partial g}{\partial \dot{x}} \Big|_{t=t_f} = 0, \quad g - \frac{\partial g}{\partial \dot{x}} \dot{x}^* \Big|_{t=t_f} = 0 \quad \Rightarrow g \Big|_{t=t_f} = 0$$

b) $x^*(t_f) = \theta(t_f)$

$$\delta x_f \approx \dot{\theta} \delta t_f$$

$$\left[\frac{\partial g}{\partial \dot{x}} (\dot{\theta} - \dot{x}^*) + g \right]_{t=t_f} = 0$$

So far, the functionals considered have contained only a single function (x) and its first derivative ($\dot{x}$). We can generalize our discussion to include functionals that may contain several independent functions and their first-derivatives.

⇒ The result are matrix version (Sec.4.3)

1) Regardless of the boundary conditions, the Euler-Lagrange equations are necessary conditions and must be satisfied.

$$\frac{\partial g}{\partial \vec{x}} (\vec{x}^*(t), \dot{\vec{x}}^*(t), t) - \frac{d}{dt} \left[\frac{\partial g}{\partial \dot{\vec{x}}} (\vec{x}^*(t), \dot{\vec{x}}^*(t), t) \right] = \vec{0}$$

2) The required boundary conditions are found from the following equation by making the appropriate substitutions for $\delta \vec{x}_f$ and δt_f .

$$\left[\frac{\partial g}{\partial \dot{\vec{x}}} (\vec{x}^*(t), \dot{\vec{x}}^*(t), t) \right]_{t=t_f}^T \delta \vec{x}_f + \left[g(\vec{x}^*(t), \dot{\vec{x}}^*(t), t) - \left\{ \frac{\partial g}{\partial \dot{\vec{x}}} (\vec{x}^*(t), \dot{\vec{x}}^*(t), t) \right\}^T \dot{\vec{x}}(t) \right]_{t=t_f} \delta t_f = 0$$

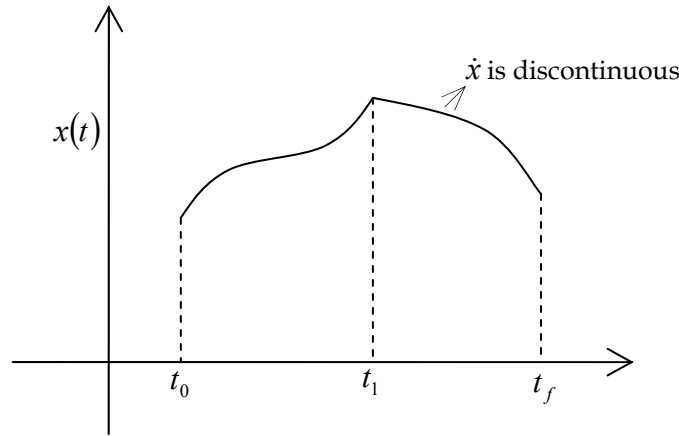
Table 4-1 DETERMINATION OF BOUNDARY-VALUE RELATIONSHIPS

<i>Problem description</i>	<i>Substitution</i>	<i>Boundary conditions</i>	<i>Remarks</i>
1. $\mathbf{x}(t_f)$, t_f both specified (Problem 1)	$\delta \mathbf{x}_f = \delta \mathbf{x}(t_f) = 0$ $\delta t_f = 0$	$\mathbf{x}^*(t_0) = \mathbf{x}_0$ $\mathbf{x}^*(t_f) = \mathbf{x}_f$	$2n$ equations to determine $2n$ constants of integration
2. $\mathbf{x}(t_f)$ free; t_f specified (Problem 2)	$\delta \mathbf{x}_f = \delta \mathbf{x}(t_f)$ $\delta t_f = 0$	$\mathbf{x}^*(t_0) = \mathbf{x}_0$ $\frac{\partial g}{\partial \dot{\mathbf{x}}}(\mathbf{x}^*(t_f), \dot{\mathbf{x}}^*(t_f), t_f) = 0$	$2n$ equations to determine $2n$ constants of integration
3. t_f free; $\mathbf{x}(t_f)$ specified (Problem 3)	$\delta \mathbf{x}_f = 0$	$\mathbf{x}^*(t_0) = \mathbf{x}_0$ $\mathbf{x}^*(t_f) = \mathbf{x}_f$ $g(\mathbf{x}^*(t_f), \dot{\mathbf{x}}^*(t_f), t_f)$ $- \left[\frac{\partial g}{\partial \dot{\mathbf{x}}}(\mathbf{x}^*(t_f), \dot{\mathbf{x}}^*(t_f), t_f) \right]^T \dot{\mathbf{x}}^*(t_f) = 0$	$(2n + 1)$ equations to determine $2n$ constants of integration and t_f
4. t_f , $\mathbf{x}(t_f)$ free and independent (Problem 4)	—	$\mathbf{x}^*(t_0) = \mathbf{x}_0$ $\frac{\partial g}{\partial \dot{\mathbf{x}}}(\mathbf{x}^*(t_f), \dot{\mathbf{x}}^*(t_f), t_f) = 0$ $g(\mathbf{x}^*(t_f), \dot{\mathbf{x}}^*(t_f), t_f) = 0$	$(2n + 1)$ equations to determine $2n$ constants of integration and t_f
5. t_f , $\mathbf{x}(t_f)$ free but related by $\mathbf{x}(t_f) = \boldsymbol{\theta}(t_f)$ (Problem 4)	$\delta \mathbf{x}_f = \frac{d\boldsymbol{\theta}}{dt}(t_f) \delta t_f^\dagger$	$\mathbf{x}^*(t_0) = \mathbf{x}_0$ $\mathbf{x}^*(t_f) = \boldsymbol{\theta}(t_f)$ $g(\mathbf{x}^*(t_f), \dot{\mathbf{x}}^*(t_f), t_f)$ $+ \left[\frac{\partial g}{\partial \dot{\mathbf{x}}}(\mathbf{x}^*(t_f), \dot{\mathbf{x}}^*(t_f), t_f) \right]^T \left[\frac{d\boldsymbol{\theta}}{dt}(t_f) - \dot{\mathbf{x}}^*(t_f) \right] = 0^\dagger$	$(2n + 1)$ equations to determine $2n$ constants of integration and t_f

$^\dagger \frac{d\boldsymbol{\theta}}{dt}$ denotes the $n \times 1$ column vector $\left[\frac{d\theta_1}{dt} \quad \frac{d\theta_2}{dt} \quad \dots \quad \frac{d\theta_n}{dt} \right]^T$.

4.4 Piecewise-Smooth Extremals

$\dot{\bar{x}}(t)$ will be discontinuous except at a finite number of times in the intervals (t_0, t_f) . At a time when $\dot{\bar{x}}$ is discontinuous, $\bar{x}(t)$ is said to have a corner.



$$\begin{aligned} \dot{x}^*(t_1^-) &\neq \dot{x}^*(t_1^+) \\ \text{but, } x^*(t_1^-) &= x^*(t_1^+) \end{aligned}$$

t_1^- and t_1^+ denoted the times just before and just after the discontinuity $\dot{\bar{x}}$.

$$J(x) = \int_{t_0}^{t_f} g(x(t), \dot{x}(t), t) dt$$

$x(t_0), x(t_f)$ are given, t_0 and t_f are also fixed

Let us look for necessary conditions that what must be satisfied by corner extremals (broken extremals)

$$\delta J = \int_{t_0}^{t_1} \left[\frac{\partial g}{\partial x} \delta x + \frac{\partial g}{\partial \dot{x}} \delta \dot{x} \right] dt + \int_{t_1}^{t_f} \left[\frac{\partial g}{\partial x} \delta x + \frac{\partial g}{\partial \dot{x}} \delta \dot{x} \right] dt$$

Assume $x(t_1)$ and t_1 are not related

$$t_0 \rightarrow t_1^-, \quad t_1^+ \rightarrow t_f$$

We can apply the same method as that in free t_f case.

$$\Rightarrow \delta x_{t_1} \text{ and } \delta t_1 \text{ are free}$$

$$\delta J = \int_{t_0}^{t_1} \left[\frac{\partial g}{\partial x} \delta x + \frac{\partial g}{\partial \dot{x}} \delta \dot{x} \right] dt - \int_{t_f}^{t_1} \left[\frac{\partial g}{\partial x} \delta x + \frac{\partial g}{\partial \dot{x}} \delta \dot{x} \right] dt$$

$$\delta x_{t_1} (= \delta x(t_1) + \dot{x}^*(t_1) \delta t_1) \quad \& \quad \delta t_1 \quad \text{are free}$$

$$\text{Remember (p.33)} \quad \delta J = \frac{\partial g}{\partial \dot{x}} \delta x_f + \left[g - \frac{\partial g}{\partial \dot{x}} \dot{x}^* \right] \delta t_f + \int_{t_0}^{t_f} \left[\frac{\partial g}{\partial x} - \frac{d}{dt} \left(\frac{\partial g}{\partial \dot{x}} \right) \right] \delta x dt = 0$$

Similary we have

$$\begin{aligned} \delta J &= \left[\frac{\partial g}{\partial \dot{x}} \delta x_{t_1} + \left(g - \frac{\partial g}{\partial \dot{x}} \dot{x} \right) \delta t_1 \right]_{t=t_1^-} + \int_{t_0}^{t_1} \left[\frac{\partial g}{\partial x} - \frac{d}{dt} \left(\frac{\partial g}{\partial \dot{x}} \right) \right] \delta x dt \\ &\quad - \left[\frac{\partial g}{\partial \dot{x}} \delta x_{t_1} + \left(g - \frac{\partial g}{\partial \dot{x}} \dot{x} \right) \delta t_1 \right]_{t=t_1^+} + \int_{t_1}^{t_f} \left[\frac{\partial g}{\partial x} - \frac{d}{dt} \left(\frac{\partial g}{\partial \dot{x}} \right) \right] \delta x dt \\ &= \left[\frac{\partial g}{\partial \dot{x}} \Big|_{t=t_1^-} - \frac{\partial g}{\partial \dot{x}} \Big|_{t=t_1^+} \right] \delta x_{t_1} + \left[\left(g - \frac{\partial g}{\partial \dot{x}} \dot{x} \right)_{t=t_1^-} - \left(g - \frac{\partial g}{\partial \dot{x}} \dot{x} \right)_{t=t_1^+} \right] \delta t_1 \\ &\quad + \int_{t_0}^{t_1} \left[\frac{\partial g}{\partial x} - \frac{d}{dt} \left(\frac{\partial g}{\partial \dot{x}} \right) \right] \delta x dt + \int_{t_1}^{t_f} \left[\frac{\partial g}{\partial x} - \frac{d}{dt} \left(\frac{\partial g}{\partial \dot{x}} \right) \right] \delta x dt \\ &= 0 \end{aligned}$$

$\delta x(t_1)$ & δt_1 are independently arbitrary, so that their coefficients each be zero.

Necessary conditions

i) Euler-Lagrange equation

$$\text{ii)} \quad \frac{\partial g}{\partial \dot{x}} \Big|_{t=t_1^-} = \frac{\partial g}{\partial \dot{x}} \Big|_{t=t_1^+}$$

$$\text{iii)} \quad g - \frac{\partial g}{\partial \dot{x}} \dot{x} \Big|_{t=t_1^-} = g - \frac{\partial g}{\partial \dot{x}} \dot{x} \Big|_{t=t_1^+}$$

ii) & iii) are Weierstrass-Erdman Corner conditions hold at t_1 , where the extremal has a corner.

If there are several times, $t_1, t_2, \dots, t_n$ when corner exist, then at each such time those corner conditions must be satisfied.

If $\delta x(t_1)$ & δt_1 are related by $x(t_1) = \theta(t_1)$

$$\delta x_{t_1} \approx \dot{\theta}(t_1) \delta t_1$$

$$\begin{aligned}
\delta J &= \frac{\partial g}{\partial \dot{x}} \dot{\theta}(t_f) \delta t_f + \left[g - \frac{\partial g}{\partial \dot{x}} \dot{x}^*(t_f) \right] \delta t_f + \int_{t_0}^{t_f} \left[\frac{\partial g}{\partial x} - \frac{d}{dt} \left(\frac{\partial g}{\partial \dot{x}} \right) \right] \delta x dt \\
&= \left[\frac{\partial g}{\partial \dot{x}} \right]_{t=t_1^-} - \left[\frac{\partial g}{\partial \dot{x}} \right]_{t=t_1^+} \dot{\theta}(t_1) \delta t_1 + \left[\left(g - \frac{\partial g}{\partial \dot{x}} \dot{x} \right)_{t=t_1^-} - \left(g - \frac{\partial g}{\partial \dot{x}} \dot{x} \right)_{t=t_1^+} \right] \delta t_1 \\
&\quad + \int_{t_0}^{t_1} \left[\frac{\partial g}{\partial x} - \frac{d}{dt} \left(\frac{\partial g}{\partial \dot{x}} \right) \right] \delta x dt + \int_{t_1}^{t_f} \left[\frac{\partial g}{\partial x} - \frac{d}{dt} \left(\frac{\partial g}{\partial \dot{x}} \right) \right] \delta x dt \\
&= \left[\frac{\partial g}{\partial \dot{x}} \dot{\theta} \right]_{t=t_1^-} + \left(g - \frac{\partial g}{\partial \dot{x}} \dot{x} \right)_{t=t_1^-} - \left[\frac{\partial g}{\partial \dot{x}} \dot{\theta} \right]_{t=t_1^+} - \left(g - \frac{\partial g}{\partial \dot{x}} \dot{x} \right)_{t=t_1^+} \delta t_1 \\
&\quad + \int_{t_0}^{t_1} \left[\frac{\partial g}{\partial x} - \frac{d}{dt} \left(\frac{\partial g}{\partial \dot{x}} \right) \right] \delta x dt + \int_{t_1}^{t_f} \left[\frac{\partial g}{\partial x} - \frac{d}{dt} \left(\frac{\partial g}{\partial \dot{x}} \right) \right] \delta x dt \\
&= 0
\end{aligned}$$

We obtain Weierstrass-Erdman condition as

$$\left[\frac{\partial g}{\partial \dot{x}} (\dot{\theta} - \dot{x}) + g \right]_{t=t_1^-} = \left[\frac{\partial g}{\partial \dot{x}} (\dot{\theta} - \dot{x}) + g \right]_{t=t_1^+}$$

We can extend the conditions for functionals having several independent functions.

Theorem:

If $\frac{\partial^2 g}{\partial \dot{x}^2} \neq 0$ for all $x, \dot{x}$ and t in the domain of g , then no $x(t)$ with a corner can extremize J .

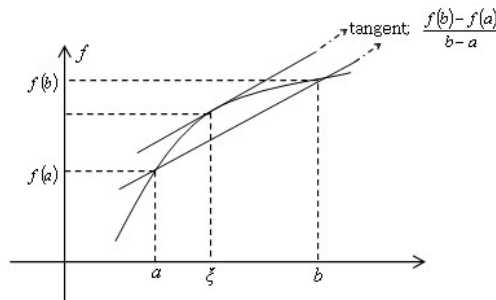
Proof.) Using mean value theorem³

³ Mean Value Theorem:

If $f(x)$ is smooth and its derivative exists in the interval (a, b) , then there exists a number ξ in (a, b) such that

$$f(b) = f(a) + f'(\xi)(b - a)$$

where $\xi = a + (b - a)\alpha$, $0 < \alpha < 1$



$$g_{\dot{x}^+} - g_{\dot{x}^-} = 0$$

$$\text{Let } f \equiv \frac{\partial g}{\partial \dot{x}}, \quad a = \dot{x}(t_1^-), \quad b = \dot{x}(t_1^+),$$

$$f(b) = \frac{\partial g}{\partial \dot{x}}(x, \dot{x}^+, t), \quad t = t_1^+$$

$$f(a) = \frac{\partial g}{\partial \dot{x}}(x, \dot{x}^-, t), \quad t = t_1^-$$

$$f' = \frac{\partial^2 g}{\partial \dot{x}^2} = g_{\dot{x}\dot{x}}$$

$$g_{\dot{x}^+} - g_{\dot{x}^-} = g_{\dot{x}\dot{x}}(x^*(t_1), \xi, t_1) [\dot{x}^*(t_1^+) - \dot{x}^*(t_1^-)]$$

where $\xi \equiv \dot{x}^*(t_1^-) + \alpha [\dot{x}^*(t_1^+) - \dot{x}^*(t_1^-)]$

If $g_{\dot{x}\dot{x}} \neq 0$ at any value of $x, \dot{x}, t$

Then $\dot{x}^*(t_1^+) - \dot{x}^*(t_1^-) = 0 \rightarrow$ no corner

$$\text{Example) } J = \int_{t_0}^{t_f} [a\dot{x}^2 + bx\dot{x} + cx^2] dt \quad a \neq 0, \quad x(0) = x_0, \quad x(t_f) = x_f$$

Show that $x^*(t)$ can have no corner

$$\frac{\partial^2 g}{\partial \dot{x}^2} = \frac{\partial}{\partial \dot{x}} [2a\dot{x} + bx] = 2a \neq 0 \quad (\because a \neq 0) \rightarrow \text{no corner!!!}$$

4.5 Constrained Extrema

Point(path) constraints

Let us determine a set of necessary conditions for a function x^* to be an extremal for a function of a form

$$J(x) = \int_{t_0}^{t_f} g(x, \dot{x}, t) dt$$

subject to $f(x(t), t) = 0$

The path constraint would be present if the admissible curves were required to lie on a specified surface.

$x(t_0), x(t_f)$ are given

Form the augmented functional by adjoining the constraining relations to use Lagrange multiplier

$$J_a(x) = \int_{t_0}^{t_f} (g + \Lambda f) dt \quad \Lambda; \text{ optimal weight function } \Lambda(t)$$

When $f=0$ is satisfied, $J_a = J$

$$\begin{aligned} \delta J_a(x) &= \int_{t_0}^{t_f} \left[\frac{\partial g}{\partial x} \delta x + \frac{\partial g}{\partial \dot{x}} \delta \dot{x} + \Lambda \frac{\partial f}{\partial x} \delta x + \delta \Lambda \cdot f \right] dt \\ &= \frac{\partial g}{\partial \dot{x}} \delta x \Big|_{t_0}^{t_f} + \int_{t_0}^{t_f} \left\{ \left[\frac{\partial g}{\partial x} - \frac{d}{dt} \left(\frac{\partial g}{\partial \dot{x}} \right) + \Lambda \frac{\partial f}{\partial x} \right] \delta x + \delta \Lambda \cdot f \right\} dt \\ &= 0 \end{aligned}$$

$$\begin{aligned} \therefore \frac{\partial g}{\partial x} - \frac{d}{dt} \left(\frac{\partial g}{\partial \dot{x}} \right) + \Lambda \frac{\partial f}{\partial x} &= 0 \quad \text{Euler-Lagrange equation} \\ f &= 0 \end{aligned}$$

Differential Equation constraints

Let us find necessary conditions for a function $x^*(t)$ to be an extremal for a functional

$$J(x) = \int_{t_0}^{t_f} g(x, \dot{x}, t) dt$$

subject to $f(x(t), \dot{x}(t), t) = 0$

$x(t_0), x(t_f)$ are given

$$J_a(x) = \int_{t_0}^{t_f} (g + \Lambda \cdot f) dt \quad \Lambda(t); \text{ optimal weight function}$$

$$\begin{aligned}
\delta J_a(x) &= \int_{t_0}^{t_f} \left[\frac{\partial g}{\partial x} \delta x + \frac{\partial g}{\partial \dot{x}} \delta \dot{x} + \Lambda \frac{\partial f}{\partial x} \delta x + \Lambda \frac{\partial f}{\partial \dot{x}} \delta \dot{x} + \delta \Lambda \cdot f \right] dt \\
&= \frac{\partial g}{\partial \dot{x}} \delta x \Big|_{t_0}^{t_f} + \Lambda \frac{\partial f}{\partial \dot{x}} \delta x \Big|_{t_0}^{t_f} + \int_{t_0}^{t_f} \left\{ \left[\frac{\partial g}{\partial x} - \frac{d}{dt} \left(\frac{\partial g}{\partial \dot{x}} \right) + \Lambda \frac{\partial f}{\partial x} - \frac{d}{dt} \left(\Lambda \frac{\partial f}{\partial \dot{x}} \right) \right] \delta x + \delta \Lambda \cdot f \right\} dt \\
&= 0
\end{aligned}$$

$$\begin{aligned}
\therefore \frac{\partial g}{\partial x} - \frac{d}{dt} \left(\frac{\partial g}{\partial \dot{x}} + \Lambda \frac{\partial f}{\partial \dot{x}} \right) + \Lambda \frac{\partial f}{\partial x} &= 0 && \text{Euler-Lagrange equation} \\
f &= 0
\end{aligned}$$

Isoperimetric constraints

We want to find $x^*(t)$ to be an extremal for a functional

$$J = \int_{t_0}^{t_f} g(x, \dot{x}, t) dt$$

$x(t_0), x(t_f)$ are given,

$$\text{subject to } \int_{t_0}^{t_f} F(x(t), \dot{x}(t), t) dt = c$$

(c is specified constant. In control problems, c can be a total fuel or energy available.)

This constraint can be put into the form of differential equation constraints by defining the new variable

$$z(t) = \int_{t_0}^t F(x(t), \dot{x}(t), t) dt$$

$$z(t_0) = 0, \quad z(t_f) = c \quad z; \text{ additional state variables}$$

$$\dot{z}(t) = F(x(t), \dot{x}(t), t)$$

$$F(x(t), \dot{x}(t), t) - \dot{z}(t) \equiv f(x(t), \dot{x}(t), t, \dot{z}) = 0, \quad f; \text{ a new differential const.}$$

Augmented functional

$$\begin{aligned}
 J_a(x) &= \int_{t_0}^{t_f} (g + \Lambda \cdot f) dt \\
 \delta J_a(x) &= \int_{t_0}^{t_f} \left[\frac{\partial g}{\partial x} \delta x + \frac{\partial g}{\partial \dot{x}} \delta \dot{x} + \Lambda \frac{\partial f}{\partial x} \delta x + \Lambda \frac{\partial f}{\partial \dot{x}} \delta \dot{x} + \Lambda \frac{\partial f}{\partial \dot{z}} \delta \dot{z} + \delta \Lambda \cdot f \right] dt \\
 &= \frac{\partial g}{\partial x} \delta x \Big|_{t_0}^{t_f} + \Lambda \frac{\partial f}{\partial x} \delta x \Big|_{t_0}^{t_f} + \Lambda \frac{\partial f}{\partial \dot{z}} \delta z \Big|_{t_0}^{t_f} \\
 &\quad + \int_{t_0}^{t_f} \left\{ \left[\frac{\partial g}{\partial x} - \frac{d}{dt} \left(\frac{\partial g}{\partial \dot{x}} + \Lambda \frac{\partial f}{\partial \dot{x}} \right) + \Lambda \frac{\partial f}{\partial x} \right] \delta x - \frac{d}{dt} \left(\Lambda \frac{\partial f}{\partial \dot{z}} \right) \delta z + \delta \Lambda \cdot f \right\} dt \\
 &= 0
 \end{aligned}$$

$$\therefore \frac{\partial g}{\partial x} - \frac{d}{dt} \left(\frac{\partial g}{\partial \dot{x}} + \Lambda \frac{\partial f}{\partial \dot{x}} \right) + \Lambda \frac{\partial f}{\partial x} = 0 \quad \text{Euler-Lagrange equation}$$

$$\frac{d}{dt} \left(\Lambda \frac{\partial f}{\partial \dot{z}} \right) = 0 \xrightarrow{f=F-\dot{z} \rightarrow f_z=-1} \frac{d}{dt} (-\Lambda) = 0 \Rightarrow \Lambda(t) = \text{const.}$$

$\Rightarrow$ Lagrange multiplier is constant

$$f = 0 \rightarrow \dot{z}(t) = F(x(t), \dot{x}(t), t)$$

(These eqs. must satisfy the boundary conditions,

$$z(t_0) = 0, \quad z(t_f) = c)$$

< Second variation & Sufficient Conditions >

- The increment of a functional

$$\Delta J \equiv J(\bar{x} + \delta \bar{x}) - J(\bar{x})$$

$$\Delta J(\bar{x}, \delta \bar{x}) = \underbrace{\delta J(\bar{x}, \delta \bar{x})}_{\text{linear term in } \delta \bar{x}} + g_1(\bar{x}, \delta \bar{x}) \cdot \|\delta \bar{x}\|$$

$$\text{If } \lim_{\|\delta \bar{x}\| \rightarrow 0} \{g_1(\bar{x}, \delta \bar{x})\} = 0$$

then, J is said to be differentiable on $\bar{x}$ and δJ is the variation of J evaluated for the function $\bar{x}$, δJ is the linear approximation to the difference in the functional J caused by $\bar{x}$, $\bar{x} + \delta \bar{x}$.

- The increment of a functional

$$\begin{aligned}
\Delta J(\bar{x}, \delta\bar{x}) &= J(\bar{x} + \delta\bar{x}) - J(\bar{x}) \\
&= J(\bar{x}) + \frac{\partial J}{\partial \bar{x}} \Big|_{\bar{x}} \delta\bar{x} + \frac{1}{2} \frac{\partial^2 J}{\partial \bar{x}^2} \Big|_{\bar{x}} (\delta\bar{x})^2 + \dots - J(\bar{x}) \\
&= \frac{\partial J}{\partial \bar{x}} \Big|_{\bar{x}} \delta\bar{x} + \frac{1}{2} \frac{\partial^2 J}{\partial \bar{x}^2} \Big|_{\bar{x}} (\delta\bar{x})^2 + H.O.T
\end{aligned}$$

where $\frac{\partial J}{\partial \bar{x}} \Big|_{\bar{x}} \delta\bar{x} \equiv \delta J(\bar{x}, \delta\bar{x})$; first variation

$$\frac{1}{2} \frac{\partial^2 J}{\partial \bar{x}^2} \Big|_{\bar{x}} (\delta\bar{x})^2 \equiv \delta^2 J; \text{ second variation}$$

- $\Delta J(\bar{x}, \delta\bar{x}) = \delta J(\bar{x}, \delta\bar{x}) + \delta^2 J(\bar{x}, \delta\bar{x}) + g_2(\bar{x}, \delta\bar{x}) \cdot \|\delta\bar{x}\|^2$
linear term in $\delta\bar{x}$

$\delta^2 J$ is a quadratic functional, and if $\lim_{\|\delta\bar{x}\| \rightarrow 0} \{g_2(\bar{x}, \delta\bar{x})\} = 0$, $J(\bar{x})$ is twice

differentiable. From now on, it will be tacitly assumed that we are dealing with functionals which are twice differentiable. The second variation of such a functional is uniquely defined.

- Theorem 1 : A necessary condition for the functional $J(\bar{x}(t))$ to have a minimum for $\bar{x}(t) = \bar{x}^*(t)$ is that
$$\delta^2 J(x(t)) \geq 0 \quad (i)$$
for $\bar{x}(t) = \bar{x}^*(t)$ and all admissible $\delta\bar{x}(t)$.
For a maximum, the sign $\geq$ in eq(i) is replaced by $\leq$.

Proof) By definition, we have

$$\begin{aligned}
\Delta J(\bar{x}, \delta\bar{x}) &= \delta J(\bar{x}, \delta\bar{x}) + \delta^2 J(\bar{x}, \delta\bar{x}) + g_2(\bar{x}, \delta\bar{x}) \cdot \|\delta\bar{x}\|^2 \\
&\quad \text{linear term in } \delta\bar{x} \\
&= \delta J(\bar{x}, \delta\bar{x}) + \delta^2 J(\bar{x}, \delta\bar{x}) + \varepsilon \|\delta\bar{x}\|^2 \quad (ii)
\end{aligned}$$

where $\varepsilon \equiv g_2(\bar{x}, \delta\bar{x})$, and $\varepsilon \rightarrow 0$ as $\|\delta\bar{x}\| \rightarrow 0$

According the fundamental theorem, $\delta J = 0$ for $\bar{x}(t) = \bar{x}^*(t)$ and all admissible $\delta\bar{x}(t)$. Hence eq(ii) becomes

$$\Delta J(\bar{x}, \delta\bar{x}) = \delta^2 J(\bar{x}, \delta\bar{x}) + \varepsilon \|\delta\bar{x}\|^2 \quad (iii)$$

Thus, for sufficiently small $\|\delta\bar{x}\|$, the sign of $\Delta J(\bar{x}, \delta\bar{x})$ will be the same as the sign of $\delta^2 J(\bar{x}, \delta\bar{x})$.

Now suppose that $\delta^2 J(\bar{x}, \delta\bar{x}_0) < 0$ for some admissible $\delta\bar{x}_0$.

Then for any $\alpha \neq 0$, no matter how small, we have

$$\delta^2 J(\bar{x}, \alpha\delta\bar{x}_0) = \alpha^2 \delta^2 J(\bar{x}, \delta\bar{x}_0) < 0$$

Hence, eq(iii) can be made negative for arbitrary small $\|\delta\bar{x}\|$.

But this is impossible, since by hypothesis $J(\bar{x}(t))$ has a minimum for $\bar{x}(t) = \bar{x}^*(t)$, i.e.,

$$\Delta J(\bar{x}, \delta\bar{x}) = J(\bar{x}^* + \delta\bar{x}) - J(\bar{x}^*) \geq 0$$

for all sufficiently small $\|\delta\bar{x}\|$

This contradiction proves the theorem. [end proof]

The condition $\delta^2 J(x(t)) \geq 0$ is necessary but of course not sufficient for the functional $J(\bar{x}(t))$ to have a minimum for a given function. To obtain a sufficient condition, we introduce the following concept: We say that a quadratic functional $\delta^2 J(\delta x(t))$ defined on some normed linear space $\mathfrak{R}$ is strongly positive if there exists a constant $k > 0$ such that

$$\delta^2 J(\delta x(t)) \geq k \|\delta x\|^2 \quad \text{for all } \delta\bar{x}(t)$$

- Theorem 2 : A sufficient condition for a functional $J(\bar{x}(t))$ to have a minimum for $\bar{x}(t) = \bar{x}^*(t)$, given that the first variation $\delta J(\delta x(t))$ vanishes for $\bar{x}(t) = \bar{x}^*(t)$, is that its second variation $\delta^2 J(\delta x(t))$ be strongly positive for $\bar{x}(t) = \bar{x}^*(t)$.

Proof) For $\vec{x}(t) = \vec{x}^*(t)$, we have $\delta J(\delta x(t)) = 0$ for all admissible $\delta \vec{x}(t)$, and hence

$$\Delta J(\delta \vec{x}) = \delta^2 J(\delta \vec{x}) + \varepsilon \|\delta \vec{x}\|^2 \quad \text{where } \varepsilon \rightarrow 0 \text{ as } \|\delta \vec{x}\| \rightarrow 0.$$

Moreover, for $\vec{x}(t) = \vec{x}^*(t)$,

$$\delta^2 J(\delta x(t)) \geq k \|\delta x\|^2 \quad \text{where } k = \text{constant} > 0.$$

Thus, for sufficiently small ε_1 , $|\varepsilon| < \frac{1}{2}k$ if $\|\delta x\| < \varepsilon_1$. It follows that

$$\Delta J(\delta \vec{x}) = \delta^2 J(\delta \vec{x}) + \varepsilon \|\delta \vec{x}\|^2 > \frac{1}{2}k \|\delta x\|^2 > 0$$

if $\|\delta x\| < \varepsilon_1$, i.e., $J(\vec{x}(t))$ has a minimum for

$\vec{x}(t) = \vec{x}^*(t)$, as asserted. [end proof]

- $J = \int_{t_0}^{t_f} g(x, \dot{x}, t) dt$ $x(t_0), x(t_f)$ are given

$$\begin{aligned} \Delta J &= \int_{t_0}^{t_f} g(x, \dot{x}, t) dt - \int_{t_0}^{t_f} g(x^*, \dot{x}^*, t) dt \\ &= \int_{t_0}^{t_f} \left\{ g(x^*, \dot{x}^*, t) + \frac{\partial g}{\partial x} \Big|_{x^*, \dot{x}^*} \delta x + \frac{\partial g}{\partial \dot{x}} \Big|_{x^*, \dot{x}^*} \delta \dot{x} \right. \\ &\quad \left. + \frac{1}{2} \left[\frac{\partial^2 g}{\partial x^2} \Big|_{x^*, \dot{x}^*} (\delta x)^2 + 2 \frac{\partial^2 g}{\partial x \partial \dot{x}} \Big|_{x^*, \dot{x}^*} \delta x \delta \dot{x} + \frac{\partial^2 g}{\partial \dot{x}^2} \Big|_{x^*, \dot{x}^*} (\delta \dot{x})^2 \right] + H.O.T \right\} dt \\ &\quad - \int_{t_0}^{t_f} g(x^*, \dot{x}^*, t) dt \end{aligned}$$

$$\delta^2 J = \int_{t_0}^{t_f} \frac{1}{2} \left[g_{xx} (\delta x)^2 + 2 g_{x\dot{x}} \delta x \delta \dot{x} + g_{\dot{x}\dot{x}} (\delta \dot{x})^2 \right] dt$$

The necessary condition for a minimum (maximum) is $\delta^2 J \geq 0$ ($\delta^2 J \leq 0$) for all variations that are admissible $\delta x(t)$ must be continuous and must have continuous first derivative (weak conditions).

i.e., $\delta J = 0, \quad \delta^2 J \geq 0 \rightarrow \text{minimum } J$ $\delta J = 0, \quad \delta^2 J \leq 0 \rightarrow \text{maximum } J$
--

< Accessory problem >

Note that $\delta^2 J = \int_{t_0}^{t_f} w(\delta x, \delta \dot{x}, t) dt$

$$\text{where } w = \frac{1}{2} [g_{xx}(\delta x)^2 + 2g_{x\dot{x}}\delta x\delta \dot{x} + g_{\dot{x}\dot{x}}(\delta \dot{x})^2]$$

$$\delta x(t_0) = \delta x(t_f) = 0, \quad \delta x(t) \text{ is arbitrary}$$

The problem of extremizing $\delta^2 J$ with respect to δx said to be the accessory problem.

Necessary condition

$$\frac{d}{dt} \left(\frac{\partial w}{\partial(\delta \dot{x})} \right) - \frac{\partial w}{\partial(\delta x)} = 0 \quad ; \text{ Jacobi differential equation}$$

Let the solution to this be $\delta v(t)$

The Jacobian equation is ($\delta x \rightarrow \delta v$, $\delta \dot{x} \rightarrow \delta \dot{v}$)

$$\begin{aligned} & \rightarrow \frac{d}{dt} [g_{\dot{x}\dot{x}}\delta \dot{v} + g_{x\dot{x}}\delta v] - (g_{x\dot{x}}\delta \dot{v} + g_{xx}\delta v) = 0 \\ & \rightarrow \frac{d}{dt} (g_{\dot{x}\dot{x}}\delta \dot{v}) + \frac{d}{dt} (g_{x\dot{x}})\delta v + g_{x\dot{x}}\delta \dot{v} - (g_{x\dot{x}}\delta \dot{v} + g_{xx}\delta v) = 0 \\ & \rightarrow \frac{d}{dt} (g_{\dot{x}\dot{x}}\delta \dot{v}) + \left[\frac{d}{dt} (g_{x\dot{x}}) - g_{xx} \right] \delta v = 0 \end{aligned} \quad \textcircled{1}$$

$$\begin{aligned} 2\delta^2 J &= \int_{t_0}^{t_f} \left[g_{\dot{x}\dot{x}}(\delta \dot{x})^2 + g_{x\dot{x}} \frac{d}{dt} (\delta x)^2 + g_{xx}(\delta x)^2 \right] dt \\ &= g_{\dot{x}\dot{x}}\delta \dot{x}\delta x \Big|_{t_0}^{t_f} + g_{x\dot{x}}(\delta x)^2 \Big|_{t_0}^{t_f} - \int_{t_0}^{t_f} \left[\frac{d}{dt} (g_{\dot{x}\dot{x}}\delta \dot{x})\delta x - \left(g_{xx} - \frac{d}{dt} g_{x\dot{x}} \right) (\delta x)^2 \right] dt \end{aligned}$$

$$\delta^2 J = -\frac{1}{2} \int_{t_0}^{t_f} \delta x \left[\frac{d}{dt} (g_{\dot{x}\dot{x}}\delta \dot{x}) - \left(g_{xx} - \frac{d}{dt} g_{x\dot{x}} \right) \delta x \right] dt \quad \textcircled{2}$$

If $\delta x \rightarrow \delta v$, $\delta^2 J = 0$ because equation $\textcircled{1}$ is satisfied.

The equation $\textcircled{2}$ can be multiplied and divided by $\delta v(t)$

$$\begin{aligned}
\delta^2 J &= -\frac{1}{2} \int_{t_0}^{t_f} \frac{\delta x}{\delta v} \left[\delta v \frac{d}{dt} (g_{xx} \delta \dot{x}) - \left(g_{xx} - \frac{d}{dt} g_{x\dot{x}} \right) \delta x \delta v \right] dt \\
&\quad \text{where } \left(g_{xx} - \frac{d}{dt} g_{x\dot{x}} \right) \delta x \delta v = \left[\left(g_{xx} - \frac{d}{dt} g_{x\dot{x}} \right) \delta v \right] \delta x = \left[\frac{d}{dt} (g_{xx} \delta \dot{v}) \right] \delta x \quad \text{from eq.(1)} \\
&= -\frac{1}{2} \int_{t_0}^{t_f} \frac{\delta x}{\delta v} \left[\delta v \frac{d}{dt} (g_{xx} \delta \dot{x}) - \frac{d}{dt} (g_{xx} \delta \dot{v}) \delta x \right] dt \\
&\quad \text{where } \frac{d}{dt} (g_{xx} \delta \dot{v} \delta x - g_{x\dot{x}} \delta \dot{x} \delta v) = \frac{d}{dt} (g_{xx} \delta \dot{v}) \delta x + (g_{xx} \delta \dot{v}) \frac{d}{dt} \delta x - \frac{d}{dt} (g_{xx} \delta \dot{x}) \delta v - (g_{xx} \delta \dot{x}) \frac{d}{dt} \delta v \\
&\quad \quad \quad = \frac{d}{dt} (g_{xx} \delta \dot{v}) \delta x - \frac{d}{dt} (g_{xx} \delta \dot{x}) \delta v \\
&= \frac{1}{2} \int_{t_0}^{t_f} \frac{\delta x}{\delta v} \frac{d}{dt} [g_{xx} (\delta \dot{v} \delta x - \delta \dot{x} \delta v)] dt \\
&= \frac{1}{2} \frac{\delta x}{\delta v} g_{xx} (\delta \dot{v} \delta x - \delta \dot{x} \delta v) \Big|_{t_0}^{t_f} - \frac{1}{2} \int_{t_0}^{t_f} g_{xx} (\delta \dot{v} \delta x - \delta \dot{x} \delta v) \frac{d}{dt} \left(\frac{\delta x}{\delta v} \right) dt \\
&\quad \text{where } (\delta \dot{v} \delta x - \delta \dot{x} \delta v) \frac{d}{dt} \left(\frac{\delta x}{\delta v} \right) = (\delta \dot{v} \delta x - \delta \dot{x} \delta v) \frac{\delta \dot{x} \delta v - \delta x \delta \dot{v}}{(\delta v)^2} = - \left(\frac{\delta x \delta \dot{v} - \delta \dot{x} \delta v}{\delta v} \right)^2 = - \left(\delta \dot{x} - \frac{\delta x \delta \dot{v}}{\delta v} \right)^2 \\
&= \frac{1}{2} \int_{t_0}^{t_f} g_{xx} \left(\delta \dot{x} - \frac{\delta x \delta \dot{v}}{\delta v} \right)^2 dt
\end{aligned}$$

For a necessary condition for a minimum (maximum), we need $\delta^2 J \geq 0$ ($\delta^2 J \leq 0$) for admissible $\delta x(t) \neq 0$ and $\delta x(t) \neq \delta v(t)$.

The condition for this is

$$\begin{array}{lll}
\therefore g_{xx} \geq 0 & \text{for minimum} & \\
g_{xx} \leq 0 & \text{for maximum} & ; \text{Legendre condition}
\end{array}$$

There might be finite number of times when $\left[\delta \dot{x} - \frac{\delta x \delta \dot{v}}{\delta v} \right]$ might not be finite.

Legendre condition (sufficient condition) is $g_{xx} \geq 0$ (or $g_{xx} \leq 0$) along every point on the reference curve (extremal).

Legendre derived it for the scalar case; it was later extended to the vector case by Clebsch and is now known as Legendre - Clebsch conditions;

Necessary conditions:

If the extremal $\vec{x}^*(t)$ satisfies the Euler - Lagrange equation and if $g(x, \dot{x}, t)$ is such that

$$g_{\ddot{x}\ddot{x}} \geq 0 \quad (g_{\ddot{x}\ddot{x}} \leq 0) \quad ; \text{ Legendre - Clebsch condition}$$

then it minimizes (maximizes) the performance index.

Chapter 5

The Variational Approach to Optimal Control Problem

Why Optimal Control?

("300 Years of Optimal Control: from the Brachistochrone to the Maximum Principle," IEEE Control Systems, June 1997, pp. 12-44)

The conventional wisdom holds that optimal control theory was born about 40 years ago in the former Soviet Union, with the work on the "Pontryagin maximum principle" by L. S. Pontryagin and his group. Some mathematicians believe that this new theory was no more than a minor addition to the classical calculus of variations, essentially involving the incorporation of inequality constraints. We believe that optimal control is significantly richer and broader than the calculus of variations, from which it differs in some fundamental ways.

The calculus of variations deals mainly with optimization problems of the following standard form

$$\begin{aligned} \text{Minimize } J &= \int_a^b L(q(t), \dot{q}(t), t) dt \\ \text{subject to } q(a) &= q_a \quad \text{and} \quad q(b) = q_b \end{aligned} \quad (1)$$

Or, equivalently, of the form

$$\begin{aligned} \text{Minimize } J &= \int_a^b L(q(t), u(t), t) dt \\ \text{subject to } q(a) &= q_a \quad \text{and} \quad q(b) = q_b \quad \text{and} \quad \dot{q}(t) = u(t) \\ \text{for } a &\leq t \leq b \end{aligned} \quad (2)$$

The distinctive feature of these problems is that the minimization of (1) takes place in the space of "all" curves, so nothing interesting happens on the level of the set of curves under consideration, and all the nontrivial features of the problem arise because of the Lagrangian L .

Optimal control problems, by contrast, involve a minimization over a set C of curves which is itself determined by some dynamical constraints. For example, C might be the set of all curves $t \rightarrow q(t)$ that satisfy a differential equation

$$\dot{q}(t) = f(q(t), u(t), t) \quad (3)$$

for some choice of the “control function” $t \rightarrow u(t)$. Even more precisely, since it may happen a member of C does not uniquely determine the control u that generates it, we should be talking about trajectory-control pairs $(q(\cdot), u(\cdot))$. So in optimal control problem there are at least two objects that give the situation interesting structure, namely, the dynamics f and the functional J to be minimized. In particular, optimal control theory contains, at the opposite extreme from the calculus of variations, problems where the Lagrangian L is 1, i.e. completely trivial, and therefore all the interesting action occurs because of the dynamics f . Such problems, in which it is desired to minimize time (i.e., the integral J of (2) with $L=1$) among all curves $t \rightarrow q(t)$ that satisfy endpoint constraints as in (2) and are solutions of (3) for some control $t \rightarrow u(t)$, are called *minimum time problems*. It is in these problems that the difference between optimal control and the calculus of variations is most clearly seen, and it is no accident that these were the problems that propelled the development of optimal control in early 1960s, and that time-optimal control is prominently represented in today’s research and in modern optimal control text books.

Bernoulli’s problem (the brachistochrone) is a true minimum time problem of the kind that is studied today in optimal control theory. Moreover, the brachistochrone problem is the first one ever to deal with a dynamical behavior and explicitly ask for the optimal selection of a path. A large part of the subsequent history of the calculus of variations can be best understood as the search for simplest and most general statement of the necessary conditions for optimality, and this statement is provided by *the maximum principle of optimal control theory*.

5.1 Necessary conditions for optimal control

To find an admissible control $\bar{u}^*(t)$ that causes the system

$$\dot{\bar{x}}(t) = \bar{f}(\bar{x}, \bar{u}, t)$$

to follow an admissible trajectory $\bar{x}^*(t)$ that extrimizes the performance measure

$$J(\bar{u}) = h(\bar{x}(t_f), t_f) + \int_{t_0}^{t_f} g(\bar{x}(t), \bar{u}(t), t) dt$$

$\bar{x}(t_0) = \bar{x}_0$ and $\bar{x}(t_f) = \bar{x}_f$ are given, t_f is fixed.

$$\bar{x} \in \mathbf{R}^n, \quad \bar{u} \in \mathbf{R}^m, \quad m \leq n$$

$\bar{u}$ is unconstrained

$\bar{x}$ is also unconstrained inside (t_0, t_f)

The augmented functional

$$J_a = h(\bar{x}(t_f), t_f) + \int_{t_0}^{t_f} \left[g + \bar{\Lambda}^T (\bar{f} - \dot{\bar{x}}) \right] dt \quad \bar{\Lambda}(t); \text{ Lagrange multipliers}$$

$$\bar{\Lambda} \in \mathbf{R}^n$$

$$\delta J_a = \frac{\partial h}{\partial \bar{x}(t_f)} \delta \bar{x}(t_f) + \int_{t_0}^{t_f} \left[\frac{\partial g}{\partial \bar{x}} \delta \bar{x} + \frac{\partial g}{\partial \bar{u}} \delta \bar{u} + \bar{\Lambda}^T \left\{ \frac{\partial \bar{f}}{\partial \bar{x}} \delta \bar{x} + \frac{\partial \bar{f}}{\partial \bar{u}} \delta \bar{u} - \delta \dot{\bar{x}} \right\} + \delta \bar{\Lambda}^T (\bar{f} - \dot{\bar{x}}) \right] dt$$

$$= -\bar{\Lambda}^T \delta \bar{x} \Big|_{t_0}^{t_f} + \int_{t_0}^{t_f} \left[\left(\frac{\partial g}{\partial \bar{x}} + \bar{\Lambda}^T \frac{\partial \bar{f}}{\partial \bar{x}} + \dot{\bar{\Lambda}}^T \right) \delta \bar{x} + \left(\frac{\partial g}{\partial \bar{u}} + \bar{\Lambda}^T \frac{\partial \bar{f}}{\partial \bar{u}} \right) \delta \bar{u} + (\bar{f} - \dot{\bar{x}})^T \delta \bar{\Lambda} \right] dt$$

$$= 0$$

Apply the boundary conditions

The differential equation is also satisfied at all times. $\rightarrow \dot{\bar{x}} = \bar{f}$

For all admissible variations $\delta \bar{x}$ and $\delta \bar{u}$, the necessary conditions are

$$\dot{\vec{\Lambda}}^T = -\frac{\partial g}{\partial \vec{x}} - \vec{\Lambda}^T \frac{\partial \vec{f}}{\partial \vec{x}} \quad ; \text{costate equation}$$

$$\frac{\partial g}{\partial \vec{u}} + \vec{\Lambda}^T \frac{\partial \vec{f}}{\partial \vec{u}} = 0$$

$2n$ differential equations ($\dot{\vec{x}}, \dot{\vec{\Lambda}}$) we have $2n$ boundary condition ($\vec{x}(t_0), \vec{x}(t_f)$)

→ Two Point Boundary Value Problem

Define Hamitonian

$$H \equiv g + \vec{\Lambda}^T \vec{f}, \quad H = H(\vec{x}(t), \vec{u}(t), \vec{\Lambda}(t), t)$$

$$J_a = h(\vec{x}(t_f), t_f) + \int_{t_0}^{t_f} [H - \vec{\Lambda}^T \dot{\vec{x}}] dt$$

$$\begin{aligned} \delta J_a &= \frac{\partial h}{\partial \vec{x}(t_f)} \delta \vec{x}(t_f) + \int_{t_0}^{t_f} \left[\frac{\partial H}{\partial \vec{u}} \delta \vec{u} + \frac{\partial H}{\partial \vec{x}} \delta \vec{x} + \frac{\partial H}{\partial \vec{\Lambda}} \delta \vec{\Lambda} - \delta \vec{\Lambda}^T \dot{\vec{x}} - \vec{\Lambda}^T \delta \dot{\vec{x}} \right] dt \\ &= -\vec{\Lambda}^T \delta \vec{x} \Big|_{t_0}^{t_f} + \int_{t_0}^{t_f} \left[\frac{\partial H}{\partial \vec{u}} \delta \vec{u} + \left(\frac{\partial H}{\partial \vec{x}} + \dot{\vec{\Lambda}}^T \right) \delta \vec{x} + \delta \vec{\Lambda}^T \left(\frac{\partial H}{\partial \vec{\Lambda}} - \dot{\vec{x}} \right) \right] dt \\ &= 0 \end{aligned}$$

Apply the boundary conditions

The differential equation is also satisfied

The condition for $J_a = 0$ for all admissible $\delta \vec{x}, \delta \vec{u}$ are

$$\frac{\partial H}{\partial \vec{u}} = 0 \quad ; \text{optimality conditions}$$

$$\dot{\vec{\Lambda}} = -\frac{\partial H^T}{\partial \vec{x}} \quad ; \text{costate equation(or adjoint equations)}$$

$$\dot{\vec{x}} = \frac{\partial H^T}{\partial \vec{\Lambda}} = \vec{f} \quad ; \text{state equations}$$

Example

$$J = \frac{1}{2} \int_0^1 \left(x(t)^2 + u(t)^2 \right) dt$$

$$\dot{x}(t) = u(t)$$

$$x(0) = 1, \quad x(1) = 0$$

$$H = g + \bar{\Lambda}^T \bar{f} = \frac{1}{2} (x^2 + u^2) + \Lambda u$$

$$\frac{\partial H}{\partial \bar{u}} = 0 \quad \rightarrow u + \Lambda = 0$$

$$\therefore u(t) = -\Lambda(t)$$

$$\dot{\Lambda} = -\frac{\partial H}{\partial \bar{x}} = -x$$

$$\ddot{x} = \dot{u} = -\dot{\Lambda} = x$$

$$x = c_1 e^t + c_2 e^{-t}$$

$$x(0) = 1 \quad \rightarrow c_1 + c_2 = 1$$

$$x(1) = 0 \quad \rightarrow c_1 e + \frac{c_2}{e} = 0$$

$$\Rightarrow c_1 = \frac{1}{1-e^2}$$

$$c_2 = \frac{-e^2}{1-e^2}$$

$$\therefore x(t)^* = \frac{1}{1-e^2} e^t - \frac{e^2}{1-e^2} e^{-t}$$

$$\dot{\Lambda}^* = -x^* = -\frac{1}{1-e^2} e^t + \frac{e^2}{1-e^2} e^{-t}$$

$$u = -\Lambda, \quad \dot{x} = u \quad \Rightarrow \quad \Lambda = -\dot{x},$$

$$u(t) = \dot{x}(t) = \frac{1}{1-e^2} e^t + \frac{e^2}{1-e^2} e^{-t}$$

$$\dot{x} = -\Lambda \quad \text{and} \quad \dot{\Lambda} = -x$$

Note: $\rightarrow \begin{Bmatrix} \dot{x}(t) \\ \dot{\Lambda}(t) \end{Bmatrix} = \begin{bmatrix} 0 & -1 \\ -1 & 0 \end{bmatrix} \begin{Bmatrix} x(t) \\ \Lambda(t) \end{Bmatrix} = A \begin{Bmatrix} x(t) \\ \Lambda(t) \end{Bmatrix}$

Solution is $\begin{Bmatrix} x(t) \\ \Lambda(t) \end{Bmatrix} = e^{At} \begin{Bmatrix} x(0) \\ \Lambda(0) \end{Bmatrix}$

Linear Two Point Boundary Value Problem

$$\begin{bmatrix} \dot{\vec{x}} \\ \dot{\vec{\Lambda}} \end{bmatrix} = A \begin{bmatrix} \vec{x} \\ \vec{\Lambda} \end{bmatrix}$$

A is $2n \times 2n$ constant matrix

$$\vec{x}(t_0) = \vec{x}_0, \quad \vec{x}(t_f) = \vec{x}_f \quad ; 2n \text{ boundary condition given}$$

$$\begin{bmatrix} \vec{x}(t) \\ \vec{\Lambda}(t) \end{bmatrix} = e^{At} \begin{bmatrix} \vec{x}(0) \\ \vec{\Lambda}(0) \end{bmatrix}$$

$$= \begin{bmatrix} \phi_{11}(t) & \phi_{12}(t) \\ \phi_{21}(t) & \phi_{22}(t) \end{bmatrix} \begin{bmatrix} \vec{x}_0 \\ \vec{\Lambda}(0) \end{bmatrix}$$

$$\vec{x}(t) = \phi_{11}(t) \vec{x}_0 + \phi_{12}(t) \vec{\Lambda}(0)$$

$$\vec{x}(t_f) = \vec{x}_f = \phi_{11}(t_f) \vec{x}_0 + \phi_{12}(t_f) \vec{\Lambda}(0)$$

$$\rightarrow \vec{\Lambda}(0) = \phi_{12}^{-1}(t_f) [\vec{x}_f - \phi_{11}(t_f) \vec{x}_0]$$

$$\Rightarrow \vec{\Lambda}(t) = e^{At} \vec{\Lambda}(0)$$

$$\vec{x}(t) = e^{At} \vec{x}_0$$

$\Rightarrow$ Control $u(t)$ can be calculated using $\vec{\Lambda}(t)$ and $\vec{x}(t)$

because equation $\frac{\partial H}{\partial \vec{u}} = 0$ contain $\vec{\Lambda}(t)$ and $\vec{x}(t)$.

< Final time specified, $\vec{x}(t_f)$ free >

$$J(\vec{u}) = h(\vec{x}(t_f), t_f) + \int_{t_0}^{t_f} g(\vec{x}(t), \vec{u}(t), t) dt$$

$$\text{subject to } \dot{\vec{x}}(t) = \vec{f}(\vec{x}(t), \vec{u}(t), t)$$

$$\vec{x}(t_0) = \vec{x}_0, \vec{x}(t_f) = \text{free, and } t_f \text{ is specified}$$

$$H \equiv g + \vec{\Lambda}^T \vec{f} = g + \vec{\Lambda}^T \dot{\vec{x}} = H(\vec{x}(t), \vec{u}(t), \vec{\Lambda}(t), t)$$

$$J_a = h(\vec{x}(t_f), t_f) + \int_{t_0}^{t_f} [H - \vec{\Lambda}^T \dot{\vec{x}}] dt$$

$$\begin{aligned} \delta J_a &= \frac{\partial h}{\partial \vec{x}(t_f)} \delta \vec{x}(t_f) + \int_{t_0}^{t_f} \left[\frac{\partial H}{\partial \vec{u}} \delta \vec{u} + \frac{\partial H}{\partial \vec{x}} \delta \vec{x} + \frac{\partial H}{\partial \vec{\Lambda}} \delta \vec{\Lambda} - \delta \vec{\Lambda}^T \dot{\vec{x}} - \vec{\Lambda}^T \delta \dot{\vec{x}} \right] dt \\ &= \left[\frac{\partial h}{\partial \vec{x}(t_f)} - \vec{\Lambda}^T(t_f) \right] \delta \vec{x}(t_f) + \int_{t_0}^{t_f} \left[\frac{\partial H}{\partial \vec{u}} \delta \vec{u} + \left(\frac{\partial H}{\partial \vec{x}} + \dot{\vec{\Lambda}}^T \right) \delta \vec{x} + \delta \vec{\Lambda}^T \left(\frac{\partial H}{\partial \vec{\Lambda}} - \dot{\vec{x}} \right) \right] dt \\ &= 0 \end{aligned}$$

$$\Rightarrow \frac{\partial H}{\partial \vec{u}} = 0, \quad \dot{\vec{\Lambda}}^T = -\frac{\partial H}{\partial \vec{x}}, \quad \dot{\vec{x}} = \frac{\partial H}{\partial \vec{\Lambda}} = \vec{f}$$

Boundary conditions

$$\begin{aligned} \vec{\Lambda}^T(t_f) &= \frac{\partial h}{\partial \vec{x}} \Big|_{t=t_f} & \text{at } t = t_f \\ \vec{x}(t_0) &= \vec{x}_0 \end{aligned}$$

Linear Two Point Boundary Value Problem

$$\begin{bmatrix} \dot{\vec{x}}(t) \\ \dot{\vec{\Lambda}}(t) \end{bmatrix} = A \begin{bmatrix} \vec{x}(t) \\ \vec{\Lambda}(t) \end{bmatrix}$$

$$\vec{x}(t_0) = \vec{x}_0, \quad \vec{x}(t_f) = \text{free}$$

$$\Rightarrow \vec{\Lambda}(t_f) = \frac{\partial h^T}{\partial \vec{x}} \Big|_{t=t_f} = \vec{\Lambda}_f$$

$$\begin{aligned} \begin{bmatrix} \vec{x}(t) \\ \vec{\Lambda}(t) \end{bmatrix} &= e^{At} \begin{bmatrix} \vec{x}(0) \\ \vec{\Lambda}(0) \end{bmatrix} \\ &= \begin{bmatrix} \phi_{11}(t) & \phi_{12}(t) \\ \phi_{21}(t) & \phi_{22}(t) \end{bmatrix} \begin{bmatrix} \vec{x}_0 \\ \vec{\Lambda}(0) \end{bmatrix} \end{aligned}$$

$$\vec{\Lambda}(t_f) = \vec{\Lambda}_f = \phi_{21}(t_f) \vec{x}_0 + \phi_{22}(t_f) \vec{\Lambda}(0)$$

$$\rightarrow \vec{\Lambda}(0) = \phi_{22}^{-1}(t_f) [\vec{\Lambda}_f - \phi_{21}(t_f) \vec{x}_0]$$

$$\therefore \vec{\Lambda}(t) = e^{At} \vec{\Lambda}(0)$$

Generally,

$$\dot{\vec{x}}(t) = \vec{f}(\vec{x}(t), \vec{u}(t), t), \quad \vec{x} \in \mathbb{R}^n$$

$$J(x) = h(\vec{x}(t_f), t_f) + \int_{t_0}^{t_f} g(\vec{x}(t), \vec{u}(t), t) dt$$

$$\vec{x}(0) = \vec{x}_0 \text{ (given)}$$

$$\vec{x}_i(t_f) = \vec{x}_f \text{ (given)}, \quad i = 1, 2, \dots, k$$

$$\vec{x}_j(t_f) = \text{free}, \quad j = k+1, k+2, \dots, n$$

If $x_j(t_f)$ = free, then

$$\Rightarrow \vec{\Lambda}_j(t_f) = \frac{\partial h^T}{\partial \vec{x}_j} \Big|_{t=t_f}$$

If all other final states are fixed,

$$x_i(t_f) = x_{if}, \quad i \neq j$$

< General Optimal Control Problem >

No inequality constraints on the states and controls

$$J(x) = h(\vec{x}(t_f), t_f) + \int_{t_0}^{t_f} g(\vec{x}, \vec{u}, t) dt$$

$$\text{subject to } \dot{\vec{x}}(t) = \vec{f}(\vec{x}, \vec{u}, t)$$

$$\vec{x}(t_0) = \text{given},$$

$$t_f \text{ and } \vec{x}(t_f) \text{ are free}$$

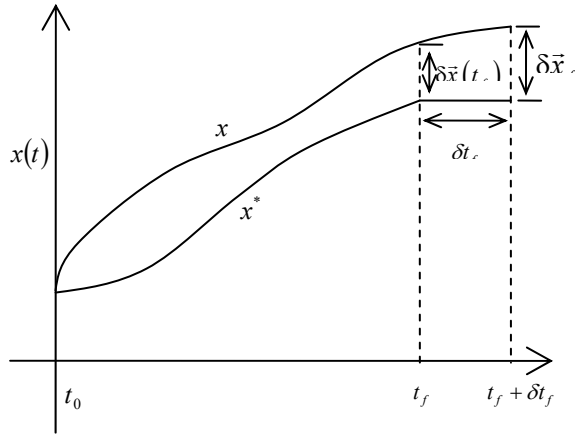
$$\vec{m}(\vec{x}(t_f), t_f) = 0 \quad m \in \mathbf{R}^k; \quad \text{terminal conditions}$$

$$J_a = h(\vec{x}(t_f), t_f) + \vec{v}(t)^T \vec{m}(\vec{x}(t_f), t_f) + \int_{t_0}^{t_f} (H(\vec{x}(t), \vec{u}(t), \vec{\lambda}(t), t) - \vec{\lambda}(t)^T \dot{\vec{x}}(t)) dt$$

$$\text{where } H \equiv g + \vec{\lambda}^T \vec{f}$$

$$\begin{aligned} \delta J_a = & \frac{\partial h}{\partial \vec{x}(t_f)} \delta \vec{x}_f + \frac{\partial h}{\partial t_f} \delta t_f + \vec{v}^T \frac{\partial \vec{m}}{\partial \vec{x}(t_f)} \delta \vec{x}_f + \vec{v}^T \frac{\partial \vec{m}}{\partial t_f} \delta t_f + \delta \vec{v}^T \vec{m}(\vec{x}(t_f), t_f) \\ & + \int_{t_0}^{t_f} \left[\frac{\partial H}{\partial \vec{u}} \delta \vec{u} + \frac{\partial H}{\partial \vec{x}} \delta \vec{x} + \frac{\partial H}{\partial \vec{\lambda}} \delta \vec{\lambda} - \vec{\lambda}^T \delta \dot{\vec{x}} - \delta \vec{\lambda}^T \dot{\vec{x}} \right] dt \\ & + \int_{t_f}^{t_f + \delta t_f} (H - \vec{\lambda}^T \dot{\vec{x}}) dt \end{aligned}$$

$$\int_{t_f}^{t_f + \delta t_f} (H - \vec{\lambda}^T \dot{\vec{x}}) dt \approx (H - \vec{\lambda}^T \dot{\vec{x}}) \Big|_{t_f^*} \delta t_f$$



$$\delta \vec{x}(t_f) = \delta \vec{x}_f - \dot{\vec{x}}^*(t_f^*) \delta t_f$$

$$\begin{aligned}
\delta J_a &= \frac{\partial h}{\partial \vec{x}(t_f)} \delta \vec{x}_f + \frac{\partial h}{\partial t_f} \delta t_f + \vec{v}^T \frac{\partial \vec{m}}{\partial \vec{x}(t_f)} \delta \vec{x}_f + \vec{v}^T \frac{\partial \vec{m}}{\partial t_f} \delta t_f - \vec{\lambda}^T \delta \vec{x}(t_f) \\
&\quad + \int_{t_0}^{t_f} \left[\frac{\partial H}{\partial \vec{u}} \delta \vec{u} + \frac{\partial H}{\partial \vec{x}} \delta \vec{x} + \dot{\vec{\lambda}}^T \delta \vec{x} + \delta \vec{\lambda}^T \left(\frac{\partial H^T}{\partial \vec{\lambda}} - \dot{\vec{x}} \right) \right] dt \\
&\quad + \left[H(t_f^*) - \vec{\lambda}^T(t_f^*) \dot{\vec{x}}(t_f^*) \right] \delta t_f \\
&= \left[\frac{\partial h}{\partial \vec{x}(t_f)} + \vec{v}^T \frac{\partial \vec{m}}{\partial \vec{x}(t_f)} - \vec{\lambda}^T(t_f) \right] \delta \vec{x}_f \\
&\quad + \left[\frac{\partial h}{\partial t_f} + \vec{v}^T \frac{\partial \vec{m}}{\partial t_f} + \vec{\lambda}^T(t_f) \dot{\vec{x}}(t_f) + H(t_f) - \vec{\lambda}^T(t_f) \dot{\vec{x}}(t_f) \right] \delta t_f \\
&\quad + \int_{t_0}^{t_f} \left[\frac{\partial H}{\partial \vec{u}} \delta \vec{u} + \left(\frac{\partial H}{\partial \vec{x}} + \dot{\vec{\lambda}}^T \right) \delta \vec{x} + \delta \vec{\lambda}^T \left(\frac{\partial H^T}{\partial \vec{\lambda}} - \dot{\vec{x}} \right) \right] dt \\
&= 0
\end{aligned}$$

Necessary conditions

$$\begin{aligned}
\frac{\partial H}{\partial \vec{u}} &= \vec{0} && \text{optimality conditions} \\
\dot{\vec{\lambda}} &= - \frac{\partial H^T}{\partial \vec{x}} && \text{costate (adjoint) equations} \\
\dot{\vec{x}} &= \frac{\partial H^T}{\partial \vec{\lambda}} = \vec{f} && \text{state equations} \\
\vec{\lambda}^T(t_f) &= \left[\frac{\partial h}{\partial \vec{x}(t_f)} + \vec{v}^T \frac{\partial \vec{m}}{\partial \vec{x}(t_f)} \right]_{t=t_f} && \text{boundary conditions} \\
H(t_f) &= - \left[\frac{\partial h}{\partial t_f} + \vec{v}^T \frac{\partial \vec{m}}{\partial t_f} \right]_{t=t_f} && \text{boundary conditions} \\
\vec{x}(t_0) &= \vec{x}_0 \quad \text{given} && \text{boundary conditions}
\end{aligned}$$

$\vec{x}(t_0)$ are given, $\vec{\lambda}(t_f)$ is determined by boundary conditions. $\vec{\lambda}(t)$ must evidently be determined as an intermediate step in finding the optimal control $\vec{u}^*(t)$ which depend on $\vec{\lambda}(t)$ through the optimality conditions $(\frac{\partial H}{\partial \vec{u}} = \vec{0})$.

< Hamiltonian >

The time derivative of the Hamiltonian;

$$\begin{aligned}\frac{dH}{dt} &= \frac{\partial H}{\partial t} + \frac{\partial H}{\partial \bar{x}} \frac{d\bar{x}}{dt} + \frac{\partial H}{\partial \bar{u}} \frac{d\bar{u}}{dt} + \frac{\partial H}{\partial \bar{\Lambda}} \frac{d\bar{\Lambda}}{dt} \\ &= H_t + H_{\bar{u}} \dot{\bar{u}} + \left(H_{\bar{x}} + \dot{\bar{\Lambda}}^T \right) \bar{f}\end{aligned}$$

$$H(\bar{x}, \bar{u}, \bar{\Lambda}, t)$$

$$H = g + \bar{\Lambda}^T \cdot \bar{f}$$

$$\bar{f} = \dot{\bar{x}}$$

If $\bar{u}(t)$ is an optimal control $H_{\bar{u}}=0$, $\dot{\bar{\Lambda}}^T = -H_{\bar{x}}$

$$\dot{H} = H_t$$

Now, in the time-invariant case, f and g are not explicit functions of t , and so neither is H .

In this situation,

$$\dot{H} = 0$$

Hence, for time-invariant systems and cost functions, the Hamiltonian is a constant on the optimal solution.

If t_f is free (or minimum time problem) and $h=0$, $\mathbf{m}=0$

$$H(t_f) = - \left[\frac{\partial h}{\partial t_f} + \bar{\mathbf{v}}^T \frac{\partial \bar{\mathbf{m}}}{\partial t_f} \right]_{t=t_f} = 0$$

$$\dot{H} = 0 \Rightarrow H(t) = c \quad \text{and} \quad H(t_f) = 0 \Rightarrow H(t) = 0$$

SUMMARY OF BOUNDARY CONDITIONS IN OPTIMAL CONTROL
(Table 5-1)

Problem	Description	Substitution	Boundary -condition equations	Remark
t_f fixed	1. $\mathbf{x}(t_f) = \mathbf{x}_f$ specified final state	$\delta \mathbf{x}_f = \delta \mathbf{x}(t_f) = \mathbf{0}$ $\delta t_f = 0$	$\mathbf{x}^*(t_0) = \mathbf{x}_0$ $\mathbf{x}^*(t_f) = \mathbf{x}_f$	$2n$ equations to determine $2n$ constraints of integration
	2. $\mathbf{x}(t_f)$ free	$\delta \mathbf{x}_f = \delta \mathbf{x}(t_f)$ $\delta t_f = 0$	$\mathbf{x}^*(t_0) = \mathbf{x}_0$ $\frac{\partial h}{\partial \mathbf{x}}(\mathbf{x}^*(t_f)) - \boldsymbol{\Lambda}^*(t_f) = \mathbf{0}$	$2n$ equations to determine $2n$ constraints of integration
	3. $\mathbf{x}(t_f)$ on the surface $\mathbf{m}(\mathbf{x}(t)) = 0$	$\delta \mathbf{x}_f = \delta \mathbf{x}(t_f)$ $\delta t_f = 0$	$\mathbf{x}^*(t_0) = \mathbf{x}_0$ $\frac{\partial h}{\partial \mathbf{x}}(\mathbf{x}^*(t_f)) - \boldsymbol{\Lambda}^*(t_f) = \sum_{i=1}^k \nu_i \left[\frac{\partial m_i}{\partial \mathbf{x}}(\mathbf{x}^*(t_f)) \right]$ $\mathbf{m}(\mathbf{x}^*(t_f)) = \mathbf{0}$	$(2n+k)$ equations to determine $2n$ constraints of integration and the variables $\nu_1, \dots, \nu_k$
t_f free	4. $\mathbf{x}(t_f) = \mathbf{x}_f$ specified final state	$\delta \mathbf{x}_f = \mathbf{0}$	$\mathbf{x}^*(t_0) = \mathbf{x}_0$ $\mathbf{x}^*(t_f) = \mathbf{x}_f$ $H(\mathbf{x}^*(t_f), \mathbf{u}^*(t_f), \boldsymbol{\Lambda}^*(t_f), t_f) + \frac{\partial h}{\partial t}(\mathbf{x}^*(t_f), t_f) = 0$	$(2n+1)$ equations to determine $2n$ constraints of integration t_f
	5. $\mathbf{x}(t_f)$ free		$\mathbf{x}^*(t_0) = \mathbf{x}_0$ $\frac{\partial h}{\partial \mathbf{x}}(\mathbf{x}^*(t_f)) - \boldsymbol{\Lambda}^*(t_f) = \mathbf{0}$ $H(\mathbf{x}^*(t_f), \mathbf{u}^*(t_f), \boldsymbol{\Lambda}^*(t_f), t_f) + \frac{\partial h}{\partial t}(\mathbf{x}^*(t_f), t_f) = 0$	$(2n+1)$ equations to determine $2n$ constraints of integration t_f
	6. $\mathbf{x}(t_f)$ on the moving $\boldsymbol{\theta}(t)$	$\delta \mathbf{x}_f = \frac{d\boldsymbol{\theta}}{dt}(t_f) \delta t_f$	$\mathbf{x}^*(t_0) = \mathbf{x}_0$ $\mathbf{x}^*(t_f) = \boldsymbol{\theta}(t_f)$ $H(\mathbf{x}^*(t_f), \mathbf{u}^*(t_f), \boldsymbol{\Lambda}^*(t_f), t_f) + \frac{\partial h}{\partial t}(\mathbf{x}^*(t_f), t_f) + \left[\frac{\partial h}{\partial \mathbf{x}}(\mathbf{x}^*(t_f), t_f) - \boldsymbol{\Lambda}^*(t_f) \right]^T \frac{d\boldsymbol{\theta}}{dt}(t_f) = 0$	$(2n+1)$ equations to determine $2n$ constraints of integration t_f
	7. $\mathbf{x}(t_f)$ on the surface $\mathbf{m}(\mathbf{x}(t)) = \mathbf{0}$		$\mathbf{x}^*(t_0) = \mathbf{x}_0$ $\frac{\partial h}{\partial \mathbf{x}}(\mathbf{x}^*(t_f)) - \boldsymbol{\Lambda}^*(t_f) = \sum_{i=1}^k \nu_i \left[\frac{\partial m_i}{\partial \mathbf{x}}(\mathbf{x}^*(t_f)) \right]$ $\mathbf{m}(\mathbf{x}^*(t_f)) = \mathbf{0}$ $H(\mathbf{x}^*(t_f), \mathbf{u}^*(t_f), \boldsymbol{\Lambda}^*(t_f), t_f) + \frac{\partial h}{\partial t}(\mathbf{x}^*(t_f), t_f) = 0$	$(2n+k+1)$ equations to determine $2n$ constraints of integration, the variables $\nu_1, \dots, \nu_k$, and t_f

	8. $\mathbf{x}(t_f)$ on the surface $\mathbf{m}(\mathbf{x}(t), t) = \mathbf{0}$		$\mathbf{x}^*(t_0) = \mathbf{x}_0$ $\frac{\partial h}{\partial \mathbf{x}}(\mathbf{x}^*(t_f)) - \boldsymbol{\Lambda}^*(t_f) = \sum_{i=1}^k \nu_i \left[\frac{\partial m_i}{\partial \mathbf{x}}(\mathbf{x}^*(t_f), t_f) \right]$ $\mathbf{m}(\mathbf{x}(t_f), t_f) = \mathbf{0}$ $H(\mathbf{x}^*(t_f), \mathbf{u}^*(t_f), \mathbf{A}^*(t_f), t_f) + \frac{\partial h}{\partial t}(\mathbf{x}^*(t_f), t_f)$ $= \sum_{i=1}^k \nu_i \left[\frac{\partial m_i}{\partial t}(\mathbf{x}^*(t_f), t_f) \right]$	$(2n+k+1)$ equations to determine $2n$ constraints of integration, the variables $\nu_1, \dots, \nu_k$, and t_f

Significance of the initial costate values

$\vec{\Lambda}(0)=?$ What do these numbers tell us?

$$J(x) = \int_{t_0}^{t_f} g(\vec{x}, \vec{u}, t) dt$$

$$\dot{\vec{x}}(t) = \vec{f}(\vec{x}, \vec{u}, t)$$

$\vec{x}(t_0)$; unspecified, t_0 is fixed

$\vec{x}(t_f)$; specified, t_f is fixed

$$H \equiv g + \vec{\Lambda}^T \vec{f}$$

$$J_a = \int_{t_0}^{t_f} (H - \vec{\Lambda}^T \dot{\vec{x}}) dt$$

$$\delta J_a = \int_{t_0}^{t_f} \left[\frac{\partial H}{\partial \vec{u}} \delta \vec{u} + \frac{\partial H}{\partial \vec{x}} \delta \vec{x} + \frac{\partial H}{\partial \vec{\Lambda}} \delta \vec{\Lambda} - \dot{\vec{\Lambda}}^T \delta \vec{x} - \delta \vec{\Lambda}^T \dot{\vec{x}} \right] dt$$

$$= -\vec{\Lambda}^T \delta \vec{x} \Big|_{t_0}^{t_f} + \int_{t_0}^{t_f} \left[\frac{\partial H}{\partial \vec{u}} \delta \vec{u} + \left(\frac{\partial H}{\partial \vec{x}} + \dot{\vec{\Lambda}}^T \right) \delta \vec{x} + \delta \vec{\Lambda}^T \left(\frac{\partial H}{\partial \vec{\Lambda}} - \dot{\vec{x}} \right) \right] dt$$

$$= \vec{\Lambda}(t_0)^T \delta \vec{x}(t_0) + \int_{t_0}^{t_f} \left[\frac{\partial H}{\partial \vec{u}} \delta \vec{u} + \left(\frac{\partial H}{\partial \vec{x}} + \dot{\vec{\Lambda}}^T \right) \delta \vec{x} + \delta \vec{\Lambda}^T \left(\frac{\partial H}{\partial \vec{\Lambda}} - \dot{\vec{x}} \right) \right] dt$$

$$= 0$$

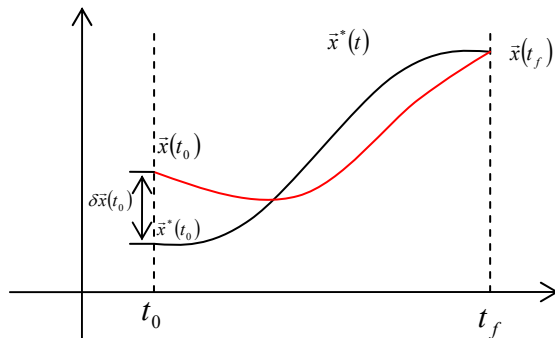
$$\Rightarrow H_{\vec{u}} = 0$$

$$\dot{\vec{\Lambda}} = -H_{\vec{x}}$$

$$\dot{\vec{x}} = H_{\vec{\Lambda}}$$

$$\delta J_a = \vec{\Lambda}(t_0)^T \delta \vec{x}(t_0)$$

transversality condition



$$\vec{x}^*(t_0) \rightarrow J^*$$

$$\vec{x}^*(t_0) + \delta \vec{x}(t_0) \rightarrow J^* + \delta J$$

$$\delta J = \delta J_a \quad \text{since all equations are satisfied.}$$

$$\delta J_a = \vec{\Lambda}(t_0)^T \delta \vec{x}(t_0) \rightarrow \vec{\Lambda}(0) = \frac{\partial J}{\partial \vec{x}(0)}$$

$\vec{\Lambda}(0)$ tells us that the sensitivity of the cost with respect to initial conditions.

5.2 Linear Quadratic Problem (LQR)

Many dynamic systems of interest are well approximated as linear. If quadratic performance indices and linear state equations are used to pose optimal problems for linear system, the optimal controls are linear feedback on the state variables.

$$\text{Minimize } J = \frac{1}{2} \vec{x}^T(t_f) L \vec{x}(t_f) + \frac{1}{2} \int_{t_0}^{t_f} [\vec{x}^T(t) Q(t) \vec{x}(t) + \vec{u}^T(t) R(t) \vec{u}(t)] dt$$

$$\text{subject to } \dot{\vec{x}}(t) = A(t) \vec{x}(t) + B(t) \vec{u}(t)$$

$$\vec{x} \in \mathfrak{R}^n, \quad \vec{u} \in \mathfrak{R}^m \quad A: (n \times n), B: (n \times m)$$

$$\vec{x}(t_0) = \vec{x}_0, \quad \vec{x}(t_f) = \text{free}, \quad t_f \text{ is fixed.}$$

Q is symmetric positive semi-definite⁴ matrix (n×n)

R is symmetric positive definite matrix (m×m)

L is symmetric positive semi-definite matrix (n×n)

The control and states are not bounded.

It is desired to maintain the state vector close to the origin without an excessive expenditure of control effort.

The Hamiltonian is

$$H = \frac{1}{2} \vec{x}^T Q \vec{x} + \frac{1}{2} \vec{u}^T R \vec{u} + \vec{\Lambda}^T (A \vec{x} + B \vec{u})$$

$$\dot{\vec{\Lambda}} = - \left(\frac{\partial H}{\partial \vec{x}} \right)^T = -A^T \vec{\Lambda} - Q \vec{x}$$

$$\frac{\partial H}{\partial \vec{u}} = 0 \Rightarrow R \vec{u} + B^T \vec{\Lambda} = 0$$

$$\Rightarrow \vec{u} = -R^{-1} B^T \vec{\Lambda},$$

The existence of R^{-1} is assured because
R is a positive definite matrix

⁴ Symmetric A matrix such that $A \in \mathbf{R}^{n \times n}$ is positive definite if $\vec{x}^T A \vec{x} > 0$ for all non-zero $x \in \mathbf{R}^n$, and symmetric A matrix such that $A \in \mathbf{R}^{n \times n}$ is positive semi-definite if $\vec{x}^T A \vec{x} \geq 0$ for all non-zero $x \in \mathbf{R}^n$.

$$\dot{\vec{x}} = A\vec{x} + B\vec{u} = A\vec{x} - BR^{-1}B^T\vec{\Lambda}, \quad \vec{x}(t_0) = \vec{x}_0, \quad \vec{\Lambda}(t_f) = \frac{\partial h}{\partial \vec{x}} \Big|_{t=t_f} = L\vec{x}(t_f)$$

Kalman define $\vec{\Lambda}(t) = [K(t)]\vec{x}(t)$, $\vec{\Lambda}(t)$ is linear function of the states.

$$\begin{aligned} \dot{\vec{\Lambda}} &= \dot{K}\vec{x} + K\dot{\vec{x}} & \dot{\vec{\Lambda}} &= -A^T\vec{\Lambda} - Q\vec{x} = -A^TK\vec{x} - Q\vec{x} \\ &= \dot{K}\vec{x} + K(A\vec{x} - BR^{-1}B^T\vec{\Lambda}) \end{aligned}$$

$$\begin{aligned} &\rightarrow \dot{K}\vec{x} + K(A\vec{x} - BR^{-1}B^TK\vec{x}) = -A^TK\vec{x} - Q\vec{x} \\ &\rightarrow [\dot{K} + KA - KBR^{-1}B^TK + A^TK + Q]\vec{x} = 0 \end{aligned}$$

This equation must be satisfied at all times.

$$\begin{aligned} \dot{K}(t) &= -K(t)A(t) - A^T(t)K(t) - Q(t) + K(t)B(t)R^{-1}(t)B^T(t)K(t) \\ &\quad ; \text{Riccati differential equation} \end{aligned}$$

Boundary conditions

$$\begin{aligned} \vec{\Lambda}(t_f) &= L\vec{x}(t_f) & \rightarrow & K(t_f)\vec{x}(t_f) = L\vec{x}(t_f) \\ &\rightarrow K(t_f) = L & ; \text{requires backward integration} \end{aligned}$$

K is n x n matrix $\rightarrow$ the system is n^2 first-order differential equations.

K is symmetric $\rightarrow$ not n^2 equation, but $\frac{n(n+1)}{2}$ first-order differential equations.

These equation can be integrated numerically.

The integration is started at $t=t_f$ and process backward in time to at $t=t_0$.

$K(t)$ is stored and $\vec{u}(t)$ is calculated as

$$\begin{aligned} \vec{u}(t) &= -R^{-1}B^T\vec{\Lambda}(t) \\ &= -R^{-1}B^TK(t)\vec{x}(t), & \leftarrow \vec{x}(t) \text{ measured or computed} \end{aligned}$$

Note $\dot{\vec{x}} = A\vec{x} + B\vec{u} = A\vec{x}(t) - BR^{-1}B^TK(t)\vec{x}(t)$, $\vec{x}(t_0) = \vec{x}_0$ (given)

integration forward $t=t_0$ to $t=t_f$. $\in \mathfrak{R}^n \Rightarrow \vec{x}(t)$

This indicated that the optimal control law is a linear, time-varying, combination of the system states.

< Special case of LQR >

If Q, R, A, B are constant(time-invariant) and L=0, t_f is free, $\vec{x}(t_f)$ is free,

Transversality conditions

$$\left(\frac{\partial h}{\partial \vec{x}} + \mathbf{v}^T \frac{\partial \vec{m}}{\partial \vec{x}} - \vec{\Lambda} \right)_{t=t_f} \delta \vec{x}_f + \left(\frac{\partial h}{\partial t} + \mathbf{v}^T \frac{\partial \vec{m}}{\partial t} + H \right)_{t=t_f} \delta t_f = 0$$

$$\vec{\Lambda}(t_f) = 0, \quad H(t_f) = 0$$

What can we say if the Hamiltonian does not explicitly depend on time.

$$H(t) = \text{constant}, \quad \text{and} \quad H(t_f) = 0$$

$$\Rightarrow H(t) = 0 = \frac{1}{2} \vec{x}^T Q \vec{x} + \frac{1}{2} \vec{u}^T R \vec{u} + \vec{\Lambda}^T (A \vec{x} + B \vec{u})$$

$$\vec{u} = -R^{-1} B^T \vec{\Lambda} = -R^{-1} B^T K \vec{x}$$

$$\vec{\Lambda} = K \vec{x} \rightarrow \vec{\Lambda}^T = \vec{x}^T K^T = \vec{x}^T K$$

K and R are symmetric matrix

$$= \frac{1}{2} \vec{x}^T \left[Q + K^T B R^{-1} R R^{-1} B^T K + 2KA - 2KBR^{-1} B^T K \right] \vec{x}$$

$$\therefore Q + 2KA - KBR^{-1} B^T K = 0$$

algebraic Riccati equation

$$\Rightarrow K(t)$$

$$\Rightarrow \vec{u}(t) = -R^{-1} B^T K(t) \vec{x}(t) \quad \text{with} \quad \vec{x}(t)$$

Note $\dot{K}(t) = 0 \Leftrightarrow K(t) = \text{const.}$ as $t_f \rightarrow \infty$

< Symetricity of R, Q, L and K matrix >

Example)

$$x_1^2 + x_1x_2 + x_2^2, \quad \vec{x} = \begin{Bmatrix} x_1 \\ x_2 \end{Bmatrix}$$

$$\rightarrow \text{Let } Q = \begin{bmatrix} 1 & 0 \\ 1 & 1 \end{bmatrix}, \quad \vec{x}^T Q \vec{x} = (x_1 \ x_2) \begin{bmatrix} 1 & 0 \\ 1 & 1 \end{bmatrix} \begin{Bmatrix} x_1 \\ x_2 \end{Bmatrix} = x_1^2 + x_1x_2 + x_2^2$$

$$\text{let } Q = \begin{bmatrix} 1 & \frac{1}{2} \\ \frac{1}{2} & 1 \end{bmatrix}, \quad \vec{x}^T Q \vec{x} = (x_1 \ x_2) \begin{bmatrix} 1 & \frac{1}{2} \\ \frac{1}{2} & 1 \end{bmatrix} \begin{Bmatrix} x_1 \\ x_2 \end{Bmatrix} = x_1^2 + x_1x_2 + x_2^2$$

Thus, Q matrix can be chosen as a symmetric matrix.

Similarly, R or L matrix can be chosen as a symmetric matrix.

The Riccati Equation:

$$\dot{K}(t) = -K(t)A(t) - A^T(t)K(t) - Q(t) + K(t)B(t)R^{-1}(t)B^T(t)K(t) \quad (*)$$

$$K(t_f) = L$$

Taking the transpose of the Riccati equation gives

$$\left[\dot{K}(t) \right]^T = \left[-K(t)A(t) - A^T(t)K(t) - Q(t) + K(t)B(t)R^{-1}(t)B^T(t)K(t) \right]^T$$

Let's omit the argument (t) for simplicity.

By using the matrix identity $[CD]^T = D^T C^T$, we have

$$\left[\dot{K} \right]^T = -A^T K^T - K^T A - Q^T + K^T B [R^{-1}]^T B^T K^T$$

Note that Q is symmetric; R is symmetric, so R^{-1} is symmetric and $[R^{-1}]^T = R^{-1}$.

Therefore,

$$\left[\dot{K} \right]^T = -A^T K^T - K^T A - Q + K^T B R^{-1} B^T K^T \quad (1)$$

We know that operations of transposition and differentiation are commutative;

$$\text{i.e. } \left[\frac{d}{dt} K \right]^T = \frac{d}{dt} [K^T]$$

Then we have

$$\frac{d}{dt} [K^T] = -A^T K^T - K^T A - Q + K^T B R^{-1} B^T K^T \quad (2)$$

The boundary condition for eq(1) is

$$K^T(t_f) = L^T = L$$

In eq(2), let $K^T \equiv M$, then

$$\begin{aligned} \dot{M} &= -A^T M - M A - Q + M B R^{-1} B^T M \\ M(t_f) &= L \end{aligned} \quad (3)$$

Note that $M(t)$ in eq(3) and $K(t)$ in the Riccati equation(eq. *) satisfy the same differential equation with the same boundary conditions.

Therefore, since the solution is unique, we conclude that

$$M(t) \equiv K^T(t) = K(t)$$

Since $K(t)$ matrix is symmetric, we need solve only for the elements of $K(t)$ which lie on or above the main diagonal. The number of terms involved is $1+2+3+\dots+n$. This sum is equal to $n(n+1)/2$.

< Linear Tracking Problem >

Let us generalize the results obtained for the linear regulator problem to the tracking problem:

If initial condition is changed a little bit.

How to control the system about reference solution.

The state equations are

$$\dot{\vec{x}}(t) = A(t)\vec{x}(t) + B(t)\vec{u}(t)$$

$\vec{x}(t_0)$ is given

$\vec{x}(t_f)$ is free, t_f is fixed.

The performance measure to be minimized is

$$J = \frac{1}{2} \left[\vec{x}(t_f) - \vec{r}(t_f) \right]^T L \left[\vec{x}(t_f) - \vec{r}(t_f) \right] + \frac{1}{2} \int_{t_0}^{t_f} \left[\left[\vec{x}(t) - \vec{r}(t) \right]^T Q(t) \left[\vec{x}(t) - \vec{r}(t) \right] + \vec{u}^T(t) R(t) \vec{u}(t) \right] dt$$

$\vec{r}(t_f)$ is known or calculated prior

$\vec{r}(t)$ is the desired or reference value of the state vector.

The Hamiltonian is

$$H = \frac{1}{2} \left\{ \left(\vec{x} - \vec{r} \right)^T Q \left(\vec{x} - \vec{r} \right) + \vec{u}^T R \vec{u} \right\} + \vec{\lambda}^T \left(A\vec{x} + B\vec{u} \right)$$

$$\frac{\partial H}{\partial \vec{u}} = \vec{0} \rightarrow R\vec{u} + \vec{\lambda}^T B = \vec{0}, \quad \vec{u} = -R^{-1} B^T \vec{\lambda}$$

$$\dot{\vec{x}} = A\vec{x} + B\vec{u} = A\vec{x} - B R^{-1} B^T \vec{\lambda}$$

$$\dot{\vec{\lambda}} = - \left(\frac{\partial H}{\partial \vec{x}} \right)^T = -Q(\vec{x} - \vec{r}) - A^T \vec{\lambda} = -Q\vec{x} - A^T \vec{\lambda} + Q\vec{r}$$

Transversality condtion

$$\left(\frac{\partial h}{\partial \vec{x}} + \vec{v}^T \frac{\partial \vec{m}}{\partial \vec{x}} - \vec{\Lambda} \right)_{t=t_f} \delta x_f + \left(\frac{\partial h}{\partial t} + \vec{v}^T \frac{\partial \vec{m}}{\partial t} + H \right)_{t=t_f} \delta t_f = 0$$

$$\vec{\Lambda}(t_f) = \frac{\partial h}{\partial \vec{x}} \Big|_{t_f} = L[\vec{x}(t_f) - \vec{r}(t_f)]$$

Let $\vec{\Lambda}(t) = K(t)\vec{x}(t) + S(t)$

$$\begin{aligned} \dot{\vec{\Lambda}} &= \dot{K}\vec{x} + K\dot{\vec{x}} + \dot{S} \\ &= \dot{K}\vec{x} + K[A\vec{x} - BR^{-1}B^T(K\vec{x} + S)] + \dot{S} \end{aligned}$$

$$\rightarrow \dot{K}\vec{x} + K[A\vec{x} - BR^{-1}B^T(K\vec{x} + S)] + \dot{S} = -Q\vec{x} - A^T[K\vec{x} + S] + Q\vec{r}$$

$$\rightarrow [\dot{K} + KA - KBR^{-1}B^TK + A^TK + Q]\vec{x} + [-KBR^{-1}B^TS + \dot{S} + A^TS - Q\vec{r}] = 0$$

The above equation must be satisfied for all $\vec{x}$ and $\vec{r}$.

Necessary conditions

$$\begin{aligned} \dot{K} &= -KA + KBR^{-1}B^TK - A^TK - Q \\ \dot{S} &= -(A^T - KBR^{-1}B^T)S + Q\vec{r} \end{aligned}$$

Boundary conditions

$$\begin{aligned} \vec{\Lambda}(t_f) &= L\vec{x}(t_f) - L\vec{r}(t_f) \\ \vec{\Lambda}(t_f) &= K(t_f)\vec{x}(t_f) + S(t_f) \\ \rightarrow K(t_f) &= L, \quad S(t_f) = -L\vec{r}(t_f) \end{aligned}$$

Note that we know $\vec{r}(t)$ completely prior,

Solve 1) about K

2) about S, using K

$$\therefore \vec{u}(t) = -R^{-1}B^T\vec{\Lambda} = -R^{-1}B^T[K\vec{x} + S]$$

$$\text{if } \vec{r}(t) = 0 \rightarrow S(t) = 0 \quad ; \text{ LQR problem}$$

< State-Dependent Riccati Equation (SDRE) Controller >

The State-Dependent Riccati Equation (SDRE) control method was suggested by James R. Cloutier. The general process of the SDRE technique is the same as that of LQR control method. However, the SDRE technique updates the dynamic model matrices **A** and **B** at each time step because these matrices in the SDRE technique are dependent on the states as shown in Figure 1. It affects the value of gain matrix **K**, so that the controller can make proper control values for the dynamic systems at any time. In order to use the SDRE control method, like LQR control method, the state-space equations like Eq. (14) are needed that can be written as below:

$$\begin{aligned}\dot{\mathbf{x}}(t) &= \mathbf{f}(\mathbf{x}(t)) + \mathbf{g}(\mathbf{x}(t))\mathbf{u}(t) \\ \dot{\mathbf{x}}(t) &= \mathbf{A}(\mathbf{x}(t))\mathbf{x}(t) + \mathbf{B}(\mathbf{x}(t))\mathbf{u}(t)\end{aligned}$$

Where $\mathbf{x}(t)$, $\mathbf{u}(t)$, $\mathbf{A}(\mathbf{x}(t))$ and $\mathbf{B}(\mathbf{x}(t))$ denote states, control, and model matrices, respectively. Note that unlike LQR control method, in the SDRE control method, dynamic model matrices $\mathbf{A}(\mathbf{x}(t))$ and $\mathbf{B}(\mathbf{x}(t))$ are not constant any more. These matrices are dependent on the states.

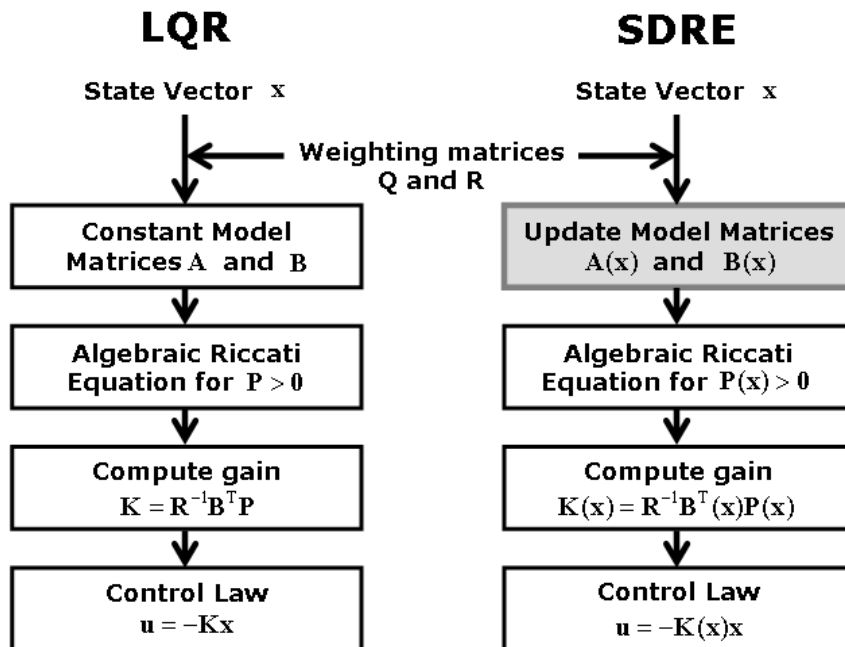


Figure 1 Description of the SDRE technique

Controllability and Observability: A system is said to be controllable at time t_0 if it is possible by means of an unconstrained control vector to transfer the system from any initial state ($\mathbf{x}(t_0)$) to any other state in a finite interval of time. A system is said to be observable at time t_0 if, with the system in state $\mathbf{x}(t_0)$, it is possible to determine this state from the observation of the output over a finite time interval.

Observability is very important because, in practice, the difficulty encountered with state feedback control is that some of the state variables are not accessible for direct measurement, with the result that it becomes necessary to estimate the unmeasurable state variables in order to construct the control signals.

$$\dot{\bar{\mathbf{x}}}(t) = A\bar{\mathbf{x}}(t) + B\bar{\mathbf{u}}(t), \quad \bar{\mathbf{y}}(t) = C\bar{\mathbf{x}}(t)$$

The system is completely observable if and only if the (n by nm matrix)

$$\begin{bmatrix} C^T & A^T C^T & \cdots & (A^T)^{n-1} C^T \end{bmatrix}$$

is of rank n or has linearly independent column vectors. This matrix is called the observability matrix.

If the system completely state controllable, then, given any initial state $\mathbf{x}(0)$, the rank of the (n by n) matrix

$$\begin{bmatrix} B & AB & \cdots & (A)^{n-1} B \end{bmatrix}$$

is of rank n.

Stability is a notion that describes whether the system will be able to follow the input command. In a nonrigorous manner, a system is said to be unstable if its output is out of control or increases without bound.

J. R. Cloutier, "State-Dependent Riccati Equation Techniques: An Overview", *Proceedings of the American Control Conference*, Albuquerque, NM, 1997, pp. 932-936

Pontryagin's Minimum Principle and Constraints

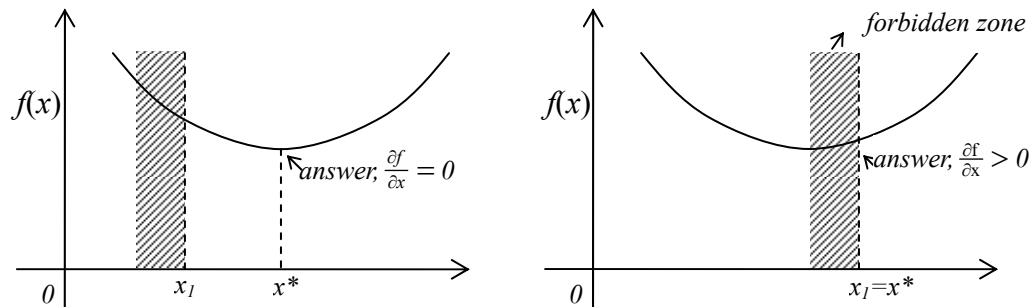
So far, we have assumed that the admissible controls and states are not constrained by any boundaries; however, in realistic systems such constraints do commonly occur. Physically realizable controls generally have magnitude limitations. State constraints often arise because of safety or structural restrictions.

If $\delta \vec{u}(t)$ is arbitrary $\rightarrow \delta J(\vec{u}^*, \delta \vec{u}) = 0$, $\vec{u}^*$ is optimal control.

If $\delta \vec{u}(t)$ is not arbitrary $\rightarrow \delta J(\vec{u}^*, \delta \vec{u}) = ?$

ex.) Min $f(x)$ function

subject to $x \geq x_1$



The first order necessary condition for minimizing $f(x)$;

$$\frac{\partial f}{\partial x} \geq 0$$

Control constraints

$$\dot{\vec{x}} = \vec{f}(\vec{x}(t), \vec{u}(t), t)$$

$$J = h(\vec{x}(t_f), t_f) + \int_{t_0}^{t_f} g(\vec{x}(t), \vec{u}(t), t) dt$$

Subject to $\vec{x}(t_0) = \vec{x}_0, \vec{x}(t_f) = \text{free}$

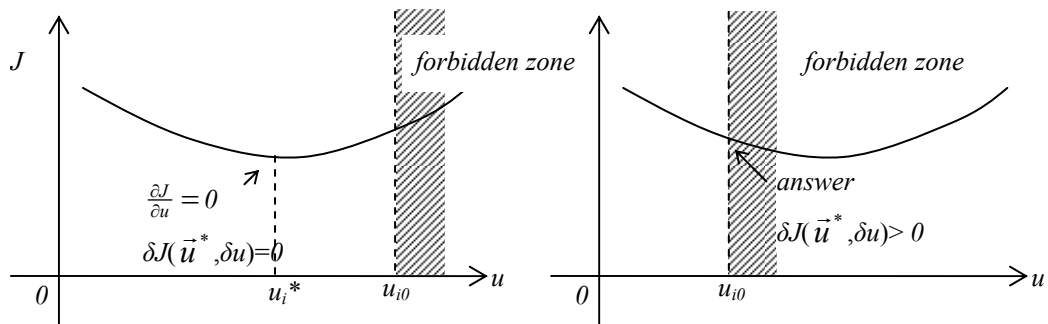
$$|u_i| \leq u_{i0}$$

Assume that boundary conditions are satisfied.

$$\delta J = \left[\frac{\partial h}{\partial \vec{x}} + \vec{v}^T \frac{\partial \vec{m}}{\partial \vec{x}} - \bar{\Lambda}^T \right]_{t=t_f} \delta \vec{x}_f + \left[\frac{\partial h}{\partial t} + \vec{v}^T \frac{\partial \vec{m}}{\partial t} + H \right]_{t=t_f} \delta t_f \\ + \int_{t_0}^{t_f} \left[\frac{\partial H}{\partial \vec{u}} \delta \vec{u} + \left(\frac{\partial H}{\partial \vec{x}} + \dot{\bar{\Lambda}}^T \right) \delta \vec{x} + \delta \bar{\Lambda}^T \left(\frac{\partial H}{\partial \bar{\Lambda}} - \dot{\vec{x}} \right) \right] dt$$

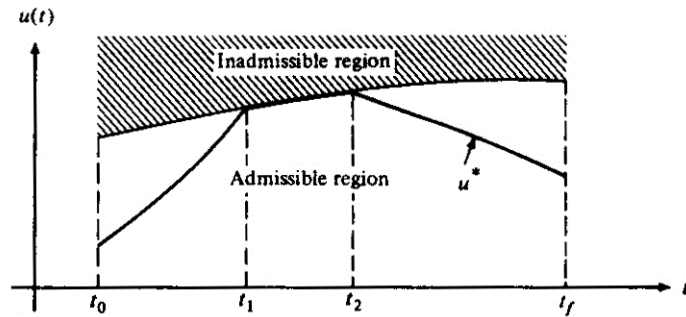
Let the state and costate differential equations be satisfied.

$$\delta J = \int_{t_0}^{t_f} \frac{\partial H}{\partial \vec{u}} \delta \vec{u} dt$$

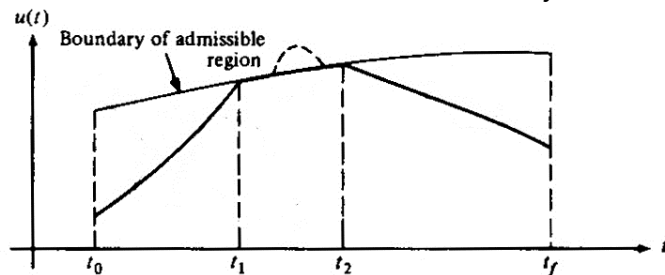


Then, $\delta J \geq 0$ for minimum J

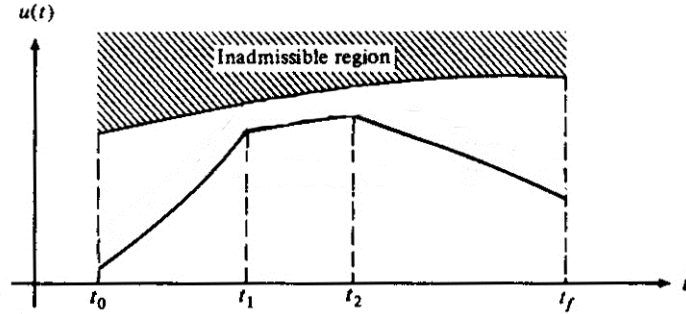
Case a) $\delta J = 0$: extremal control that lies on a boundary



Case b) $\delta J > 0$: extremal control that is constrained by a boundary



Case c) $\delta J = 0$: extremal control lies within the boundary during any portion of time interval $[t_0, t_f]$



$\Rightarrow$ Thus, case a), b), c) yields $\underline{\delta J(u^*, \delta u) \geq 0}$ for the problem with control constraints.

$$\begin{aligned}
 \Delta H &= H(\bar{x}^*, \bar{u}^* + \delta \bar{u}, \bar{\Lambda}^*, t) - H(\bar{x}^*, \bar{u}^*, \bar{\Lambda}^*, t) \\
 &= H(\bar{u}^*) + \frac{\partial H}{\partial \bar{u}} \Big|_{\bar{u}^*} \delta \bar{u} + \frac{1}{2} \frac{\partial^2 H}{\partial \bar{u}^2} \Big|_{\bar{u}^*} (\delta \bar{u})^2 + (H.O.T.) - H(\bar{u}^*) \\
 &\cong \frac{\partial H}{\partial \bar{u}} \Big|_{\bar{u}^*} \delta \bar{u} \quad \begin{array}{l} \delta u \text{ is sufficiently small} \\ \rightarrow (\delta u)^2 \text{ can be neglect!!} \end{array}
 \end{aligned}$$

$$\begin{aligned}
 \delta J &= \int_{t_0}^{t_f} \frac{\partial H}{\partial \bar{u}} \delta \bar{u} dt \\
 &\approx \int_{t_0}^{t_f} \Delta H dt \\
 &= \int_{t_0}^{t_f} \left[H(\bar{x}^*, \bar{u}^* + \delta \bar{u}, \bar{\Lambda}^*, t) - H(\bar{x}^*, \bar{\Lambda}^*, \bar{u}^*, t) \right] dt \\
 &\geq 0
 \end{aligned}$$

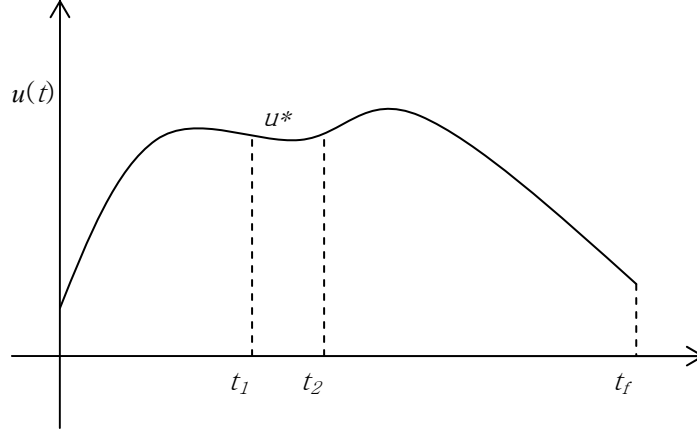
For all admissible $\delta \bar{u}$, this can be happen, if

$$\therefore H(\bar{x}^*, \bar{u}^* + \delta \bar{u}, \bar{\Lambda}^*, t) \geq H(\bar{x}^*, \bar{u}^*, \bar{\Lambda}^*, t) \text{ at every time.}$$

Proof.) Contradict the above result.

$$H(x^*, u^* + \delta u, \Lambda^*, t) < H(x^*, u^*, \Lambda^*, t) \text{ in } [t_1, t_2]$$

where $[t_1, t_2]$ is an arbitrary small, but nonzero, time interval.



$$u = u^* \quad 0 \leq t < t_1$$

$$u = u^* \quad t_2 < t < t_f$$

$$u = u^* + \delta u \quad t_1 \leq t \leq t_2$$

we can make

$$\begin{aligned} \delta J &= \int_{t_0}^{t_f} [H(x^*, \Lambda^*, u, t) - H(x^*, \Lambda^*, u^*, t)] dt \\ &= \int_{t_0}^{t_1} [H(x^*, \Lambda^*, u^*, t) - H(x^*, \Lambda^*, u^*, t)] dt \\ &\quad + \int_{t_1}^{t_2} [H(x^*, \Lambda^*, u, t) - H(x^*, \Lambda^*, u^*, t)] dt \\ &\quad + \int_{t_2}^{t_f} [H(x^*, \Lambda^*, u^*, t) - H(x^*, \Lambda^*, u^*, t)] dt \\ &= \int_{t_1}^{t_2} [H(x^*, \Lambda^*, u, t) - H(x^*, \Lambda^*, u^*, t)] dt \\ &< 0 \end{aligned}$$

This violate the optimality condition; $\delta J \geq 0$, since $[t_1, t_2]$ can be anywhere in $[0, t_f)$ it is clear that if $H(x^*, \Lambda^*, u^* + \delta u, t) < H(x^*, \Lambda^*, u^*, t)$ for any $t \in [0, t_f)$ then, it is always possible to construct an admissible control which makes $\Delta J < 0 (\approx \delta J < 0)$, thus contracting the optimality of the control $u^*(t)$ ($\delta J \geq 0$).

Thus a necessary condition for $\vec{u}^*$ to minimize the functional J is

$$\therefore H(x^*, u^* + \delta u, \Lambda^*, t) \geq H(x^*, u^*, \Lambda^*, t)$$

for all $t \in [0, t_f]$ and for all admissible control.

This means that an optimal control must minimize the Hamiltonian
 $\Rightarrow$ Pontryagin's minimum principle

In terms of the Hamiltonian

$$H \equiv g(\vec{x}, \vec{u}, t) + \vec{\Lambda}^T \vec{f}(\vec{x}, \vec{u}, t)$$

Necessary conditions for $\vec{u}^*(t)$ to be an optimal control are

$$\dot{\vec{x}} = \frac{\partial H}{\partial \vec{\Lambda}}$$

$$\dot{\vec{\Lambda}} = -\frac{\partial H}{\partial \vec{x}}$$

$$H(\vec{x}^*(t), \vec{\Lambda}^*(t), \vec{u}(t), t) \geq H(\vec{x}^*(t), \vec{\Lambda}^*(t), \vec{u}^*(t), t)$$

$$\left[\frac{\partial h}{\partial \vec{x}} + \nu^T \frac{\partial m}{\partial \vec{x}} - \vec{\Lambda}^T \right]_{t=t_f} \delta x_f + \left[H + \frac{\partial h}{\partial t} + \nu^T \frac{\partial m}{\partial t} \right]_{t=t_f} \delta t_f = 0$$

(If $\vec{u}$ is constrained, $\frac{\partial H}{\partial u} \neq 0$)

Pontryagin's minimum principle

If $\vec{x}^*(t)$, $\vec{\Lambda}^*(t)$ and $\vec{u}^*(t)$ are extremal states, costates and controls, respectively. Then, the optimal control (constrained or unconstrained control) must minimize the Hamiltonian.

$$\vec{u}^*(t) = \min_{\vec{u} \in \Omega} \arg H(\vec{x}^*(t), \vec{\Lambda}^*(t), \vec{u}(t), t)$$

The minimum principle can be also applied to problems in which the admissible controls are not bounded ($\Rightarrow \frac{\partial H}{\partial u} = 0$).

(Example) minimum-time problem

$$J = \int_0^{t_f} 1 dt = t_f \quad , t_f \text{ is free}$$

$$\begin{aligned} \dot{x}_1 &= x_2 & x_1(0) &= \text{given} , & x_1(t_f) &= 0 \\ \dot{x}_2 &= u & x_2(0) &= \text{given} , & x_2(t_f) &= 0 \\ |u| &\leq u_0 \end{aligned}$$

The problem is to transfer the system from initial state $\vec{x}_0$ to the origin in minimum time.

The Hamiltonian

$$H = 1 + \Lambda_1 x_2 + \Lambda_2 u \quad \Lambda^T f = (\Lambda_1 \quad \Lambda_2) \begin{pmatrix} \dot{x}_1 \\ \dot{x}_2 \end{pmatrix}$$

$$\left[\frac{\partial h}{\partial \vec{x}} + \nu^T \frac{\partial m}{\partial \vec{x}} - \bar{\Lambda}^T \right]_{t=t_f} \delta x_f + \left[H + \frac{\partial h}{\partial t} + \nu^T \frac{\partial m}{\partial t} \right]_{t=t_f} \delta t_f = 0$$

$$\rightarrow H(t_f) = 0$$

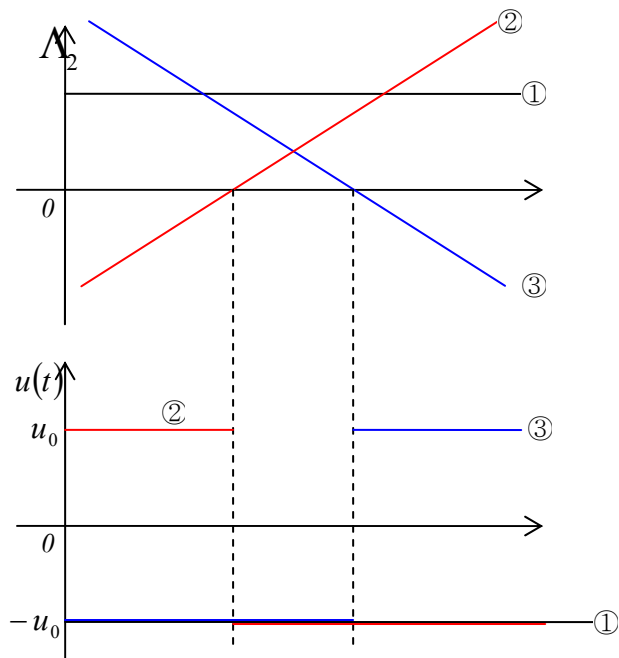
$$\begin{aligned} \dot{H} = 0 & \Rightarrow H(t) = \text{const.} & \Rightarrow H(t) = 0 \\ & H(t_f) = 0 \end{aligned}$$

$$\dot{\Lambda}_1 = -\frac{\partial H}{\partial x_1} = 0 \quad \rightarrow \Lambda_1 = \text{const.}$$

$$\dot{\Lambda}_2 = -\frac{\partial H}{\partial x_2} = -\Lambda_1 \quad \rightarrow \Lambda_2 = -\Lambda_1 t + \Lambda_{20}$$

$$\vec{u}^*(t) = \min_{u \in R} \arg H(x^*(t), \Lambda^*(t), u(t), t)$$

$$\text{If } \begin{aligned} \Lambda_2 &> 0 & u &= -u_0 \\ \Lambda_2 &< 0 & u &= u_0 \end{aligned} \quad u = -u_0 \text{sign}(\Lambda_2)$$



If $u_0 = 1$

$$\begin{aligned}\dot{x}_1 &= x_2 \\ \dot{x}_2 &= 1\end{aligned}$$

or

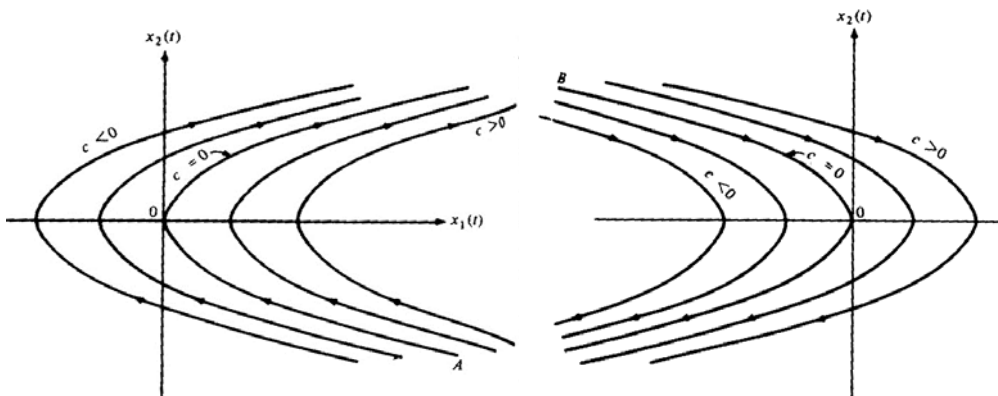
$$\begin{aligned}\dot{x}_1 &= x_2 \\ \dot{x}_2 &= -1\end{aligned}$$

$$\Rightarrow \begin{aligned}dx_1 &= x_2 dt \\ dx_2 &= \pm dt\end{aligned}$$

$$\rightarrow \frac{dx_1}{dx_2} = \frac{x_2}{\pm 1}$$

$$\rightarrow \pm \int dx_1 = \int x_2 dx_2$$

$$\rightarrow \pm x_1 = \frac{1}{2} x_2^2 + c$$



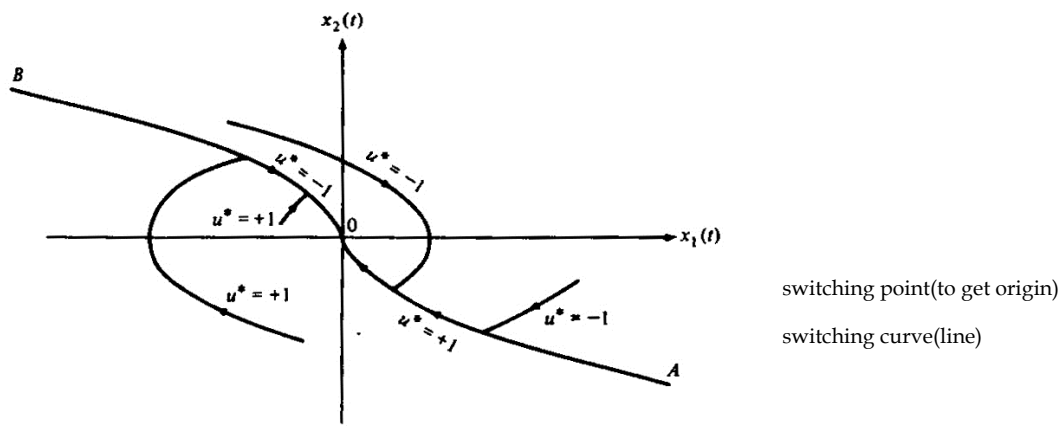
t_f is integrated time from start and to ②

$$x_1(t) = \frac{1}{2}x_2^2(t) + c_1 \quad \text{for } u = 1$$

$$x_1(t) = -\frac{1}{2}x_2^2(t) + c_2 \quad \text{for } u = -1$$

If $x_2(t) > 0$, switching curve $x_1(t) = -\frac{1}{2}x_2^2(t)$

If $x_2(t) < 0$, switching curve $x_1(t) = \frac{1}{2}x_2^2(t)$



Let $c_1 = c_2 = 0$, we find the equation of the switching curve(line)

$$x_1(t) = -\frac{1}{2}x_2(t)|x_2(t)|$$

$$s \equiv x_1 + \frac{1}{2}x_2|x_2|$$

We have obtained the optimal control law, that is, the optimal control at every time t is known as a function of the state variables (x_1, x_2)

Notice that

$S(\mathbf{x}(t)) > 0$ implies $\mathbf{x}(t)$ lies above the switching curve (A-O-B)

$S(\mathbf{x}(t)) < 0$ implies $\mathbf{x}(t)$ lies below the switching curve (A-O-B)

$S(\mathbf{x}(t)) = 0$ implies $\mathbf{x}(t)$ lies on the switching curve (A-O-B)

Thus, in terms of this switching function, the optimal control law is

$$\begin{aligned} u^*(t) &= -1, & \text{for } S(\mathbf{x}(t)) > 0 \\ &= +1, & \text{for } S(\mathbf{x}(t)) < 0 \\ &= -1, & \text{for } S(\mathbf{x}(t)) = 0 \text{ and } x_2(t) > 0 \\ &= +1, & \text{for } S(\mathbf{x}(t)) = 0 \text{ and } x_2(t) < 0 \\ &= 0, & \text{for } \mathbf{x}(t) = 0 \end{aligned}$$

< Problems with inequality constraints for a Function >

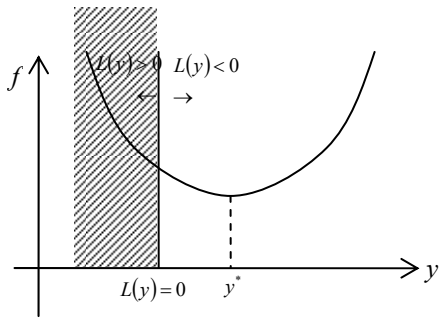
Function

Minimizing $f(y)$

Subject to $L(y) \leq 0$: inequality constraints.

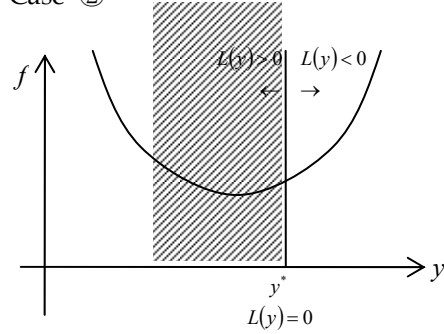
f and L are vectors of different dimension.

Case ①



$$y^* : L(y^*) < 0, \quad \frac{\partial f}{\partial y} = 0$$

Case ②



$$y^* : L(y^*) = 0, \quad \frac{\partial f}{\partial y} > 0$$

$$\rightarrow \left. \frac{\partial L}{\partial y} \right|_{y^*} = 0$$

Case ① : Constraint is not effective and can be ignored.

Case ② : Constraint is effective.

$$df = \left. \frac{\partial f}{\partial y} \right|_{y^*} dy \geq 0, \quad (1) \text{ } dy : \text{small perturbation about } y^*$$

$$dL = \left. \frac{\partial L}{\partial y} \right|_{y^*} dy \leq 0 \quad (2)$$

In order that eq(1) and eq(2) be consistent, it is clearly necessary that either.

$$\text{sign} \left(\frac{\partial f}{\partial y} \right) = -\text{sign} \left(\frac{\partial L}{\partial y} \right) \quad \text{or} \quad \frac{\partial f}{\partial y} = 0$$

These two possibilities may be expressed in one relation

$$\frac{\partial f}{\partial y} + \lambda \frac{\partial L}{\partial y} = 0, \quad \lambda \geq 0$$

$$\text{If } \begin{cases} \frac{\partial f}{\partial y} = 0, & \text{then } \lambda = 0 \\ \frac{\partial f}{\partial y} \neq 0, & \text{then } \lambda > 0 \end{cases}$$

The two cases may be treated analytically by adjoining the $f(y)$ and $L(y)$.

Define $K(y, \lambda) = f(y) + \lambda L(y)$

λ : Lagrange multiplier, inequality constraint: $L(y)$

Then, necessary conditions for minimum with respect to y becomes

$$\begin{aligned} \frac{\partial K}{\partial y} = 0 & \quad \leftrightarrow \quad \frac{\partial f}{\partial y} + \lambda \frac{\partial L}{\partial y} = 0 \\ \text{and } L(y) & \leq 0 \end{aligned}$$

where $\lambda > 0$, $L(y) = 0$ active constraints, case ②

$$\left(\frac{\partial f}{\partial y} > 0, \frac{\partial L}{\partial y} < 0, \text{sign}\left(\frac{\partial f}{\partial y}\right) = -\text{sign}\left(\frac{\partial L}{\partial y}\right) \right)$$

$\lambda = 0$, $L(y) < 0$ inactive constraints, case ①

$$\left(\frac{\partial f}{\partial y} = 0, \lambda = 0 \right)$$

Note: For a maximum, the sign of λ must be changed.

Note: We use this concept described in this section, to numerically find optimal solutions. This concept does not violate the Pontryagin's Minimum Principle.

< Inequality constraints on the control variables >

Inequality constraints

$$C(u, t) \leq 0 \quad (1)$$

Define $H^0 = g + \Lambda f$ (2)

$$\begin{aligned} \delta J &= \int_{t_0}^{t_f} H_u^0 \delta u \, dt \\ &= \int_{t_0}^{t_f} \delta H^0 \, dt \end{aligned}$$

when all state and costate necessary conditions are satisfied.

We must have $\delta J \geq 0$ for all admissible δu

$$\begin{aligned} \delta H^0 &= \frac{\partial H^0}{\partial u} \delta u \geq 0 \\ \delta C &= \frac{\partial C}{\partial u} \delta u \leq 0 \quad (\text{similar to the previous section for function}) \end{aligned}$$

Define $H = g + \Lambda^T f + \mu^T C$ (3)

The necessary condition on H is

$$\frac{\partial H}{\partial u} = \frac{\partial g}{\partial u} + \Lambda^T \frac{\partial f}{\partial u} + \mu^T \frac{\partial C}{\partial u} = 0 \quad (4)$$

$$\begin{aligned} \text{where } \mu &\geq 0 & C &= 0 \Rightarrow \text{eq(1) determines } u(t) \text{ at constrained arc} \\ \mu &= 0 & C &< 0 \Rightarrow \text{eq(4) determines } u(t) \end{aligned}$$

$$\dot{\Lambda} = -\frac{\partial H}{\partial x} = -\frac{\partial g}{\partial x} - \Lambda^T \frac{\partial f}{\partial x}$$

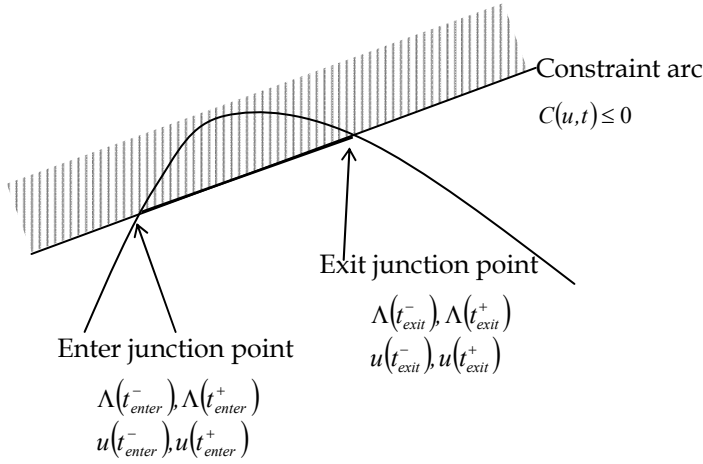
$$\dot{x} = f$$

and Boundary Conditions

In solving a particular problem, constrained and unconstrained arcs must be pieced together to satisfy all the necessary conditions.

At the junction points of constrained and unconstrained arcs, the control may or may not be continuous; if it is discontinuous, the point is called a corner.

These seem to be no a prior method for determining the existence of corners. If $u(t)$ is continuous a junction point, it follows from the continuity of Λ , H_u and H that $\mu(t)$ is continuous across the junction point.



< Inequality constraints on function of the control and state variable >

Inequality constraints

$$C(x, u, t) \leq 0 \quad (1)$$

$$H = g + \Lambda^T f + \mu^T C \quad (2)$$

$$\begin{array}{ll} \text{where } \mu \geq 0 & C = 0 \\ \mu = 0 & C < 0 \end{array}$$

$$\dot{\Lambda} = -\frac{\partial H}{\partial x} = -\frac{\partial g}{\partial x} - \Lambda^T \frac{\partial f}{\partial x} - \mu^T \frac{\partial C}{\partial x}, \quad C = 0 \quad (3)$$

$$\dot{\Lambda} = -\frac{\partial H}{\partial x} = -\frac{\partial g}{\partial x} - \Lambda^T \frac{\partial f}{\partial x} \quad C < 0 \quad (4)$$

$$\dot{x} = f$$

and Boundary Conditions

$$\frac{\partial H}{\partial u} = \frac{\partial g}{\partial u} + \Lambda^T \frac{\partial f}{\partial u} + \mu^T \frac{\partial C}{\partial u} = 0 \quad (5)$$

$$\begin{array}{lll} \text{where } \mu = 0 & C < 0 & \rightarrow \text{equation (5) determines } u(t) \\ \mu \geq 0 & C = 0 & \rightarrow \text{equation (1) determines } u(t) \text{ at constrained arc} \\ & & \text{equation (5) determines } \mu(t) \\ & & \mu \text{ is needed for equation (3)} \\ & & \text{eq.(3) \&eq. (5) } \rightarrow \mu, \Lambda \text{ for constrained arc} \end{array}$$

< Inequality constraints on function of the state variable >

State inequality constraints

$$S(x, t) \leq 0 \quad (1)$$

We take successive total time derivative of (1) and substitute $f(x, u, t)$ for $\dot{x}$; until we obtain an expression that is explicitly dependent on u . If q^{th} time derivatives are required, we shall call eq (1) a q^{th} -order state variable inequality constraint.

Now $S^{(q)}(x, u, t)$, the q^{th} total time derivative of S , plays the same role as $C(x, u, t)$ in previous section.

$$\begin{array}{l} H = g + \Lambda^T f + \mu^T S^{(q)}(x, u, t) \\ \text{where } S^{(q)} = 0 \text{ on the constraint boundary } S = 0 \\ \mu = 0 \text{ off the constraint boundary } S < 0 \end{array} \quad (2)$$

Tangency constraint apply to the path leaving the constant boundary

$$N(x, t) \equiv \begin{Bmatrix} S(x, t) \\ S^{(1)}(x, t) \\ \vdots \\ S^{(q-1)}(x, t) \end{Bmatrix} = 0 \quad (3)$$

Equation (3) form a set of interior boundary conditions. Consequently the $\Lambda(t)$ are, in general, discontinuous at junction points between constrained and unconstrained arcs.

Necessary conditions

$$\frac{\partial H}{\partial u} = \frac{\partial g}{\partial u} + \Lambda^T \frac{\partial f}{\partial u} + \mu^T \frac{\partial S^{(q)}}{\partial u} = 0 \quad (4)$$

$$\text{where } \mu = 0 \quad S^{(q)} < 0$$

$$\mu \geq 0 \quad S^{(q)} = 0$$

→ equation (4) determines $u(t)$

→ $S^{(q)} = 0$ determines $u(t)$ at constrained arc
equation (4) determines $u(t)$

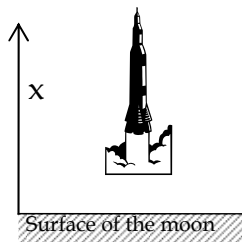
$$\dot{\Lambda} = -\frac{\partial H}{\partial x} = -\left(\frac{\partial g}{\partial x} + \Lambda^T \frac{\partial f}{\partial x} + \mu^T \frac{\partial S^{(q)}}{\partial x} \right) \quad (5)$$

$$\dot{x} = f$$

and Boundary Conditions

Example: minimum-time problem

A lunar rocket in the terminal phase of a minimum-time will make a soft landing on the surface of the moon.



m ; spacecraft mass

$-M \leq \dot{m} \leq 0$; spacecraft mass rate, control variable

g ; gravitational constant in surface of the moon

T ; thrust ($T = -k\dot{m}$)

where k is constant relative exhaust velocity of gas

$$F=ma \Rightarrow$$

$$m(t)\ddot{x}(t) = -gm(t) + T(t)$$

$$= -gm(t) - k\dot{m}(t)$$

Let $x_1 = x$, $x_2 = \dot{x}$, $x_3 = m$, $u = \dot{m}$

$$\dot{x}_1(t) = x_2(t)$$

$$\dot{x}_2(t) = -g - \frac{k}{x_3(t)} u(t)$$

$$\dot{x}_3(t) = u(t)$$

$$J = \int_{t_0}^{t_f} 1 dt$$

$$\begin{aligned} \rightarrow H &= 1 + \vec{\Lambda}^T \vec{f} \\ &= 1 + \Lambda_1 x_2 + \Lambda_2 \left(-g - \frac{k}{x_3} u \right) + \Lambda_3 u \\ &= 1 + \Lambda_1 x_2 - \Lambda_2 g + \left(-\Lambda_2 \frac{k}{x_3} + \Lambda_3 \right) u \end{aligned}$$

Optimal control

$$H(\Lambda^*, x^*, u^*) \leq H(\Lambda^*, x^*, u)$$

All possible $u(t)$

$$u(t) = \begin{cases} 0 & , & -\Lambda_2 \frac{k}{x_3} + \Lambda_3 < 0 \\ -M & , & -\Lambda_2 \frac{k}{x_3} + \Lambda_3 > 0 \\ \text{undetermined} & , & -\Lambda_2 \frac{k}{x_3} + \Lambda_3 = 0, \text{ for a finite time} \end{cases}$$

To obtain an explicit for $u^*(t)$, we have to solve a nonlinear two-point boundary-value problem using the numerical technique described in the previous sections.

Minimum time problem has a “bang-bang” control effect; that is the optimal control between its maximum and minimum admissible value.

< Control variable equality constraints >

Control variable equality constraint

$$C(u, t) = 0 \tag{1}$$

$u(t)$ is the m -dimensional control vector $m \geq 2$, C is n vector function.

Hamiltonian is defined as

$$H = g + \bar{\Lambda}^T \vec{f} + \bar{\mu}^T \vec{C}$$

Optimality condition

$$\frac{\partial H}{\partial \vec{u}} = \frac{\partial g}{\partial \vec{u}} + \bar{\Lambda}^T \frac{\partial \vec{f}}{\partial \vec{u}} + \bar{\mu}^T \frac{\partial \vec{C}}{\partial \vec{u}} = 0 \quad (2)$$

Equation (1) and equation (2) represents $m + n$ equations for determining in control vector and the n function $\bar{\mu}(t)$.

All other equations remain the same.

$$\dot{\bar{\Lambda}} = -\frac{\partial H}{\partial \vec{x}} = -\frac{\partial g}{\partial \vec{x}} - \bar{\Lambda}^T \frac{\partial \vec{f}}{\partial \vec{x}}$$

$$\dot{x} = f$$

Boundary Conditions

< Equality constraints on functions of the control and state variables >

$$C(\vec{x}, \vec{u}, t) = 0 \quad (1)$$

where $\frac{\partial C}{\partial \vec{u}} \neq 0$ for any u

$$\vec{u} \in \mathbf{R}^m$$

C can be a scalar or vector function

Hamiltonian

$$H = g + \bar{\Lambda}^T \vec{f} + \bar{\mu}^T \vec{C}$$

Optimality condition

$$\frac{\partial H}{\partial \vec{u}} = \frac{\partial g}{\partial \vec{u}} + \bar{\Lambda}^T \frac{\partial \vec{f}}{\partial \vec{u}} + \bar{\mu}^T \frac{\partial \vec{C}}{\partial \vec{u}} = 0 \quad (2)$$

Equation (1) and equation (2) represents $m + n$ equations for determining in control vector and the n function $\vec{\mu}(t)$

Costate equation

$$\dot{\Lambda}^T = -\frac{\partial H}{\partial \vec{x}} = -\frac{\partial g}{\partial \vec{x}} - \Lambda^T \frac{\partial f}{\partial \vec{x}} - \mu^T \frac{\partial \bar{C}}{\partial \vec{x}} \quad (3)$$

All other equations remain the same.

< Equality constraints on functions of the state variables >

Equality constraint

$$S(\vec{x}, t) = 0 \quad (1)$$

$$\dot{S}(\vec{x}, \vec{u}, t) = 0$$

Equation (1) can be added a boundary condition at $t = 0$, or $t = t_f$.

Or, we can eliminate one component of $\vec{x}$ in terms of the remaining $(n-1)$ component using equation (1)

If $\dot{S}$ does not explicitly involve controls, we may take another time derivative and substitute $\dot{x} = f$, until some explicit dependence on controls does occur.

$$S^{(q)}(\vec{x}, \vec{u}, t) = 0 \quad (2) \quad (\text{qth-order state variable equality constraint})$$

plays the role of a control variable equality constraint

$$\text{where } S^{(q)} = \frac{d^q S}{dt^q}$$

$$\begin{bmatrix} S(\vec{x}, t) \\ S^1(\vec{x}, t) \\ \vdots \\ S^{(q-1)}(\vec{x}, t) \end{bmatrix} = \mathbf{0} \quad (3)$$

In addition, we must either eliminate q components of $\vec{x}$ in terms of the remaining $(n-q)$ components using equation(3).

Or add equation (3) as a set of boundary conditions $t = 0$ or $t = t_f$

or $S(\vec{x}, t) = 0, \quad S^{(q)}(\vec{x}, \vec{u}, t) = 0$

$$\begin{aligned} H &= g + \Lambda^T f + \mu^T S^{(q)}(x, u, t) \\ \frac{\partial H}{\partial u} &= \frac{\partial g}{\partial u} + \Lambda^T \frac{\partial f}{\partial u} + \mu^T \frac{\partial S^{(q)}}{\partial u} = 0 \\ \dot{\Lambda} &= -\frac{\partial H}{\partial x} = -\frac{\partial g}{\partial x} - \Lambda^T \frac{\partial f}{\partial x} - \mu^T \frac{\partial S^{(q)}}{\partial x} \end{aligned} \quad (4)$$

All other equations remain the same

Equation (2) and equation (4) determine $\vec{u}(t), \quad \vec{\mu}(t)$

Example 1: minimum-energy problem

$$\begin{aligned} \dot{x}_1 &= x_2 \\ \dot{x}_2 &= u_1 \\ \dot{x}_3 &= x_4 \\ \dot{x}_4 &= u_2 \end{aligned}$$

State (equality) constraint

$$S(\vec{x}) = x_1^2 + x_3^2 - 1 = 0$$

$$J = \frac{1}{2} \int_{t_0}^{t_f} (u_1^2 + u_2^2) dt$$

t_f is fixed, $\vec{x}(t_f)$ are free, $\vec{x}(0)$ are given

Path constraint

$$\dot{S}(\vec{x}) = 2x_1\dot{x}_1 + 2x_3\dot{x}_3 = 2x_1x_2 + 2x_3x_4 = 0$$

$$\ddot{S}(\vec{x}) = 2(\dot{x}_1x_2 + x_1\dot{x}_2 + \dot{x}_3x_4 + x_3\dot{x}_4) = 2(x_2^2 + x_1u_1 + x_4^2 + x_3u_2) = 0$$

$$\rightarrow \text{constraint} \rightarrow x_2^2 + x_1u_1 + x_4^2 + x_3u_2 = 0 \quad (1)$$

Hamiltonian

$$H = \frac{1}{2}(u_1^2 + u_2^2) + \Lambda_1 x_2 + \Lambda_2 u_1 + \Lambda_3 x_4 + \Lambda_4 u_2 + \mu(x_2^2 + x_1 u_1 + x_4^2 + x_3 u_2)$$

Optimal control

$$\frac{\partial H}{\partial u_1} = u_1 + \Lambda_2 + \mu x_1 = 0 \quad \rightarrow u_1 = -(\Lambda_2 + \mu x_1) \quad (2)$$

$$\frac{\partial H}{\partial u_2} = u_2 + \Lambda_4 + \mu x_3 = 0 \quad \rightarrow u_2 = -(\Lambda_4 + \mu x_3) \quad (3)$$

Substitute equation (2), equation (3) $\rightarrow$ equation (1)

$$\begin{aligned} x_2^2 + x_1 u_1 + x_4^2 + x_3 u_2 &= 0 \\ \rightarrow x_2^2 + x_4^2 - x_1(\Lambda_2 + \mu x_1) - x_3(\Lambda_4 + \mu x_3) &= 0 \\ \rightarrow x_2^2 + x_4^2 - x_1 \Lambda_2 - x_3 \Lambda_4 - \mu(x_1^2 + x_3^2) &= 0 \\ \rightarrow \underbrace{(x_1^2 + x_3^2)}_{=1} \mu &= x_2^2 + x_4^2 - x_1 \Lambda_2 - x_3 \Lambda_4 \\ \rightarrow \mu &= x_2^2 + x_4^2 - x_1 \Lambda_2 - x_3 \Lambda_4 \end{aligned} \quad (4)$$

$$\dot{\Lambda}(t) = -\frac{\partial H}{\partial x(t)} = -\left(\frac{\partial g}{\partial x} + \Lambda^T \frac{\partial f}{\partial x} + \mu^T \frac{\partial S^{(2)}}{\partial x} \right) \quad (5)$$

Boundary conditions

$x(0)$ given $\rightarrow$ 4 boundary conditions

$x(t_f)$ should be on $S(\vec{x}(t_f)) = 0$

$$\rightarrow x_1^2(t_f) + x_3^2(t_f) - 1 = 0$$

$$\begin{aligned} x(t_f) \text{ is free} \quad \rightarrow \quad \frac{\partial h}{\partial \vec{x}}(\vec{x}(t_f), t_f) + \nu^T \frac{\partial \vec{m}}{\partial \vec{x}} - \Lambda(t_f) &= 0 \\ (h \text{ is 0 in this problem}) \end{aligned}$$

$$\Rightarrow \bar{\Lambda}(t_f) = v^T \frac{\partial \vec{m}}{\partial \vec{x}} = v^T \frac{\partial S}{\partial \vec{x}} \quad \text{at } t_f$$

The following 5 eqs. at t_f can numerically determine 4 boundary conditions($\bar{\Lambda}(t_f)$) and v .

$$\Lambda_1(t_f) = 2vx_1(t_f), \quad \Lambda_2(t_f) = 0, \quad \Lambda_3(t_f) = 2vx_3(t_f), \quad \Lambda_4(t_f) = 0$$

$$x_1^2(t_f) + x_3^2(t_f) - 1 = 0$$

Example2: minimum-fuel problem

$$\dot{x}(t) = u(t)$$

The system is to be transferred from an arbitrary initial state x_0 to the origin.

Performance measure

$$J = \int_{t_0}^{t_f} |u(t)| dt \quad \text{where } t_f \text{ is free}$$

$$|u(t)| \leq 1.0$$

Find optimal control using the Pontryagin's Minimum principle.

Hamiltonian $H = |u(t)| + \Lambda u(t)$

Costate equation $\dot{\Lambda}(t) = -\frac{\partial H}{\partial x} = 0$
 $\Lambda(t) = C_1 (\text{const.})$

Optimal control

$$u^*(t) = \begin{cases} 1 & , & C_1 < -1 \\ 0 & , & -1 < C_1 < 1 \\ -1 & , & C_1 > 1 \\ \text{undetermined} & , & C_1 = -1 \text{ or } C_1 = 1 \text{ for a finite time} \end{cases}$$

< A weighted combination of time and fuel >

$$J = \int_{t_0}^{t_f} (w + |u(t)|) dt$$

where t_f is free

$w > 0$ is chosen to weight the relative importance of time and fuel.

Example: minimum-energy problem

$$\dot{x}(t) = -ax(t) + u(t) \quad a > 0$$

The system is to be transferred from an arbitrary initial state x_0 to the origin by a constraint minimizes the performance index.

$$J(u) = \int_{t_0}^{t_f} [w + u^2(t)] dt \quad \text{where } t_f \text{ is free, } w > 0$$

The admissible controls are constrained by

$$|u(t)| \leq 1.0$$

Find optimal control using the Pontryagin's Minimum principle.

$$\text{Hamiltonian} \quad H = w + u^2(t) - \Lambda ax(t) + \Lambda u(t)$$

$$\text{Costate equation} \quad \dot{\Lambda}(t) = -\frac{\partial H}{\partial x} = a\Lambda(t) \quad \rightarrow \quad \Lambda(t) = C_1 e^{at}$$

Unbounded control case (i.e., $|u(t)| < 1.0$)

$$\frac{\partial H}{\partial u} = 0 = 2u + \Lambda$$

$$\Rightarrow u^* = -\frac{1}{2}\Lambda \quad \Leftrightarrow \quad |\Lambda(t)| < 2 \quad \text{when } |u(t)| < 1.0$$

If $|\Lambda(t)| \geq 2$, then the control that minimized H is

$$u^*(t) = \begin{cases} 1 & , \quad \Lambda(t) \leq -2 \\ -1 & , \quad \Lambda(t) \geq 2 \end{cases}$$

thus,

$$u^*(t) = \begin{cases} 1 & , \quad \Lambda(t) \leq -2 \\ -\frac{1}{2}\Lambda(t) & , \quad -2 < \Lambda(t) < 2 \\ -1 & , \quad \Lambda(t) \geq 2 \end{cases}$$

Note:

In minimum-time problems, the optimal control is “bang-bang”

In minimum-fuel problems, the optimal control is “bang-off-bang”

In minimum-energy problems, the optimal control is a continuous function.

< Singular intervals >

If there is a time interval $[t_1, t_2]$ of finite duration which the necessary condition provides no information about the relationship between $x^*(t)$, $\Lambda^*(t)$ and $u^*(t)$, then we say that the problem is singular or the control is singular control, or the state on that interval is singular arc, the interval $[t_1, t_2]$ is a singular interval.

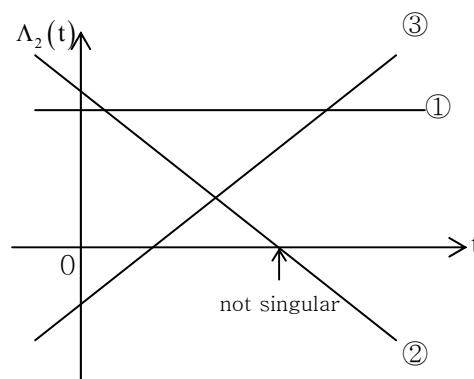
Example 1: $J = \int_{t_0}^{t_f} 1 dt$

$$\dot{x}_1(t) = x_2(t), \quad \dot{x}_2(t) = u(t)$$

$x_1(0), x_2(0)$ are given.

$x_1(t_f), x_2(t_f)$ are given.

$$|u(t)| \leq 1.0$$



$$H = 1 + \Lambda_1 x_2 + \Lambda_2 u$$

$$u = -\text{sign}(\Lambda_2)$$

$$\dot{\Lambda}_1 = -\frac{\partial H}{\partial x_1} = 0$$

$$\dot{\Lambda}_2 = -\frac{\partial H}{\partial x_2} = -\Lambda_1$$

$$\Lambda_2(t) = -\Lambda_1 t + \Lambda_{20}$$

Singular interval cannot occur for this problem. How can we know this problem has singular control or not?

Let Λ_2 be zero for a finite time

$$\Lambda_2 = 0 \rightarrow \dot{\Lambda}_2 = 0 \Rightarrow \Lambda_1 = 0 \rightarrow H(t) = 1 : \text{this is not possible.}$$

In general,

$$H(t_f) = -\frac{\partial h}{\partial t} - \nu^T \frac{\partial m}{\partial t} \quad \text{at } t = t_f$$

$$= 0$$

$$H(t) = \text{const.}$$

$$\Rightarrow H(t) = 0$$

So, Λ_2 cannot be zero for a finite time.

$\Rightarrow$ thus, singular intervals cannot exist.

Example 2: $J = \frac{1}{2} \int_{t_0}^{t_f} x_1^2 dt$

$$\dot{x}_1 = x_2 + u, \quad \dot{x}_2 = -u$$

$$x_1(0), x_2(0), t_f \text{ are given.}$$

$$x_1(t_f), x_2(t_f) \text{ are given}$$

u is unbounded

$$H = \frac{1}{2} x_1^2 + \Lambda_1(x_2 + u) - \Lambda_2 u$$

$$\begin{aligned}\dot{\Lambda}_1(t) &= -\frac{\partial H}{\partial x_1} = -x_1 \\ \dot{\Lambda}_2(t) &= -\frac{\partial H}{\partial x_2} = -\Lambda_1 \\ \frac{\partial H}{\partial u} &= 0 \rightarrow \Lambda_1 - \Lambda_2 = 0\end{aligned}$$

when we cannot determine the control from $\frac{\partial H}{\partial u} = 0$ explicitly, then the control is said to be singular. So, singular intervals exist for this problem.

Assume, $\Lambda_1 - \Lambda_2 = 0$ for a finite time

$$\begin{aligned}\dot{\Lambda}_1 - \dot{\Lambda}_2 &= 0 \rightarrow -x_1 + \Lambda_1 = 0 & \Rightarrow \Lambda_1 = x_1 \\ \ddot{\Lambda}_1 - \ddot{\Lambda}_2 &= 0 \rightarrow -\dot{x}_1 + \dot{\Lambda}_1 = -(x_2 + u) - x_1 = 0 & \Rightarrow u = -(x_1 + x_2)\end{aligned}$$

The singular interval is called a singular arc of second order.

< Discontinuities in the state variables at interior points >

Suppose that with the general optimal programming problem, we have a set of interior boundary conditions,

$$N(\vec{x}(t_1), t_1) = 0 \quad (1)$$

Where t_1 is some intermediate time, $t_0 < t_1 < t_f$, and N is a q -component vector function. We now have a three-point boundary-value problem instead of two-point boundary-value problem. Now, eq(1) represents a set of terminal constraints for the part of the path from t_0 to t_1 .

Example:

$$\begin{aligned}\dot{x} &= u \\ x(0) &= x_0 \quad x(1) = 0 \\ J &= \int_0^1 g(x, u) dt\end{aligned}$$

Derive the necessary conditions for optimality if x is suddenly changed at a time $0 < t_1 < 1$ as follows;

$$x(t_1^+) = x(t_1^-) + 1$$

Is the Hamiltonian continuous or not?

$$J = \int_0^{t_1^-} g(x, u) dt + \int_{t_1^+}^1 g(x, u) dt$$

We can obtain necessary conditions by using Lagrange multiplier functions $\Lambda(t)$ and constant Lagrange multiplier ν .

$$J_a = \nu(x(t_1^+) - x(t_1^-) - 1) + \int_0^{t_1^-} [g(x, u) + \Lambda f - \Lambda \dot{x}] dt + \int_{t_1^+}^1 [g(x, u) + \Lambda f - \Lambda \dot{x}] dt$$

Let's define Hamiltonian $H = g(x, u) + \Lambda f$ and $\Phi \equiv \nu(x(t_1^+) - x(t_1^-) - 1)$

$$\begin{aligned} \delta J_a &= J(x + \delta x, u + \delta u, \Lambda + \delta \Lambda) - J(x, u, \Lambda) \\ &= \Phi[x(t_1^-) + \delta x(t_1^-), x(t_1^+) + \delta x(t_1^+), \delta t_1] - \Phi[x(t_1^-), x(t_1^+), t_1] \\ &\quad + \int_0^{t_1^-} [H(x + \delta x, u + \delta u, \Lambda + \delta \Lambda) - H(x, u, \Lambda) - \{\Lambda(\dot{x} + \delta \dot{x}) + \dot{x} \delta \Lambda - \Lambda \dot{x}\}] dt + [H(t_1^-) - \Lambda \dot{x}(t_1^-)] \delta t_1 \\ &\quad + \int_{t_1^+}^1 [H(x + \delta x, u + \delta u, \Lambda + \delta \Lambda) - H(x, u, \Lambda) - \{\Lambda(\dot{x} + \delta \dot{x}) + \dot{x} \delta \Lambda - \Lambda \dot{x}\}] dt - [H(t_1^+) - \Lambda \dot{x}(t_1^+)] \delta t_1 \\ &= \frac{\partial \Phi}{\partial x(t_1^-)} \delta x_{t_1^-} + \frac{\partial \Phi}{\partial x(t_1^+)} \delta x_{t_1^+} + \frac{\partial \Phi}{\partial t_1} \delta t_1 \\ &\quad + \int_0^{t_1^-} \left[\frac{\partial H}{\partial x} \delta x + \frac{\partial H}{\partial u} \delta u + \frac{\partial H}{\partial \Lambda} \delta \Lambda - \dot{x} \delta \Lambda - \Lambda \delta \dot{x} \right] dt + (H - \Lambda \dot{x})_{t_1^-} \delta t_1 \\ &\quad + \int_{t_1^+}^1 \left[\frac{\partial H}{\partial x} \delta x + \frac{\partial H}{\partial u} \delta u + \frac{\partial H}{\partial \Lambda} \delta \Lambda - \dot{x} \delta \Lambda - \Lambda \delta \dot{x} \right] dt - (H - \Lambda \dot{x})_{t_1^+} \delta t_1 \end{aligned}$$

Using integration by part

$$\begin{aligned}
\delta J_a &= \frac{\partial \Phi}{\partial x(t_1^-)} \delta x_{t_1^-} + \frac{\partial \Phi}{\partial x(t_1^+)} \delta x_{t_1^+} + \frac{\partial \Phi}{\partial t_1} \delta t_1 \\
&\quad - \Lambda \delta x \Big|_0^{t_1^-} + \int_0^{t_1^-} \left[\left(\frac{\partial H}{\partial x} + \dot{\Lambda} \right) \delta x + \frac{\partial H}{\partial u} \delta u + \left(\frac{\partial H}{\partial \Lambda} - \dot{x} \right) \delta \Lambda \right] dt + (H - \Lambda \dot{x})_{t_1^-} \delta t_1 \\
&\quad - \Lambda \delta x \Big|_{t_1^+}^1 + \int_{t_1^+}^1 \left[\left(\frac{\partial H}{\partial x} + \dot{\Lambda} \right) \delta x + \frac{\partial H}{\partial u} \delta u + \left(\frac{\partial H}{\partial \Lambda} - \dot{x} \right) \delta \Lambda \right] dt - (H - \Lambda \dot{x})_{t_1^+} \delta t_1
\end{aligned}$$

Note : $\delta x_{t_1^\pm} = \delta x(t_1^\pm) + \dot{x}(t_1^\pm) \delta t_1 \rightarrow \delta x(t_1^\pm) = \delta x_{t_1^\pm} - \dot{x}(t_1^\pm) \delta t_1$

$$-\Lambda \delta x \Big|_0^{t_1^-} \Rightarrow -[\Lambda \delta x(t_1^-) - \Lambda \delta x(0)] = -\Lambda [\delta x_{t_1^-} - \dot{x}(t_1^-) \delta t_1]$$

$$-\Lambda \delta x \Big|_{t_1^+}^1 \Rightarrow -[\Lambda \delta x(1) - \Lambda \delta x(t_1^+)] = \Lambda [\delta x_{t_1^+} - \dot{x}(t_1^+) \delta t_1]$$

$$\begin{aligned}
\delta J_a &= \left[\frac{\partial \Phi}{\partial t_1} + \Lambda \dot{x}(t_1^-) - \Lambda \dot{x}(t_1^+) + H(t_1^-) - \Lambda \dot{x}(t_1^-) - H(t_1^+) + \Lambda \dot{x}(t_1^+) \right] \delta t_1 \\
&\quad + \left[\frac{\partial \Phi}{\partial x(t_1^-)} - \Lambda(t_1^-) \right] \delta x_{t_1^-} + \left[\frac{\partial \Phi}{\partial x(t_1^+)} + \Lambda(t_1^+) \right] \delta x_{t_1^+} \\
&\quad + \int_0^{t_1^-} \left[\left(\frac{\partial H}{\partial x} + \dot{\Lambda} \right) \delta x + \frac{\partial H}{\partial u} \delta u + \left(\frac{\partial H}{\partial \Lambda} - \dot{x} \right) \delta \Lambda \right] dt + \int_{t_1^+}^1 \left[\left(\frac{\partial H}{\partial x} + \dot{\Lambda} \right) \delta x + \frac{\partial H}{\partial u} \delta u + \left(\frac{\partial H}{\partial \Lambda} - \dot{x} \right) \delta \Lambda \right] dt
\end{aligned}$$

$\delta J = 0$ by the fundamental theorem of Calculus of variation

We obtain the necessary conditions

$$\frac{\partial \Phi}{\partial t_1} + H(t_1^-) = H(t_1^+) \Rightarrow \text{if } \frac{\partial \Phi}{\partial t_1} = 0 \rightarrow H(t_1^-) = H(t_1^+)$$

$$\Lambda(t_1^-) = \frac{\partial \Phi}{\partial x(t_1^-)}$$

$$\Lambda(t_1^+) = -\frac{\partial \Phi}{\partial x(t_1^+)}$$

$$\dot{\Lambda} = -\frac{\partial H}{\partial x}$$

$$\frac{\partial H}{\partial u} = 0$$

$$\dot{x} = \frac{\partial H}{\partial \Lambda} = f$$

< Interior-point Constraints >

In some problems there will be discontinuities in the state variables as well as discontinuities of the system equations at interior points.

Furthermore, the performance index and the constraints may be functions of the state and/or time at several discrete points. A general class of such problems involves choosing $u(t)$ to minimize

$$J = \phi \left[x(t_0^-), x(t_0^+), \dots, x(t_N^-), x(t_N^+); t_0, \dots, t_N \right] + \sum_{i=1}^N \int_{t_{i-1}^+}^{t_i^-} L^{(i)}(x(t), u(t), t) dt \quad (1)$$

subject to constraints

$$\dot{x} = f^{(i)}(x(t), u(t), t) \quad , \quad t_{i-1} < t < t_i \quad i = 1, 2, \dots, N \quad (2)$$

$$\psi^{(j)} \left[x(t_0^-), x(t_0^+), \dots, x(t_N^-), x(t_N^+); t_0, \dots, t_N \right] = 0 \quad j=0, 1, 2, \dots, N \quad (3)$$

Where $x(t_i^-)$ signifies the state vector just before $t=t_i$ and $x(t_i^+)$ signifies the state vector just after $t=t_i$.

Necessary conditions for a minimum are derived by adjoining equation (2) and equation (3) to equation (1) by Lagrange multiplier function $\lambda(t)$ and constant Lagrange multiplier $\nu^{(i)}$ respectively

$$J_a = \phi + \sum_{j=0}^N \left[\nu^{(j)} \right]^T \psi^{(j)} + \sum_{i=1}^N \int_{t_{i-1}^+}^{t_i^-} \left[L^{(i)} + \lambda^T f^{(i)} - \lambda^T \dot{x} \right] dt \quad (4)$$

$$\text{Define } \Phi = \phi + \sum_{j=0}^N \left[\nu^{(j)} \right]^T \psi^{(j)} \quad (5)$$

$$H^{(i)} \equiv L^{(i)} + \lambda^T f^{(i)} \quad i = 1, 2, \dots, N \quad (6)$$

The first variation of equation (4), using the usual integration by part, is then

$$\begin{aligned}
\delta J_a = & \sum_{i=0}^N \left[\frac{\partial \Phi}{\partial t_i} \delta t_i + \frac{\partial \Phi}{\partial x(t_i^-)} \delta x_{t_i}^- + \frac{\partial \Phi}{\partial x(t_i^+)} \delta x_{t_i}^+ \right] \\
& + \sum_{i=1}^N \left\{ \left[H^{(i)} - \lambda^T \dot{x} \right]_{t=t_i^-} \delta t_i - \left[H^{(i+1)} - \lambda^T \dot{x} \right]_{t=t_i^+} \delta t_i \right\} \\
& + \sum_{i=1}^N \left\{ \left[-\lambda \delta x \right]_{t_i^+}^{t_i^-} + \int_{t_{i-1}^+}^{t_i^-} \left[\left(\dot{\lambda}^T + \frac{\partial H^{(i)}}{\partial x} \right) \delta x + \frac{\partial H^{(i)}}{\partial u} \delta u + \left(\frac{\partial H^{(i)}}{\partial \Lambda} - \dot{x} \right) \delta \Lambda \right] dt \right\} \\
= & 0
\end{aligned} \tag{7}$$

Eliminating $\delta x(t_i^\pm)$ from equation (7) using the relations

$$\delta x_{t_i}^\pm = \delta x(t_i^\pm) + \dot{x}(t_i^\pm) \delta t_i, \quad \delta x(t_i^\pm) = \delta x_{t_i}^\pm - \dot{x}(t_i^\pm) \delta t_i \tag{8}$$

We obtain by regrouping

$$\begin{aligned}
\delta J_a = & \sum_{i=0}^N \left[\frac{\partial \Phi}{\partial t_i} + H^{(i)}(t_i^-) - H^{(i+1)}(t_i^+) + \Lambda \dot{x}(t_i^-) - \Lambda \dot{x}(t_i^+) - \Lambda \dot{x}(t_i^-) + \Lambda \dot{x}(t_i^+) \right] \delta t_i \\
& + \sum_{i=1}^N \left\{ \left[\frac{\partial \Phi}{\partial x(t_i^-)} - \lambda^T(t_i^-) \right] \delta x_{t_i}^- \right\} + \sum_{i=0}^{N-1} \left\{ \left[\frac{\partial \Phi}{\partial x(t_i^+)} + \lambda^T(t_i^+) \right] \delta x_{t_i}^+ \right\} \\
& + \sum_{i=1}^N \int_{t_{i-1}^+}^{t_i^-} \left[\left(\dot{\lambda}^T + \frac{\partial H^{(i)}}{\partial x} \right) \delta x + \frac{\partial H^{(i)}}{\partial u} \delta u + \left(\frac{\partial H^{(i)}}{\partial \Lambda} - \dot{x} \right) \delta \Lambda \right] dt = 0
\end{aligned} \tag{9}$$

where it is to be noted that $H^{(0)} = H^{(N+1)} \equiv 0$

We obtain the following necessary conditions

$$\dot{\lambda}^T = -\frac{\partial H^{(i)}}{\partial x}, \quad , \quad t_{i-1}^+ < t < t_i^- \quad , \quad i=1, \dots, N \tag{10}$$

$$\lambda^T(t_i^-) = \frac{\partial \Phi}{\partial x(t_i^-)}, \quad , \quad i = 1, \dots, N \tag{11}$$

$$\lambda^T(t_i^+) = -\frac{\partial \Phi}{\partial x(t_i^+)}, \quad , \quad i = 0, \dots, N-1 \tag{12}$$

And let us choice $t_i (i = 0, \dots, N)$ to satisfy (transvrsality conditions)

$$\frac{\partial \Phi}{\partial t_i} + H^{(i)}(t_i^-) - H^{(i+1)}(t_i^+) = 0 \quad , \quad i = 0, \dots, N \quad (13)$$

If t_i is specified, the corresponding relation in equation (13) is not necessary since $\delta t_i = 0$ in equation (9). Similarly, if $x(t_0^+)$ is specified, we have $\delta x_0^+ = 0$ in equation (9) and equation (12), with $i = 0$ is not necessary.

We must also choose $\nu^{(j)}$ to satisfy the conditions $\psi^{(j)} = 0$ in equation (13).

Again, an optimality condition must be made to justify setting

$$\frac{\partial H^{(i)}}{\partial u} = 0 \quad (14)$$

in equation (9) since $\delta u(t)$ is arbitrary but must produce δx_i^- , δx_i^+ , δt_i consistent with

$$\delta \psi^{(j)} = \sum_{i=0}^N \left[\frac{\partial \psi^{(j)}}{\partial t_i} \delta t_i + \frac{\partial \psi^{(j)}}{\partial x(t_i^-)} \delta x_i^- + \frac{\partial \psi^{(j)}}{\partial x(t_i^+)} \delta x_i^+ \right] = 0 \quad , \quad j = 0, \dots, N \quad (15)$$

Equations (10), (11), (12) and (14) are necessary conditions and equation (13) is a set of transversality conditions.

< Hamilton's principle in Mechanics >

The motion of a conservative system, from time t_0 to t_f is such that the integral

$$J = \int_{t_0}^{t_f} L(u, q) dt \quad (1)$$

has a stationary value(=Hamiltonian principle)

where $L = T(u, q) - V(q)$ = the Lagrangian of the system

T = kinetic energy of the system

V = potential energy of the system

q = generalized coordinate vector (state of the system)

$\dot{q} = u$ = generalized velocity vector.

Optimal control theory

The Hamiltonian: $H = L + \Lambda^T f \quad (2)$

Consequently, the Euler-Lagrange equations are

$$\dot{\Lambda}^T = -\frac{\partial H}{\partial q} = -\frac{\partial L}{\partial q} \quad (3)$$

$$0 = \frac{\partial H}{\partial u} = \frac{\partial L}{\partial u} + \Lambda^T \quad (4)$$

Nature chooses $\dot{q}(=u)$ as a control value.

combining these last two vector equations

$$0 = \frac{d}{dt} \left(\frac{\partial L}{\partial u} \right) + \frac{d}{dt} (\Lambda^T) \quad (5)$$

$$\frac{d}{dt} \left(\frac{\partial L}{\partial \dot{q}} \right) - \frac{\partial L}{\partial q} = 0 \quad (6)$$

Lagrange equations of motion for a conservation system.

if L is not explicit function of time, the Hamiltonian is constant.

$$\begin{aligned}
 H &= L + A^T u \\
 &= L - \frac{\partial L}{\partial u} u \\
 &= T - V - \frac{\partial L}{\partial u} u \\
 &= \text{const.}
 \end{aligned}$$

T is a homogeneous quadratic form in u ($T = \frac{1}{2}mv^2 = \frac{1}{2}m\dot{q}^2 = \frac{1}{2}mu^2$)

$$\frac{\partial T}{\partial u} u = 2T$$

Hence, we have

$$\begin{aligned}
 H &= T - V - 2T \\
 &= -(T + V) \\
 &= \text{const.}
 \end{aligned}$$

That is, the kinetic energy plus potential energy is constant during the time.

6. Numerical Determination of Optimal Control

In problems with linear plant dynamics and quadratic performance (linear regulator and linear tracking systems), it was found that it is possible to obtain the optimal control law by numerically integrating a matrix differential equation of the Riccati type. Optimal control laws were also determined for several other simple examples by applying Pontryagin's minimum principle.

In general, however, the variational approach leads to a nonlinear two-point boundary-value problem (TPBVP) that cannot be solved analytically to obtain the optimal control law, or even an optimal open-loop control.

Assuming that the state and control variables are not constrained by any boundaries, that the final time t_f is fixed, and that $\mathbf{x}(t_f)$ is free.

$$\dot{\mathbf{x}}(t) = \frac{\partial H}{\partial \boldsymbol{\lambda}} \quad (1)$$

$$\dot{\boldsymbol{\lambda}}(t) = -\frac{\partial H}{\partial \mathbf{x}} \quad (2)$$

$$\mathbf{0} = \frac{\partial H}{\partial \mathbf{u}} \quad (3)$$

$$\mathbf{x}(t_0) = \mathbf{x}_0 \quad (4)$$

$$\boldsymbol{\lambda}(t_f) = \frac{\partial h}{\partial \mathbf{x}(t_f)} \quad (5)$$

An initial guess is used to obtain the solutions to a problem in which one or more of the five necessary and boundary conditions (1) through (5) is not satisfied. This solution is then used to adjust the initial guess in an attempt to make the next solution come closer to satisfying all of the conditions. If these steps are repeated and the iterative procedure converges, the conditions (1) through (5) will eventually be satisfied.

There are two main categories for numerical method to solve the TPBVP.

1) Direct method

Guess control $u(t)$

$$\dot{\mathbf{x}}(t) = \frac{\partial H}{\partial \boldsymbol{\lambda}}$$

$$\dot{\boldsymbol{\lambda}}(t) = -\frac{\partial H}{\partial \mathbf{x}}$$

Modify control: $\mathbf{u}^{k+1}(t) = \mathbf{u}^k(t) + \alpha \mathbf{u}(t)$

2) Indirect method

Guess initial costates $\boldsymbol{\lambda}(t_0)$

$$\dot{\mathbf{x}}(t) = \frac{\partial H}{\partial \boldsymbol{\lambda}}$$

$$\dot{\boldsymbol{\lambda}}(t) = -\frac{\partial H}{\partial \mathbf{x}}$$

$$\mathbf{0} = \frac{\partial H}{\partial \mathbf{u}} \text{ or Minimum principle}$$

Update initial costates $\boldsymbol{\lambda}^{k+1}(t_0) = \boldsymbol{\lambda}^k(t_0) + \delta \boldsymbol{\lambda}(t_0)$ by using sensitivity matrix at final time.

< Function Minimization >

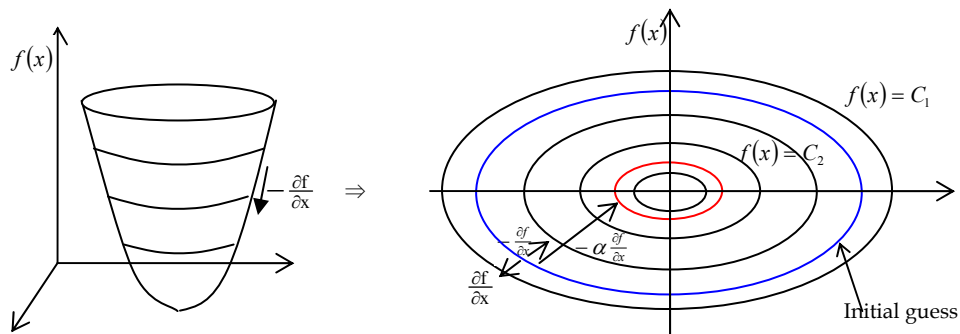
1) 1st order Gradient method (the Steepest Descent Algorithms)

$$\text{Min } f(x)$$

$$x^{k+1} = x^k - \alpha \frac{\partial f}{\partial x}$$

$\alpha > 0$; step size, α is determined using one-dimension line search

$\frac{\partial f}{\partial x}$; gradient(Jacobian)



If $df(x) = \frac{\partial f}{\partial x} \Delta x = 0$, then f has a minimum

$$\left. \frac{\partial f}{\partial x} \right|_{x=x^*} = 0 \text{ at a relative minimum}$$

Convergence is guaranteed if $\alpha \approx 0$ (many iterations)

If function is a particular (so nonlinear) $\rightarrow$ oscillation

2) 2nd order Gradient method (Gauss-Newton)

$$f(x^k) \approx f(x^*) + \underbrace{\left. \frac{\partial f}{\partial x} \right|_{x=x^k}}_{G^k, \text{Jacobian}} \Delta x + \frac{1}{2} \Delta x^2 \underbrace{\left. \frac{\partial^2 f}{\partial x^2} \right|_{x=x^k}}_{H^k, \text{Hessian}}$$

x^k ; current guess

$$\min f(x) \text{ with respect to } \Delta x \quad (\Rightarrow \frac{\partial f}{\partial \Delta x} = 0)$$

$$\Rightarrow \min [G^k \Delta x + \frac{1}{2} \Delta x^T H^k \Delta x]$$

$$\frac{\partial [G^k \Delta x + \frac{1}{2} \Delta x^T H^k \Delta x]}{\partial \Delta x}$$

$$\rightarrow G^k + H^k \Delta x = 0$$

$$\rightarrow \Delta x = -[H^k]^{-1} G^k$$

$$\therefore \Delta x^{k+1} = x^k - \alpha [H^k]^{-1} G^k \quad \text{step size } \alpha > 0$$

To see the descent property of this method, look at the convection to the formation.

$$\Delta f \approx \frac{\partial f}{\partial x} \Delta x = -G^k [H^k]^{-1} G^k < 0$$

if H^k is positive definite.

$\Leftrightarrow \Delta f$ is negative

$\rightarrow f(x)$ is decreased by each step.

The Hessian has to be positive definite.

Problems 1) for many nonlinear problems, H may not be positive definite.

2) H is expensive to evaluate. $\rightarrow$ Use Quasi-Newton method
(approximate the inverse of H directly)

< Simplest Optimal Control Problem (Direct method) >

$$J = h(\vec{x}(t_f), t_f) + \int_0^{t_f} g(\vec{x}, \vec{u}, t) dt$$

$\vec{x}(0)$ given, $\vec{x}(t_f)$ free, t_f is fixed

$$\begin{aligned} \delta J_a = & \left[\frac{\partial h}{\partial \vec{x}(t_f)} - \bar{\Lambda}^T(t_f) \right] \delta \vec{x}(t_f) \\ & + \int_0^{t_f} \left[\left(\dot{\bar{\Lambda}}^T + \frac{\partial H}{\partial \vec{x}} \right)^T \delta \vec{x} + \left(\frac{\partial H}{\partial \vec{u}} \right)^T \delta \vec{u} + \left(\frac{\partial H}{\partial \bar{\Lambda}} - \dot{\vec{x}} \right)^T \delta \bar{\Lambda} \right] dt \end{aligned}$$

Numerical procedure

- 1) Guess a time history for $\vec{u}(t)$
- 2) Integrate the state equations forward in time.

$$\dot{\vec{x}} = \vec{f}(\vec{x}, \vec{u}, t)$$

- 3) Note down the value of $\vec{x}(t)$
- 4) Integrate the costate equations backward in time.

$$\dot{\bar{\Lambda}}^T(t_f) = -\frac{\partial H}{\partial \vec{x}} \quad \text{with} \quad \bar{\Lambda}^T(t_f) = \frac{\partial h}{\partial \vec{x}(t_f)}$$

note down the value of $\frac{\partial H}{\partial \vec{u}}$ and evaluate $\left\| \frac{\partial H}{\partial \vec{u}}(t) \right\|$. If

$\left\| \frac{\partial H}{\partial \vec{u}}(t) \right\| < \gamma$ (preselected positive constant), terminate the iteration procedure

- 5) Compute $\delta \vec{u}^k(t) = -\alpha \frac{\partial H}{\partial \vec{u}}(t) \rightarrow$ means $\delta \vec{u}^k(t)$ makes H go to be minimized with respect to $\vec{u}$. ($\frac{\partial H}{\partial \vec{u}}(t)$: direction of increasing H by $\delta \vec{u}^k(t)$),

$$\text{Note : } \delta J_a = \int_0^{t_f} \frac{\partial H}{\partial \vec{u}}(t) \delta \vec{u}^k(t) dt = \int_0^{t_f} \frac{\partial H}{\partial \vec{u}}(t) [-\alpha \frac{\partial H}{\partial \vec{u}}(t)] dt < 0$$

- 6) Update $\vec{u}(t)^{k+1} = \vec{u}(t)^k + \delta \vec{u}^k(t)$, α ; step size
- 7) Go to step 2)

< Optimization with terminal constraint specified at t_f >

Using a Gradient Method

$$J = \phi(x(t_f)) + \int_{t_0}^{t_f} L(x, u, t) dt$$

where t_f is specified

$$\dot{x} = f(x, u, t), \quad x(t_0) = x_0$$

$$\psi(x(t_f)) = 0 \quad ; \text{ terminal constraint}$$

Necessary conditions for a stationary path

Hamiltonian

$$H = L + \Lambda^T f$$

$$\Phi = \phi(x(t_f)) + v^T \Psi(x(t_f))$$

$$\dot{\Lambda} = -\frac{\partial H}{\partial x} = -\frac{\partial L}{\partial x} - \frac{\partial f^T}{\partial x} \Lambda$$

$$\Lambda(t_f) = \frac{\partial \Phi}{\partial x(t_f)} = \left[\frac{\partial \phi}{\partial x} + v^T \frac{\partial \Psi}{\partial x} \right]_{t=t_f}$$

$$\frac{\partial H}{\partial u} = \frac{\partial L}{\partial u} + \Lambda^T \frac{\partial f}{\partial u} = 0$$

This forms a TPBVP for $x(t)$ and $\Lambda(t)$ where we must also find v to satisfy the terminal constraints $\psi(t_f) = 0$

$\psi(x(t_f))$: $q \times 1$ column matrix of terminal conditions

v : $q \times 1$ column matrix of constants

$\dot{x}$: n first order differential equations of motion

Λ : $n \times 1$ column matrix of costates

u : $m \times 1$ column matrix of controls

A modified (augmented) performance index.

$$J_a = \phi(x(t_f)) + v^T \psi(x(t_f)) + \int_{t_0}^{t_f} [L(x, u) + \Lambda^T (f - \dot{x})] dt$$

$$\begin{aligned} \delta J_a &= \frac{\partial \phi}{\partial x_f} \delta x_f + v^T \frac{\partial \psi}{\partial x_f} \delta x_f \\ &+ \int_{t_0}^{t_f} \left[\left(\frac{\partial L}{\partial x} + \Lambda^T \frac{\partial f}{\partial x} \right) \delta x + \left(\frac{\partial L}{\partial u} + \Lambda^T \frac{\partial f}{\partial u} \right) \delta u - \Lambda^T \delta \dot{x} + \delta \Lambda (f - \dot{x}) \right] dt \end{aligned}$$

$$\begin{aligned} \delta J_a &= \left[\frac{\partial \phi}{\partial x_f} + v^T \frac{\partial \psi}{\partial x_f} - \Lambda(t_f) \right] \delta x_f \\ &+ \int_{t_0}^{t_f} \left[\left(\frac{\partial L}{\partial x} + \Lambda^T \frac{\partial f}{\partial x} + \dot{\Lambda}^T \right) \delta x + \left(\frac{\partial L}{\partial u} + \Lambda^T \frac{\partial f}{\partial u} \right) \delta u + \delta \Lambda (f - \dot{x}) \right] dt \end{aligned}$$

$$\begin{aligned} \phi &\rightarrow \Lambda^\phi \\ \psi &\rightarrow \Lambda^\psi \end{aligned}$$

$$\begin{aligned} \Phi(t_f) &= \phi(x(t_f)) + v^T \psi(x(t_f)) \\ &\rightarrow \Lambda = \Lambda^\phi + v^T \Lambda^\psi \end{aligned}$$

$$\begin{aligned} \delta J_a &= \left[\frac{\partial \phi}{\partial x_f} + v^T \frac{\partial \psi}{\partial x_f} - \Lambda^\phi(t_f) - v^T \Lambda^\psi(t_f) \right] \delta x_f \\ &+ \int_{t_0}^{t_f} \left[\left(\frac{\partial L}{\partial x} + (\Lambda^\phi + v^T \Lambda^\psi) \frac{\partial f}{\partial x} + (\dot{\Lambda}^\phi + v^T \dot{\Lambda}^\psi)^T \right) \delta x \right. \\ &\quad \left. + \left(\frac{\partial L}{\partial u} + (\Lambda^\phi + v^T \Lambda^\psi)^T \frac{\partial f}{\partial u} \right) \delta u + \delta \Lambda (f - \dot{x}) \right] dt \end{aligned}$$

$$\text{Let } \Lambda^\phi(t_f) = \frac{\partial \phi}{\partial x_f}, \quad \Lambda^\psi(t_f) = \frac{\partial \psi}{\partial x_f} \quad \Rightarrow \quad \text{the terms of } \delta x_f \text{ goes to zero.}$$

$$\text{Let } \dot{\Lambda}^\phi(t) = -\frac{\partial L}{\partial x} - \Lambda^\phi \frac{\partial f}{\partial x}, \quad \dot{\Lambda}^\psi(t) = -\Lambda^\psi \frac{\partial f}{\partial x} \quad \Rightarrow \quad \text{the terms of } \delta x \text{ goes to zero.}$$

$$\text{Let } \dot{x} = f \quad \Rightarrow \quad \text{the terms of } \delta \Lambda \text{ goes to zero.}$$

$$\begin{aligned}
\Rightarrow \delta J_a &= \int_{t_0}^{t_f} \left(\frac{\partial L}{\partial u} + \Lambda^\phi \frac{\partial f}{\partial u} + v^T \Lambda^\psi \frac{\partial f}{\partial u} \right) \delta u dt \\
&\equiv \delta J + v^T \delta \psi \\
\text{where } \delta J &\equiv \int_{t_0}^{t_f} \left(\frac{\partial L}{\partial u} + \Lambda^\phi \frac{\partial f}{\partial u} \right) \delta u dt, \quad \delta \psi \equiv \int_{t_0}^{t_f} \left(\Lambda^\psi \frac{\partial f}{\partial u} \right) \delta u dt
\end{aligned}$$

Let's define $H_u^\phi \equiv L_u + \Lambda^\phi f_u$, $H_u^\psi \equiv \Lambda^\psi f_u$

$$\delta J_a = \int_{t_0}^{t_f} (H_u^\phi + v^T H_u^\psi) \delta u dt$$

For every iteration, δJ_a should be less than zero ($\delta J_a < 0$) $\rightarrow$ decreasing J

Thus, δu is chosen to be

$$\delta u = -k [H_u^\phi + v^T H_u^\psi]$$

$$\text{Note : } \delta J_a = \int_{t_0}^{t_f} (H_u^\phi + v^T H_u^\psi) \delta u dt = -k \int_{t_0}^{t_f} (H_u^\phi + v^T H_u^\psi)^T (H_u^\phi + v^T H_u^\psi) dt < 0$$

Then,

$$\begin{aligned}
\delta \psi &= \int_{t_0}^{t_f} \left(\Lambda^\psi \frac{\partial f}{\partial u} \delta u \right) dt = -k \int_{t_0}^{t_f} \left(\Lambda^\psi \frac{\partial f}{\partial u} (H_u^\phi + v^T H_u^\psi) \right) dt \\
&= -k \int_{t_0}^{t_f} ((H_u^\phi + v^T H_u^\psi)^T H_u^\psi) dt = -k(g + vQ)
\end{aligned}$$

$$\text{Let } g \equiv \int_{t_0}^{t_f} [H_u^\phi]^T H_u^\psi dt \quad (q \times 1)$$

$$Q \equiv \int_{t_0}^{t_f} [H_u^\psi]^T H_u^\psi dt \quad (q \times q)$$

We want to derive $\delta \psi$ to zero until the procedure is converged.

$$\begin{aligned}
\delta \psi &\approx -\eta \psi \\
-\eta \psi &= -k(g + vQ) \\
v &= \left(-\frac{g}{Q} + \frac{\eta \psi}{kQ} \right)
\end{aligned}$$

$$\begin{aligned}
\delta u &= -k \left[H_u^\phi + \left(-\frac{g}{Q} + \frac{\eta \psi}{kQ} \right) H_u^\psi \right] \\
&= -k \left[H_u^\phi - \frac{g}{Q} H_u^\psi \right] - k \frac{\eta \psi}{kQ} H_u^\psi \\
\therefore \delta u &= -k \left[H_u^\phi + \nu' H_u^\psi \right] - \eta \left(H_u^\psi \right)^T Q^{-1} \psi \\
&\quad \text{where } \nu' = -\frac{g}{Q}
\end{aligned}$$

A Gradient Algorithm (a Direct method)

- 1) Guess $\bar{u}(t)$
- 2) Find $\bar{x}(t)$ by forward integration
- 3) Evaluate $\phi, \psi, \frac{\partial \phi}{\partial x}, \frac{\partial \psi}{\partial x}$
- 4) Find H_u^ϕ, H_u^ψ backward integration.
- 5) Determine $\nu' = -Q^{-1}g$
- 6) $\delta u(t) = -k \left[H_u^\phi(t) + \nu' H_u^\psi(t) \right] - \eta \left[H_u^\psi(t) \right]^T Q^{-1} \psi$
- 7) If $\|\delta u(t)\| < \varepsilon$, then stop
- 8) $u(t) = u(t) + \delta u(t)$
- 9) Go to step 2).

Example:

$$\dot{x}_1 = x_2$$

$$\dot{x}_2 = -x_1 + x_1^3 + u, \quad x_1(0) = 1, \quad x_2(0) = 1$$

$$J = \frac{1}{2} [x_1^2(1) + x_2^2(1)] + \frac{1}{2} \int_0^1 (x_1^2 + x_2^2 + u^2) dt$$

The Hamiltonian

$$H = \frac{1}{2} (x_1^2 + x_2^2 + u^2) + \lambda_1 x_2 + \lambda_2 (-x_1 + x_1^3 + u)$$

The costate equation

$$\dot{\lambda}_1 = -\frac{\partial H}{\partial x_1} = -x_1 + \lambda_2 (1 - 3x_1^2)$$

$$\dot{\lambda}_2 = -\frac{\partial H}{\partial x_2} = -x_2 - \lambda_1$$

$$\lambda_1(1) = \frac{\partial h}{\partial x_1(1)} = x_1(1), \quad \lambda_2(1) = \frac{\partial h}{\partial x_2(1)} = x_2(1)$$

$$\frac{\partial H}{\partial u} = u + \lambda_2$$

Gradient method for Direct strategy

- 1) Guess the control $\vec{u}(t)$
- 2) Integrate the state equation from $t = 0$ to $t = 1$.

And store the resulting $x_1(t), x_2(t)$

- 3) Calculate $\lambda_1(1), \lambda_2(1)$ as the substituting $x_1(1), x_2(1)$ from step 2).

Using $\lambda_1(1), \lambda_2(1)$ as the initial conditions and $x_1(t), x_2(t)$,

integrate the costate equations backward on time. And evaluate

$$\frac{\partial H}{\partial u} = u + \lambda_2 \text{ and store this value.}$$

- 4) If $\left| \frac{\partial H}{\partial u} \right| \geq \varepsilon$ (tolerance)

$$\text{Let } u^{\text{new}}(t) = u^{\text{old}}(t) - \alpha \frac{\partial H}{\partial u}(t)$$

- 5) Repeat above procedure 2) ~ 4) until $\left| \frac{\partial H}{\partial u} \right| \leq \varepsilon$ (tolerance)

< A Gradient Algorithm for Parameter Optimization with Equality Constraint >

Function minimization

$$\begin{aligned} \text{Min } f(x) \\ \text{with the constraint } L(x) = 0 \end{aligned}$$

$$f(x + \Delta x) \sim f(x) + \frac{\partial f}{\partial x} \Delta x + \frac{1}{2} (\Delta x)^T \frac{\partial^2 f}{\partial x^2} \Delta x$$

$$f(x + \Delta x) - f(x) = \Delta f(x) = f_x \Delta x + \frac{1}{2} (\Delta x)^T f_{xx} \Delta x$$

$$\Rightarrow \text{Min } \Delta f(x) = f_x \Delta x + \frac{(\Delta x)^T \Delta x}{2k} \quad (1)$$

$$\text{subject to } dL = L_x \Delta x = -\eta L \quad (2)$$

when k, η are small constant.

To solve this linear-quadratic problem, we adjoin the constraint (2) to the performance index with a Lagrange multiplier vector α .

$$\Rightarrow \text{Min } \Delta f(x) = f_x \Delta x + \frac{(\Delta x)^T \Delta x}{2k} + \alpha^T (\eta L + L_x \Delta x) \quad (3)$$

For $d(\Delta f) = 0$ with arbitrary $d(\Delta x)$, it is clearly necessary that

$$f_x + \frac{\Delta x}{k} + \alpha^T L_x = 0 \quad (4)$$

$$\rightarrow \Delta x = -k(f_x + \alpha^T L_x) \quad (5)$$

Substituting equation (5) into (2), we may solve for α^T

$$L_x \left[-k(f_x + \alpha^T L_x) \right] = -\eta L$$

$$\alpha^T = \frac{\eta L}{L_x k L_x^T} - \frac{f_x}{L_x}$$

$$\begin{aligned}
&= -\underbrace{\frac{f_x}{L_x}}_{\equiv \Lambda} + \frac{\eta L}{k} \underbrace{\frac{1}{L_x L_x^T}}_{\equiv Q} \\
&= \Lambda + \frac{\eta}{k} L Q^{-1}
\end{aligned} \tag{6}$$

$$\begin{aligned}
\text{let } L_x^* &= \frac{1}{L_x}, & \Lambda &= -f_x L_x^* \\
Q &= L_x L_x^T, & L_x^* &= L_x^T Q^{-1}
\end{aligned} \tag{7}$$

Substituting equation (6) and equation (7) into equation (5) gives

$$\begin{aligned}
(\Delta x)^T &= -k \left[f_x + \left(\Lambda + \frac{\eta}{k} L Q^{-1} \right) L_x \right] \\
&= -k \underbrace{(f_x + \Lambda L_x)}_{\equiv H_x} - \eta \underbrace{L Q^{-1} L_x}_{L_x^*} \\
&= -k H_x - \eta L L_x^*
\end{aligned} \tag{8}$$

$$\text{where } H \equiv f + \Lambda^T L, \quad H_x \equiv f_x + \Lambda^T L_x$$

When starting with a possibly poor guess, one can put $k = 0$ and pick $\eta < 1$ to limit the size of Δx , so that the linear approximation is reasonable. Over a few steps, gradually increase η to 1, and a feasible solution ($L \sim 0$) should be obtained, after which $\eta = 1$ can be used with $k \neq 0$ to minimize f .

A straight forward numerical method is to guess an initial x and then take small steps Δx in the direction of $-H_x^T =$ the direction of steepest descent. This will lead to a local minimum if one exists. If there are several local minima, this method may not find the global minimum. Choosing the step size requires some experience. If the steps are too large, the minimum may be overshoot, while if the step size are too small, it takes many steps to reach the minimum.

< Optimization with terminal constraint at open final time >
(Numerical solution with Gradient method)

$$\text{Minimize } J = \phi(x(t_f), t_f) + \int_0^{t_f} L(x, u, t) dt \quad (1)$$

$$\dot{x} = f(x, u), \quad x(t_0) = x_0$$

$$\psi(x(t_f), t_f) = 0 \quad t_f \text{ is free}$$

$$\begin{aligned} J_a &= \phi + v^T \psi + \int_0^{t_f} [L + \Lambda^T (f - \dot{x})] dt \\ &= \phi + v^T \psi + \int_0^{t_f} [H - \Lambda^T \dot{x}] dt \end{aligned}$$

$$\text{where } H = L + \Lambda^T f$$

Remember, for t_f is fixed

$$\begin{aligned} \delta J_a &= \delta J_1 + v^T \delta \psi \\ \text{where } \delta J_1 &= \int_{t_0}^{t_f} H_u^\phi \delta u dt, \quad \delta \psi = \int_{t_0}^{t_f} H_u^\psi \delta u dt \end{aligned}$$

For t_f is free, holding the same notation

$$\begin{aligned} \Delta J_1 &= \dot{\phi} \Delta t_f + \int_{t_0}^{t_f} H_u^\phi \Delta u dt + \frac{1}{2k} \left[\int_{t_0}^{t_f} (\Delta u)^T \Delta u dt + (\Delta t_f)^2 \right] \\ \Delta \psi &= \dot{\psi} \Delta t_f + \int_{t_0}^{t_f} H_u^\psi \Delta u dt \end{aligned}$$

This algorithm may be interpreted as finding a small variation in the control function $\Delta u(t)$, and a small changes in Δt_f to minimize a linear approximation of ΔJ subject to a quadratic penalty on $\Delta u(t)$ and on Δt_f and to a linear approximation to coming closer to the terminal constant $\psi = 0$,

namely

$$\min_{\Delta u(t), \Delta t_f} \Delta J \approx \dot{\phi} \Delta t_f + \int_{t_0}^{t_f} H_u^\phi \Delta u dt + \frac{1}{2k} \left[\int_{t_0}^{t_f} (\Delta u)^T \Delta u dt + (\Delta t_f)^2 \right] \quad (3)$$

$$\text{subject to } \Delta \psi = \dot{\psi} \Delta t_f + \int_{t_0}^{t_f} H_u^\psi \Delta u dt = -\eta \psi \quad (4)$$

where all the coefficient are evaluated at the current estimate of u and t_f and $0 \leq \eta \leq 1$. k is a scalar step-size parameter. To solve this LQ problem, we adjoin the constraint (4) to the performance index (3) with a Lagrange multiplier vector $\bar{v}$

$$\Delta \bar{J} \equiv \Delta J + \bar{v}^T \left(\eta \psi + \dot{\psi} \Delta t_f + \int_{t_0}^{t_f} H_u^\psi \Delta u dt \right) \quad (5)$$

Now take a differential of $\Delta \bar{J}$

$$\begin{aligned} \Delta \bar{J} &\equiv \Delta \bar{J}(\Delta u, \Delta t_f) \\ d(\Delta \bar{J}) &= \frac{\partial(\Delta \bar{J})}{\partial(\Delta t_f)} d(\Delta t_f) + \frac{\partial(\Delta \bar{J})}{\partial(\Delta u)} d(\Delta u) \\ &= \left(\dot{\phi} + \bar{v}^T \dot{\psi} + \frac{\Delta t_f}{k} \right) d(\Delta t_f) + \int_{t_0}^{t_f} \left[H_u^\phi + \bar{v} H_u^\psi + \frac{\Delta u^T}{k} \right] d(\Delta u) dt \quad (6) \\ &= 0 \end{aligned}$$

For a stationary solution, the coefficient of $d(\Delta t_f)$ and $d(\Delta u)$ must vanish, then,

$$\Delta u = -k \left(H_u^\phi + \bar{v}^T H_u^\psi \right)^T \quad (7)$$

$$\Delta t_f = -k \left(\dot{\phi} + \bar{v}^T \dot{\psi} \right) \quad (8)$$

Substituting equations (7) and (8) into equation (4) we may solve for $\bar{v}$

$$\begin{aligned}
& -k\dot{\psi}(\dot{\phi} + \bar{v}^T \dot{\psi}) - \int_{t_0}^{t_f} \left[H_u^\psi k (H_u^\phi + \bar{v}^T H_u^\psi) \right] dt = -\eta \psi \\
& k\dot{\psi}\dot{\phi} + k\dot{\psi}\bar{v}^T \dot{\psi} + \int_{t_0}^{t_f} \left[kH_u^\psi H_u^\phi + kH_u^\psi \bar{v}^T H_u^\psi \right] dt = \eta \psi \\
& k\dot{\psi}\dot{\phi} + k\dot{\psi}\bar{v}^T \dot{\psi} + k \int_{t_0}^{t_f} H_u^\psi H_u^\phi dt + k\bar{v}^T \int_{t_0}^{t_f} H_u^\psi H_u^\psi dt = \eta \psi \\
& \rightarrow k\bar{v}^T \underbrace{\left(\dot{\psi}\dot{\psi} + \int_{t_0}^{t_f} H_u^\psi H_u^\psi dt \right)}_Q + k \underbrace{\left(\dot{\psi}\dot{\phi} + \int_{t_0}^{t_f} H_u^\psi H_u^\phi dt \right)}_{=g} = \eta \psi \\
& \rightarrow \bar{v}^T = \frac{-kg + \eta \psi}{kQ} = \frac{-g}{Q} + \frac{\eta Q^{-1} \psi}{k} \\
& \quad = \frac{-g}{Q} + \frac{\eta Q^{-1} \psi}{k} = v + \frac{\eta Q^{-1} \psi}{k}, \quad v \equiv \frac{-g}{Q}
\end{aligned} \tag{9}$$

Substituting $\bar{v}^T$ into equation (7) and equation (8) gives the desired changes

$$\therefore \Delta u = -k(H_u^\phi + v^T H_u^\psi)^T - \eta(H_u^\psi)^T Q^{-1} \psi$$

$$\therefore \Delta t_f = -k(\dot{\phi} + v^T \dot{\psi}) - \eta \dot{\psi}^T Q^{-1} \psi$$

Summary

Numerical Gradient Algorithms (Direct method)

- 1) Choice k and η
- 2) Guess t_f and $\vec{u}(t)$ $t_0 \leq t \leq t_f$
- 3) Forward integration $\dot{x} = f(x, u, t)$ and store $x(t)$.
- 4) Evaluate $\phi, \psi, \dot{\phi}, \dot{\psi}$ at t_f

$$\text{and determine } \Lambda^\phi(t_f) = \phi_x^T(x(t_f), t_f)$$

$$\Lambda^\psi(t_f) = \psi_x^T(x(t_f), t_f)$$

5) Backward integration for costate equation and store H_u^ϕ , H_u^ψ for

$$t_0 \leq t \leq t_f$$

$$\dot{\Lambda}^\phi = -L_x - f_x \Lambda^\phi, \quad \left(H_u^\phi\right)^T = -L_u - f_u \Lambda^\phi$$

$$\dot{\Lambda}^\psi = -f_x \Lambda^\phi, \quad \left(H_u^\psi\right)^T = -f_u \Lambda^\psi$$

6) Determine $\nu = -Q^{-1}g$

$$\text{where } g = \dot{\psi} \dot{\phi} + \int_{t_0}^{t_f} H_u^\psi \left(H_u^\phi\right)^T dt$$

$$Q = \dot{\psi} \dot{\psi}^T + \int_{t_0}^{t_f} H_u^\psi \left(H_u^\psi\right)^T dt$$

7) Compute $\Delta u(t)$ for $t_0 \leq t \leq t_f$

$$\Delta u = -k \left(H_u^\phi(t) + \nu^T H_u^\psi(t) \right)^T - \eta \left(H_u^\psi(t) \right)^T Q^{-1} \psi$$

8) Compute $\Delta t_f = -k \left(\dot{\phi} + \nu^T \dot{\psi} \right) - \eta \dot{\psi} Q^{-1} \psi$

9) Compute $\delta u_{arg} \equiv \sqrt{\frac{1}{t_f} \int_0^{t_f} (\Delta u^T \Delta u) dt}$

10) If $\delta u_{arg} < tol_1$ and $\Delta t_f < tol_2$, then stop.

11) Compute new $u(t)$ and t_f

$$u^{new}(t) = u^{old}(t) + \Delta u(t)$$

$$t_f^{new} = t_f^{old} + \Delta t_f$$

12) Go to step 3)

Solutions of TPBVP Using Indirect method

Shooting method (Quasi-linearization method)

$$J = h(x(t_f), t_f) + \int_0^{t_f} L(x, u, t) dt$$

$$\dot{x} = f(x, u, t)$$

$$x(0) = \text{given}, \quad x(t_f) = \text{given}$$

$$\dot{\lambda} = -\frac{\partial H}{\partial x} = g(x, u, \lambda, t)$$

$$\frac{\partial H}{\partial u} = 0 \quad \rightarrow u = u(x, \lambda)$$

Substitute $u = u(x, \lambda)$ into $\dot{x} = f(x, u, t)$ and $\dot{\lambda} = g(x, u, \lambda, t)$, then we have

$$\dot{x} = f(x, \lambda, t) \tag{1}$$

$$\dot{\lambda} = g(x, \lambda, t) \tag{2}$$

$$\text{Let } X = \begin{Bmatrix} x \\ \lambda \end{Bmatrix}$$

$$\dot{X} = \begin{Bmatrix} \dot{x} \\ \dot{\lambda} \end{Bmatrix} = \begin{Bmatrix} f \\ g \end{Bmatrix} \equiv F(X, t) \tag{3}$$

The set of differential equations eq. (3) are linearized about a nominal solution $X^k(t)$ for k^{th} trial solution (i.e., the right-hand side of eq (3) is expanded in a Taylor's series and only first-order terms are retained.). The linearized equations are given by

$$\begin{aligned} \dot{X}^k(t) + \Delta \dot{X}^k(t) &= F(X^k, t) + \left. \frac{\partial F(X, t)}{\partial X} \right|_{X^k} \Delta X^k \\ \Delta \dot{X}^k(t) &= \left. \frac{\partial F(X, t)}{\partial X} \right|_{X^k} \Delta X^k + \underbrace{\left[F(X^k, t) - \dot{X}^k(t) \right]}_{\equiv d(t)} \end{aligned} \tag{4}$$

$$\text{Let } Y = \begin{Bmatrix} \Delta x^k \\ \Delta \lambda^k \end{Bmatrix} = \{\Delta X^k\}$$

$$\frac{\partial F}{\partial X} \Big|_{X^k} \equiv A(t) \quad \text{then}$$

$$\dot{Y}(t) = A(t)Y(t) + d(t), \quad d(t) \text{ is going to zero.} \quad (5)$$

Approximate

$$\dot{Y}(t) = A(t)Y(t) \quad (6)$$

This is a linear equation then

$$Y(t) = \Phi(t, t_0)Y(t_0) \quad (7)$$

$$\text{Integrate the eq (5) forward in time using } Y(t_0) = \begin{bmatrix} \Delta x_1(t_0) \\ 0 \\ \vdots \\ 0 \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \\ \vdots \\ 0 \end{bmatrix}$$

Then, what we get is the first column of $\Phi(t_f, t_0)$ as following.

$$Y(t_f) = \Phi(t_f, t_0)Y(t_0) = \begin{bmatrix} \Phi_{11} & \Phi_{12} & \cdots & \Phi_{12n} \\ \Phi_{21} & \ddots & & \vdots \\ \vdots & & \ddots & \Phi_{2n-1, 2n} \\ \Phi_{2n1} & \cdots & \Phi_{2n, 2n-1} & \Phi_{2n, 2n} \end{bmatrix} \begin{bmatrix} 1 \\ 0 \\ \vdots \\ 0 \end{bmatrix} = \begin{bmatrix} \Phi_{11} \\ \Phi_{21} \\ \vdots \\ \Phi_{2n1} \end{bmatrix}$$

Similarly to get the second column of $\Phi(t_f, t_0)$,

$$Y(t_0) = \begin{bmatrix} 0 \\ \Delta x_2(t_0) \\ \vdots \\ 0 \end{bmatrix} = \begin{bmatrix} 0 \\ 1 \\ 0 \\ \vdots \\ 0 \end{bmatrix} \Rightarrow \begin{bmatrix} \Phi_{11} & \Phi_{12} & \cdots & \Phi_{12n} \\ \Phi_{21} & \ddots & & \vdots \\ \vdots & & \ddots & \Phi_{2n-1, 2n} \\ \Phi_{2n1} & \cdots & \Phi_{2n, 2n-1} & \Phi_{2n, 2n} \end{bmatrix} \begin{bmatrix} 0 \\ 1 \\ 0 \\ \vdots \\ 0 \end{bmatrix} = \begin{bmatrix} \Phi_{12} \\ \Phi_{22} \\ \vdots \\ \Phi_{2n2} \end{bmatrix}$$

Iteration

Finally we have $Y(t_f) = \begin{bmatrix} \phi_{11} & \phi_{12} \\ \phi_{21} & \phi_{22} \end{bmatrix} = \Phi(t_f, t_0)$ 2n by 2n matrix

$$\begin{bmatrix} \Delta x(t_f) \\ \Delta \lambda(t_f) \end{bmatrix}_{2n \times 1} = \begin{bmatrix} \phi_{11} & \phi_{12} \\ \phi_{21} & \phi_{22} \end{bmatrix}_{2n \times 2n} \begin{bmatrix} \Delta x(t_0) \\ \Delta \lambda(t_0) \end{bmatrix}_{2n \times 1}$$

$$\Delta x(t_f) = \phi_{11} \Delta x(t_0) + \phi_{12} \Delta \lambda(t_0)$$

$$\rightarrow \Delta \lambda(t_0) = \phi_{12}^{-1} [\Delta x(t_f) - \phi_{11} \Delta x(t_0)]$$

$$\therefore \lambda^{k+1}(t_0) = \lambda^k(t_0) + \Delta \lambda^k(t_0)$$

If we integrate the original state and costate equations forward in time, with $x(t_0)$ given and $\lambda(0) = \lambda^{k+1}(t_0)$, we will get a new solution to linearize about x, λ . Then we solve another linear TPBVP with convergence ($d(t) = 0$)

Algorithm

- 1) Guess the initial costate condition

$$\lambda^k(t_0)$$

- 2) Using $\frac{\partial H}{\partial u} = 0$ equation, control $u(t)$ is calculated in terms of x, λ and u is eliminated in $\dot{x}, \dot{\lambda}$ equations.

$$\begin{aligned} \text{Set } \dot{x}^k &= f(x^k, \lambda^k, t) \\ \dot{\lambda}^k &= g(x^k, \lambda^k, t) = -\frac{\partial H}{\partial x} \end{aligned}$$

- 3) Integrate $\dot{x}^k = f, \dot{\lambda}^k = g$ forward in time using $x(t_0)$ and guessed $\lambda^k(0)$.
- 4) Check $\Delta x(t_f) < \gamma$ (tolerance)

If $\Delta x(t_f) \equiv x^k(t_f) - x(t_f)$, $x^k(t_f)$: calculated, $x(t_f)$: given

If $\Delta x(t_f) < \gamma$, stop, else go to 5

5) Linearize about $x^k(t)$, $\lambda^k(t)$

$$\Delta \dot{x}^k = f'(x^k, \lambda^k, \Delta x^k, \Delta \lambda^k, t)$$

$$\Delta \dot{\lambda}^k = g'(x^k, \lambda^k, \Delta x^k, \Delta \lambda^k, t)$$

6) Obtain the state transition matrix by perturbing $\begin{bmatrix} \Delta x^k(t_0) \\ \Delta \lambda^k(t_0) \end{bmatrix}$

7) Get $\Delta \lambda^k(t_0) = \phi_{12}^{-1} [\Delta x(t_f) - \phi_{11} \Delta x(t_0)]$

$$\lambda^{k+1}(t_0) = \lambda^k(t_0) + \Delta \lambda^k(t_0)$$

Goto step 2)

Shooting method for optimization of Dynamic System with free parameters (constant)

$$\dot{x} = f(x, u, p, t)$$

$x(0)$ and $x(t_f)$ are given

$$J = h(x(t_f), t_f) + \int_0^{t_f} g(x, u, p, t) dt$$

$$\begin{aligned} \delta J = & \left[\frac{\partial h}{\partial x(t_f)} - \Lambda(t_f) \right] \delta x_f \\ & + \int_0^{t_f} \left[\frac{\partial H}{\partial u} \delta u + \left(\frac{\partial H}{\partial x} + \dot{\Lambda} \right) \delta x + \left(\frac{\partial H}{\partial \Lambda} - \dot{x} \right) \delta \Lambda + \frac{\partial H}{\partial p} \delta p \right] dt \end{aligned}$$

$$\delta J = \int_0^{t_f} \frac{\partial H}{\partial p} \delta p dt = 0$$

$$\rightarrow \delta p \int_0^{t_f} \frac{\partial H}{\partial p} dt = 0 \quad \delta p : \text{constant at initial time}$$

$$\rightarrow \int_0^{t_f} \frac{\partial H}{\partial p} dt = 0$$

$$\text{Let } \dot{z} = -\frac{\partial H}{\partial p}$$

$$\int_0^{t_f} \frac{\partial H}{\partial p} dt \rightarrow -[z(t_f) - z(0)] = 0$$

After using the optimality condition $\frac{\partial H}{\partial u} = 0$, we eliminate u in terms of Λ, x

$$\dot{x} = f(x, u, p, t) \quad , \quad x \in \mathbf{R}^n$$

$$\dot{\Lambda} = -\frac{\partial H}{\partial x} \quad , \quad \Lambda \in \mathbf{R}^n$$

$$\dot{z} = -\frac{\partial H}{\partial p} \quad , \quad p \in \mathbf{R}^m$$

$$z(0) = 0, \quad z(t_f) = 0 \quad , \quad z \in \mathbf{R}^m$$

We are looking for values of $\Lambda(0), p$

$$\dot{\Lambda} = g'(x, \Lambda, t) \quad \leftarrow \partial H / \partial u = 0, \quad u = u(x, \Lambda)$$

$$X_1 = \begin{Bmatrix} x \\ z \end{Bmatrix}, \quad X_2 = \{\Lambda\}$$

$$X = \begin{Bmatrix} X_1 \\ X_2 \end{Bmatrix} = \begin{Bmatrix} x \\ z \\ \Lambda \end{Bmatrix}$$

$$\dot{X} = \begin{Bmatrix} \dot{x} \\ \dot{z} \\ \dot{\Lambda} \end{Bmatrix} \equiv F(X, p, t)$$

1) Guess the initial value of $\Lambda(0)$ and p

2) Linearize the state-costate equation about the nominal solution generated from forward integration of the nonlinear equation.

$$\Delta \dot{X} = \frac{\partial F}{\partial X} \Delta X + \frac{\partial F}{\partial p} \Delta p$$

$$\Delta X(t_f) = \Phi \Delta X(0) + \psi \Delta p$$

where Φ is a $(2n+m) \times (2n+m)$ matrix

ψ is a $(2n+m) \times m$ matrix

$$\Phi_{ij} = \frac{\partial \Delta X_i(t_f)}{\partial \Delta X_j(0)}, \quad \psi_{ij} = \frac{\partial \Delta X_i(t_f)}{\partial \Delta p_j}$$

Method A: Finite Difference Method

Change the first state variable slightly from its nominal value.

Hold all other quantities at their nominal values

$$x_1(0) = x_1^N(0) + \varepsilon$$

$$x_i(0) = x_i^N(0), \quad i \neq 1$$

Integrate the $\dot{x}, \dot{\Lambda}, \dot{z}$ equation forward in time.

Note the value of $X(t_f)$

$$\Phi_{ij} = \frac{X(t_f) - X^N(t_f)}{\varepsilon}$$

To get ψ_{ij} , do the same thing with p

Method B:

$$\Delta X(0) = \begin{bmatrix} 1 \\ 0 \\ \vdots \\ 0 \end{bmatrix}, \quad \Delta p = \begin{bmatrix} 0 \\ 0 \\ \vdots \\ 0 \end{bmatrix}$$

Integrate $\Delta \dot{X}$ to get $\Delta X(t_f)$

$$\Rightarrow \Delta x(t_f) = \Phi_{i1}$$

Repeating yields Φ_{ij} matrix.

Similarly, follow the same step with Δp to get ψ_{ij} matrix.

Method B gives more accurate results than F.D.M, but requires more work (analytical partial derivative).

3) After finding Φ and ψ , we are ready to solve for the missing $\Delta X(0)$ and Δp .

$$\begin{bmatrix} \Delta X_1(t_f) \\ \Delta \Lambda(t_f) \end{bmatrix} = \begin{bmatrix} \phi_{11} & \phi_{12} \\ \phi_{21} & \phi_{22} \end{bmatrix} \begin{bmatrix} \Delta X_1(0) \\ \Delta \Lambda(0) \end{bmatrix} + \begin{bmatrix} \psi_1 \\ \psi_2 \end{bmatrix} \Delta p$$

$$\begin{aligned} \Delta X_1(t_f) &= \phi_{11} \Delta X_1(0) + \phi_{12} \Delta \Lambda(0) + \psi_1 \Delta p \\ &= [\phi_{12} \quad \psi_1] \begin{bmatrix} \Delta \Lambda(0) \\ \Delta p \end{bmatrix} \end{aligned}$$

$$\therefore \begin{bmatrix} \Delta \Lambda(0) \\ \Delta p \end{bmatrix} = [\phi_{12} \quad \psi_1]^{-1} \Delta X_1(t_f)$$

Example: Solve the following Optimal control Problem numerically using a shooting method.

$$\text{Minimize } J = \int_0^{t_f} (1 + u^2) dt$$

$$\text{Subject to } \dot{x} = -x + x^3 + u$$

$$x(0) = 1, \quad x(t_f) = 0, \quad t_f \text{ is free}$$

Let's convert to fixed time problem.

$$\text{Define } t = p^2 \tau$$

$$dt = p^2 d\tau$$

where τ is non dimensional time $\tau \in [0,1]$

$$\frac{d}{dt} = \frac{1}{p^2} \frac{d}{d\tau}$$

$$A' \equiv \frac{d}{d\tau} A$$

$$\text{Then, } \dot{x} = -x + x^3 + u \Leftrightarrow x' = (-x + x^3 + u)p^2$$

Let free parameter p be independent variable

$$\text{then, } p' = 0$$

$$\text{and } J = \int_0^1 (1 + u^2) p^2 d\tau$$

The Hamiltonian is

$$H = (1 + u^2) p^2 + \lambda_1 (-x + x^3 + u) p^2 + \lambda_2 0$$

$$H(t_f) = \frac{\partial h}{\partial t_f} = 0$$

$$\lambda_1' = -\frac{\partial H}{\partial x} = -\lambda_1 p^2 (-1 + 3x^2) = \lambda_1 p^2 (1 - 3x^2)$$

$$\lambda_2' = -\frac{\partial H}{\partial p} = -2p(1 + u^2) - 2p\lambda_1 (-x + x^3 + u)$$

$$0 = \frac{\partial H}{\partial u} = p^2 (2u + \lambda_1) \quad , p \neq 0$$

$$\rightarrow u = -\frac{\lambda_1}{2}$$

$$\begin{array}{l} \text{State equation} \\ x' = (-x + x^3 - \frac{1}{2} \lambda_1) p^2 \\ p' = 0 \end{array}$$

$$\begin{array}{l} \text{Costate equation} \\ \lambda_1' = \lambda_1 p^2 (1 - 3x^2) \\ \lambda_2' = -2p(1 + \frac{1}{4} \lambda_1^2) - 2p\lambda_1 (-x + x^3 - \frac{1}{2} \lambda_1) \end{array}$$

$$\begin{array}{l} p ; \text{ parameter} \xrightarrow{\delta J=0} \int_0^1 \frac{\partial H}{\partial p} d\tau = 0 \\ z(0) = z(1) = 0 \\ z' = -\frac{\partial H}{\partial p} \end{array}$$

Let $\lambda_1 = \lambda$, $\lambda_2 = z$

Then, we obtain

$$x' = \left(-x + x^3 - \frac{1}{2}\lambda\right)p^2 \quad (1-1)$$

$$z' = -2p\left(1 + \frac{1}{4}\lambda^2\right) - 2p\lambda\left(-x + x^3 - \frac{1}{2}\lambda\right) \quad (1-2)$$

$$\lambda' = \lambda p^2(1 - 3x^2) \quad (1-3)$$

$$\begin{aligned} x(0) &= 1, & x(1) &= 0 \\ z(0) &= 0, & z(1) &= 0 \end{aligned}$$

We are looking for values of $\lambda(0)$ and p

- 1) Guess the values of $\lambda(0)$ and p
- 2) Linearize the state-costate equations (1-1),(1-2),(1-3) about the nominal solution generated from forward integration of the nonlinear equations.

$$X \equiv \begin{bmatrix} x \\ z \\ \lambda \end{bmatrix}$$

$$X' = F(X, p)$$

$$\Delta X' = \frac{\partial F}{\partial X} \Delta X + \frac{\partial F}{\partial p} \Delta p$$

$$\begin{aligned} \begin{bmatrix} \Delta x' \\ \Delta z' \\ \Delta \lambda' \end{bmatrix} &= \begin{bmatrix} p^2(-1 + 3x^2) & 0 & -\frac{1}{2}p^2 \\ 2p\lambda(1 - 3x^2) & 0 & -2px + 2px^3 + p\lambda \\ -6\lambda p^2 x & 0 & p^2(1 - 3x^2) \end{bmatrix} \begin{bmatrix} \Delta x \\ \Delta z \\ \Delta \lambda \end{bmatrix} \\ &\quad + \begin{bmatrix} 2p(-x + x^3 - \frac{1}{2}\lambda) \\ -2 + \frac{1}{2}\lambda^2 + 2\lambda x - 2\lambda x^3 \\ 2p\lambda(1 - 3x^2) \end{bmatrix} \Delta p \end{aligned} \quad (2)$$

$$\Delta X(1) = \Phi \Delta X(0) + \Psi \Delta p \quad (3)$$

Let $\Delta X_i(0) = \varepsilon$ ($i=1$) and all other $\Delta X_j(0) = 0$ ($i \neq j$) and $\Delta p = 0$

Integrate equation (2) then, by equation (3)

$$\Delta X(1) = \Phi(*, i) \varepsilon$$

Doing this for all known $\Delta X_i(0)$, we obtain

$$\Delta X(1) = [\Phi(*, 1), \Phi(*, 2), \Phi(*, 3)] \Delta X(0)$$

Similarly follow the same step with Δp also, then we get Ψ

$$\Delta X(1) = \Psi \Delta p$$

After finding Φ and Ψ , we are ready to solve for missing $\Delta X(0)$ and Δp .

$$\begin{bmatrix} \Delta x(1) \\ \Delta z(1) \\ \Delta \lambda(1) \end{bmatrix} = \begin{bmatrix} \phi_{11} & \phi_{12} & \phi_{13} \\ \phi_{21} & \phi_{22} & \phi_{23} \\ \phi_{31} & \phi_{32} & \phi_{33} \end{bmatrix} \begin{bmatrix} \Delta x(0) \\ \Delta z(0) \\ \Delta \lambda(0) \end{bmatrix} + \begin{bmatrix} \psi_1 \\ \psi_2 \\ \psi_3 \end{bmatrix} \Delta p$$

$$\begin{bmatrix} \Delta x(1) \\ \Delta z(1) \end{bmatrix} = \begin{bmatrix} \phi_{11} & \phi_{12} \\ \phi_{21} & \phi_{22} \end{bmatrix} \begin{bmatrix} \Delta x(0) \\ \Delta z(0) \end{bmatrix} + \begin{bmatrix} \phi_{13} \\ \phi_{23} \end{bmatrix} \Delta \lambda(0) + \begin{bmatrix} \psi_1 \\ \psi_2 \end{bmatrix} \Delta p$$

$$\therefore \begin{bmatrix} \Delta \lambda(0) \\ \Delta p \end{bmatrix} = \begin{bmatrix} \phi_{13} & \psi_1 \\ \phi_{23} & \psi_2 \end{bmatrix}^{-1} \begin{bmatrix} \Delta x(1) \\ \Delta z(1) \end{bmatrix}$$

$$\therefore \lambda^{new}(0) = \lambda^{old}(0) + \Delta \lambda(0)$$

$$p^{new} = p^{old} + \Delta p$$

Repeat above procedure until

$$\Delta x(1) = x_g(1) - x_c(1) = 0$$

$$\Delta z(1) = z_g(1) - z_c(1) = 0$$

The answer, $p=0.9822 \rightarrow t_f = p^2 = 0.9648$

Parameter Optimization

$$\dot{x} = f(x, u, t)$$

$$x(0), x(t_f); \text{ given}$$

$$|u| \leq 1, G(x) \leq 0$$

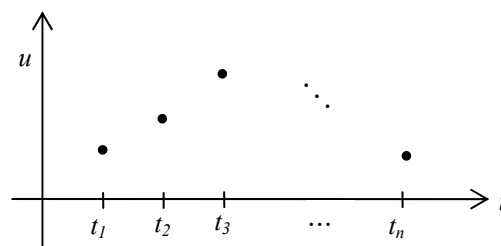
$$J = \int_0^{t_f} g(x, u, t) dt$$

Let $u(t)$ be parameterized by

- 1) Polynomials in time
- 2) Splines in time
- 3) Discrete control vectors

$$u = \sum_{i=0}^N p_i t^i$$

N is fixed



Discretization $\rightarrow$ “sub-optimal solutions”; not exact optimal solutions, approximate solutions (higher performance index)

$$J = J(u_i), u_i \text{ are the parameters}$$

$$x(t_f) \Rightarrow x(t_f, u_i)$$

$$|u(t_i)| \leq |u_i| \leq 1$$

$$G(x) \Rightarrow G(u_i) \leq 0$$

Polynomial Approximation Method

It has been shown that a polynomial of sufficient high degree can be used to approximate a continuous function to essentially arbitrary accuracy. We denote each polynomial in the series as a basis function.

The vector $\vec{x}(t)$ in the TPBVP is approximated by

$$\vec{x}(t) = [\Phi(t)]A \quad (1)$$

$$\text{where } \Phi(t) = [\phi_1(t), \phi_2(t), \dots, \phi_{NB}(t)]^T$$

$$A = [A_1^T(t), A_2^T(t), \dots, A_{NB}^T(t)]^T$$

$$A_i = [a_{i1}, a_{i2}, \dots, a_{iNB}]^T$$

NB is the number of basis function used. Each of the basis function $\phi_i(t)$ represents an $(i-1)^{th}$ degree polynomials in t , forming linearly independent, complete set.

Each of the a_{ij} represents a coefficient such that the i^{th} variable $x_i(t)$ is represented as

$$x_i(t) = \sum_{j=1}^{NB} a_{ij} \phi_j(t) \quad , \quad i = 1, 2, \dots, N$$

The $\phi_j(t)$ are specified function of time (typically orthogonal polynomials) and their derivative usually can be obtained through simple recurrence relations.

Differentiating Equation (1)

$$\dot{\vec{x}}(t) = [\dot{\Phi}(t)]A \quad (2)$$

Consider the linear systems of equations

$$\dot{\vec{x}}(t) = \vec{F}(t)\vec{x} + \vec{D}(t) \quad (3)$$

with boundary conditions

$$\mathbf{u}\vec{x}(0) = \alpha \quad (4)$$

$$\mathbf{v}\vec{x}(t_f) = \beta \quad (5)$$

$\mathbf{u}, \mathbf{v}$; known $N \times N$ matrix

α, β ; known constant $N \times 1$ vector

Substituting equation (1) and equation (2) into equation (3)

$$\dot{\Phi}A = F\Phi A + D \quad (6)$$

$$\underbrace{[\dot{\Phi} - F\Phi]}_{\equiv P_1}A = D$$

$$P_1(t)A = D \quad (7)$$

When P_1 is an $[N \times (N \times NB)]$ known time varying matrix, equation (7) can be evaluated at a number NS of sample points at times t_i on $[0, T]$ and thereby establish enough equations to determine the coefficient vector A . We must ensure satisfaction of boundary conditions equation (1) and equation (4), equation (5) yields the two boundary conditions

$$P_2 A = \alpha \quad (8)$$

$$P_3 A = \beta \quad (9)$$

$$\text{where } P_2 = \mathbf{u}\Phi(0), \quad P_3 = \mathbf{v}\Phi(t_f)$$

Equation (7) together eq (8), eq (9) yields a merged linear system (An Algebraic Equation)

$$PA = Q \quad (10)$$

$$\text{where } P = \begin{bmatrix} P_1(t_1) \\ P_1(t_2) \\ P_1(t_3) \\ \vdots \\ P_1(t_{NS}) \\ P_2 \\ P_3 \end{bmatrix}, \quad Q = \begin{bmatrix} D(t_1) \\ D(t_2) \\ D(t_3) \\ \vdots \\ D(t_{NS}) \\ \alpha \\ \beta \end{bmatrix}$$

$$P; [N \times (NS + 1)] \times [N \times NB]$$

$$Q; [N \times (NS + 1)] \times 1$$

Collocation method; NS is selected such that P is a square matrix, i.e., $NS = NB - 1$, then the coefficient a_{ij} in A are determined uniquely by solving the linear system equation (10). Thus the solution $\bar{x}(t)$ can be obtained from equation (1). This solution satisfies the differential equation only at the sample points, however the boundary conditions are satisfied exactly. As NB is increased sufficiently, the average error at intermediate points typically tends to zero.

The Nonlinear Programming Problem

Optimization problem involves constraints. If both the performance index and constraints are linear functions of the variables, the optimization problem is referred to as a linear programming problem. When nonlinear terms are present in the constraints or in performance index the problem is referred to as a nonlinear programming problem.

Consider the problem of extremizing the nonlinear performance index p

$$p = f(\bar{x}) \tag{1}$$

Where the real x_i of $\vec{x} = [x_1, x_2, \dots, x_n]^T$ are constrained by the following sets of equations;

$$g_i(\vec{x}) = C_i \quad , \quad i = 1, 2, \dots, k < n \quad (2)$$

$$g_i(\vec{x}) \leq C_i \quad , \quad i = k+1, k+2, \dots, m \quad (3)$$

All C_i are constraints, and all $g_i(\vec{x})$ are assumed to be of class C^1 and real values.

A) Unconstrained cases

$$\min f(\vec{x})$$

1) First-order Gradient method

$$\vec{x}^{new} = \vec{x}^{old} - \alpha \frac{\partial f}{\partial \vec{x}}$$

2) Second order Gradient Method

$$f(\vec{x}^k) \approx f(\vec{x}^*) + \underbrace{\underline{G}^k}_{\text{Jacobian}} \Delta \vec{x}^k + \frac{1}{2} \Delta \vec{x}^k \underbrace{\underline{H}^k}_{\text{Hessian}} \Delta \vec{x}^k$$

$\vec{x}^k$; current guess

Minimize $\Delta f \approx f(\vec{x}^k) - f(\vec{x}^*)$ with respect to $\Delta \vec{x}^k$

$$\rightarrow \frac{\partial \Delta f}{\partial \Delta \vec{x}} = 0$$

$$G^k + H^k \Delta \vec{x}^k = 0 \rightarrow \Delta \vec{x}^k = -[H^k]^{-1} G^k$$

$$\vec{x}^{k+1} = \vec{x}^k - \alpha [H^k]^{-1} G^k$$

Check decent

$$\Delta f(\vec{x}) \approx \frac{\partial f}{\partial \vec{x}} \Delta \vec{x}^k = -\alpha G^k [H^k]^{-1} G^k < 0$$

If H^k is positive definite, it is expensive to evaluate $[H^k]^{-1}$

3) Quasi-Newton Method

; Approximate the inverse of the Hessian matrix.

The search direction is still along the negative gradient

$$\Delta \vec{x}^k \equiv -A^k G^k$$

where $A^k \approx [H^k]^{-1}$ approximation to the inverse of the Hessian matrix.

$$A^{k+1} = A^k + \Delta A^k, \quad A^0 = I$$

We want to update A^k such that

$$\Delta \vec{x}^k = A^{k+1} \underbrace{[G^{k-1} - G^k]}_{=\Delta G^k} \Rightarrow \underbrace{A^{k+1}}_{A^k + \Delta A^k} = \left[\frac{\Delta G^k}{\Delta \vec{x}^k} \right]^{-1} \text{ looks like inverse Hessian}$$

$$\Delta \vec{x}^k = A^{k+1} \Delta G^k = (A^k + \Delta A^k) \Delta G^k$$

$$\Delta A^k \Delta G^k = \Delta \vec{x}^k - A^k \Delta G^k$$

$$\rightarrow \Delta A^k = \frac{\Delta \vec{x}^k}{\Delta G^k} - \frac{A^k \Delta G^k}{\Delta G^k}$$

The general solution to the above equation is

$$\Delta A^k = \frac{\Delta \vec{x}^k \vec{y}^T}{\vec{y}^T \Delta G^k} - \frac{A^k \Delta G^k \vec{z}^T}{\vec{z}^T \Delta G^k}$$

for arbitrary $\vec{y}$ & $\vec{z}$

DFP (Davidan, Flerher, Dowell) suggest

$$\vec{y} = \Delta \vec{x}^k, \quad \vec{z} = A^k \Delta G^k$$

$$\therefore A^{k+1} = A^k + \Delta A^k$$

$$\Delta A^k = \frac{\Delta \vec{x}^k (\Delta \vec{x}^k)^T}{(\Delta \vec{x}^k)^T \Delta G^k} - \frac{A^k \Delta G^k (\Delta G^k)^T (A^k)^T}{(\Delta G^k)^T (A^k)^T \Delta G^k}$$

This formula relate symmetry and positive definition of A^{k+1} in the absence of round-off-errors.

B) Constrained (equality) minimization cases

1) Gradient projection

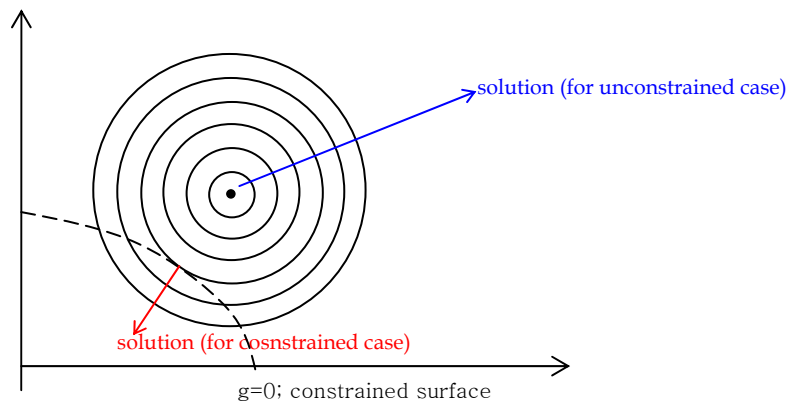
$$\begin{aligned} \text{Min } f(\vec{x}), \quad \vec{x} \in \mathbf{R}^N \\ \text{subject to } \vec{g}(\vec{x}) = 0, \quad \vec{g} \in \mathbf{R}^M, \quad M < N \end{aligned}$$

Necessary condition

$$\text{Lagrangian } L = f + \lambda^T g$$

$$\frac{\partial L}{\partial \vec{x}} = 0 = \underbrace{\frac{\partial f}{\partial \vec{x}} + \frac{\partial \vec{g}^T}{\partial \vec{x}} \lambda}_{\text{projected gradient}}$$

$$\Delta \vec{x}^k = -\alpha \frac{\partial L^k}{\partial \vec{x}}$$



Starting with a feasible solution-

Step 1) Assume that we have a feasible solution

A feasible solution satisfied the constraints

$$\text{Define } \frac{\partial \mathbf{g}^T}{\partial \vec{x}} \equiv A, \quad A^T = \left[\frac{\partial g_i}{\partial \vec{x}_j} \right]_{M \times N \text{ matrix}}$$

$$\text{Solve } \frac{\partial L}{\partial \vec{x}} = \frac{\partial f}{\partial \vec{x}} + \frac{\partial \mathbf{g}^T}{\partial \vec{x}} \lambda = 0$$

$$A\lambda = -\frac{\partial f}{\partial \vec{x}}$$

$$\rightarrow \lambda = -\left(A^T A\right)^{-1} A^T \frac{\partial f}{\partial \vec{x}}$$

$$\text{Thus } \frac{\partial L}{\partial \vec{x}} = \left[I - A\left(A^T A\right)^{-1} A^T \right] \frac{\partial f}{\partial \vec{x}}$$

$$\Delta \vec{x}^k = -\alpha \frac{\partial L^k}{\partial \vec{x}} = \left[I - A\left(A^T A\right)^{-1} A^T \right] \frac{\partial f}{\partial \vec{x}}$$

Step 2) Get back to the constraint ($g(\vec{x}^{k+1}) = 0$)

linearize the constraint

$$g(\vec{x}^k + \Delta \vec{x}^k) = g(\vec{x}^k) + A\Delta \vec{x}^k = 0$$

$$A\Delta \vec{x}^k = -g(\vec{x}^k) \quad \text{approximate solution}$$

$$\Delta \vec{x}^k = -\alpha \left(A^T A\right)^{-1} A^T g(\vec{x}^k)$$

Repeat Step 2) and 3) until convergence.

2) Multiplier penalty problem or augmented Lagrangian problem

Consider: Min x , subjected to $x = 1$

$$L = x + \frac{1}{2}\sigma(x-1)^2$$

where $\frac{1}{2}\sigma(x-1)^2$; known penalty weight ($\sigma > 0$)

$$\frac{\partial L}{\partial x} = 0 \Rightarrow 1 + \sigma(x-1) = 0$$

$$x = 1 - \frac{1}{\sigma}$$

To get correct answer, $\sigma^* \rightarrow \infty, x^* \rightarrow 1$

Min $f(\vec{x})$, subjected to $g(\vec{x}) = 0, g(\vec{x}) \in \mathbf{R}^M$

Construct the augmented Lagrangian as follows,

$$\phi_\sigma(\vec{x}, \theta) = f(\vec{x}) + \frac{1}{2} \sum_{i=1}^m \sigma_i [g_i(\vec{x}) - \theta_i]^2$$

σ_i and θ_i are parameters

$$\phi_\sigma = f(\vec{x}) + \frac{1}{2} \sum_{i=1}^m \sigma_i g_i^2(\vec{x}) - \sum_{i=1}^m \sigma_i g_i(\vec{x}) \theta_i + \frac{1}{2} \sum_{i=1}^m \sigma_i \theta_i^2$$

Since we are optimizing with respect to $\vec{x}$, neglect $\frac{1}{2} \sum_{i=1}^m \sigma_i \theta_i^2$ in the above equation and define $\Lambda_i = \sigma_i \theta_i$

θ_i are parameters that artificially shift the origin. The idea is to increase σ_i (finite) and decrease θ_i so that Λ_i is finite.

Theorem: If $\vec{x}^*$ and $\vec{\lambda}^*$ satisfy second order sufficient conditions

$$\left[\frac{\partial \phi}{\partial x} = 0, \frac{\partial \phi}{\partial \lambda} = 0; Y \frac{\partial^2 \phi}{\partial x^2} Y > 0, \text{ such that } \frac{\partial g}{\partial x} Y = 0 \text{ for any } Y \neq 0 \right], \text{ then there}$$

exists $\bar{\sigma} > 0$, such that for any $\sigma > \bar{\sigma}$, $\vec{x}^*$ is strict relative minimum of

$$\phi_\sigma(\vec{x}, \vec{\lambda}^*)$$

The new problem is Min ϕ_σ with respect to $\vec{x}$

$$\phi_\sigma(\vec{x}, \vec{\Lambda}) = f(\vec{x}) - \vec{\Lambda}^T g(\vec{x}) + \frac{1}{2} \sum_{i=1}^m \sigma_i g_i^2(\vec{x}) \quad \sigma_i; \text{const.}$$

$$\vec{x}^{k+1} = \vec{x}^k - \frac{\partial \phi_\sigma}{\partial \vec{x}^k}$$

$$\vec{\Lambda}^{k+1} = \vec{\Lambda}^k - \frac{\partial \phi_\sigma}{\partial \vec{\Lambda}^k} = \vec{\Lambda} + g(\vec{x}^k)$$

3) Quadratic Programming

$$\text{Min } f(\vec{x}) = \frac{1}{2} \vec{x}^T Q \vec{x} + c^T \vec{x}, \quad Q; \text{ symmetric positive definite}$$

$$\text{subject to } g(\vec{x}) = A\vec{x} - b = 0$$

(linear equality constraints)

Necessary condition

$$L(\vec{x}, \vec{\Lambda}) = f(\vec{x}) + \vec{\Lambda}^T (A\vec{x} - b)$$

$$\frac{\partial L}{\partial \vec{x}} = 0 = Q\vec{x} + c + \vec{\Lambda}^T A$$

$$\frac{\partial L}{\partial \vec{\Lambda}} = 0 = A\vec{x} - b$$

$$\begin{bmatrix} Q & A^T \\ A & 0 \end{bmatrix} \begin{bmatrix} \vec{x} \\ \vec{\Lambda} \end{bmatrix} = \begin{bmatrix} -\vec{c} \\ b \end{bmatrix}$$

$$\begin{bmatrix} Q & A^T \\ A & 0 \end{bmatrix}^{-1} = \begin{bmatrix} H & P^T \\ P & S \end{bmatrix}$$

$$H = Q^{-1} - Q^{-1} A^T (A Q^{-1} A^T)^{-1} A Q^{-1}$$

$$P = (A Q^{-1} A^T)^{-1} A Q^{-1}$$

$$S = (A Q^{-1} A^T)^{-1}$$

$$\therefore \begin{bmatrix} \vec{x} \\ \vec{\Lambda} \end{bmatrix} = \begin{bmatrix} H & P^T \\ P & S \end{bmatrix} \begin{bmatrix} -\vec{c} \\ b \end{bmatrix}$$

$$\begin{aligned}\bar{\mathbf{x}} &= -\mathbf{H}\bar{\mathbf{c}} + \mathbf{P}^T \mathbf{b} \\ \bar{\Lambda} &= -\mathbf{P}\mathbf{c} + \mathbf{S}\mathbf{b}\end{aligned}$$

4) Sequential Quadratic programming (SQP)

$$\text{Min } f(\bar{\mathbf{x}}) \quad \text{subject to } g_i(\bar{\mathbf{x}}) = 0, \quad i = 1, 2, \dots, m$$

$$f(\bar{\mathbf{x}}^k + \Delta \bar{\mathbf{x}}^k) \approx f(\bar{\mathbf{x}}^k) + \nabla f(\bar{\mathbf{x}}^k) \Delta \bar{\mathbf{x}}^k + \frac{1}{2} \Delta \bar{\mathbf{x}}^k H^k \Delta \bar{\mathbf{x}}^k$$

$$g_i(\bar{\mathbf{x}}^k + \Delta \bar{\mathbf{x}}^k) \approx g_i(\bar{\mathbf{x}}^k) + \nabla g_i(\bar{\mathbf{x}}^k) \Delta \bar{\mathbf{x}}^k$$

sub-problem (solved by Quadratic Programming)

$$\begin{aligned}\text{Min } \nabla f(\bar{\mathbf{x}}^k) \Delta \bar{\mathbf{x}}^k + \frac{1}{2} \Delta \bar{\mathbf{x}}^k H^k \Delta \bar{\mathbf{x}}^k \\ \text{subject to } g_i(\bar{\mathbf{x}}^k) + \nabla g_i(\bar{\mathbf{x}}^k) \Delta \bar{\mathbf{x}}^k = 0 \\ i = 1, 2, \dots, m\end{aligned}$$

solve for $\Delta \bar{\mathbf{x}}^k$ and $\bar{\Lambda}^k$ using Gradient Projection or Quadratic programming.

C) Constrained (inequality) minimization cases

1) Augmented Lagrangian

$$\text{Min } f(\bar{\mathbf{x}}), \text{ subjected to } g(\bar{\mathbf{x}}) \geq 0, \quad g(\bar{\mathbf{x}}) \in \mathbf{R}^M$$

Construct the augmented Lagrangian as follows,

$$\phi_\sigma(\bar{\mathbf{x}}, \theta) = f(\bar{\mathbf{x}}) + \frac{1}{2} \sum_{i=1}^m \sigma_i [g_i(\bar{\mathbf{x}}) - \theta_i]^2$$

σ_i and θ_i are parameters

$$\phi_\sigma = f(\vec{x}) + \frac{1}{2} \sum_{i=1}^m \sigma_i g_i^2(\vec{x}) - \sum_{i=1}^m \sigma_i g_i(\vec{x}) \theta_i + \frac{1}{2} \sum_{i=1}^m \sigma_i \theta_i^2$$

Define $\Lambda_i = \sigma_i \theta_i > 0$

The new problem is Min ϕ_σ with respect to $\vec{x}$

$$\phi_\sigma(\vec{x}, \vec{\Lambda}) = f(\vec{x}) - \vec{\Lambda}^T \vec{g}(\vec{x}) + \frac{1}{2} \sum_{i=1}^m \sigma_i g_i^2(\vec{x}) \quad \sigma_i; \text{const.}$$

$$\vec{x}^{k+1} = \vec{x}^k - \alpha \frac{\partial \phi_\sigma}{\partial \vec{x}^k}$$

$$\Lambda_i^{k+1} = \Lambda_i^k - \min \left[\sigma_i g_i(\vec{x}^k), \Lambda_i^k \right]$$

If $g_i > 0 \rightarrow \Lambda_i = 0$

: i.e., i th constraint inactive

2) Kuhn Tucker Method

Min $f(\vec{x})$, subjected to $g(\vec{x}) \geq 0$, $g(\vec{x}) \in \mathbf{R}^M$

Lagrangian : $L = f(x) + \Lambda^T g$

$$\text{using } \frac{\partial L}{\partial x} = 0, \quad \text{solve for } x$$

$$\Lambda_i > 0, \quad \text{if } g_i = 0$$

$$\Lambda_i = 0, \quad \text{if } g_i > 0$$

If constraints are violated, the cost should be increased ($\Lambda_i > 0$). Check the sign of Λ_i in order to know if the answer is optimal.